

Canonical GHZ configurations - cluster P4 - helicity flipping

species: electron
polarization cluster: P4
source cluster configurations: 81
canonical configurations collected: 39

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity flipping
classification: complete-chain R_flip/conserved > 1
per-page labels: isolated flip/conserved and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 54 configurations
- 4 components: 8 configurations
- 5 components: 3 configurations
- 6 components: 14 configurations
- 7 components: 2 configurations

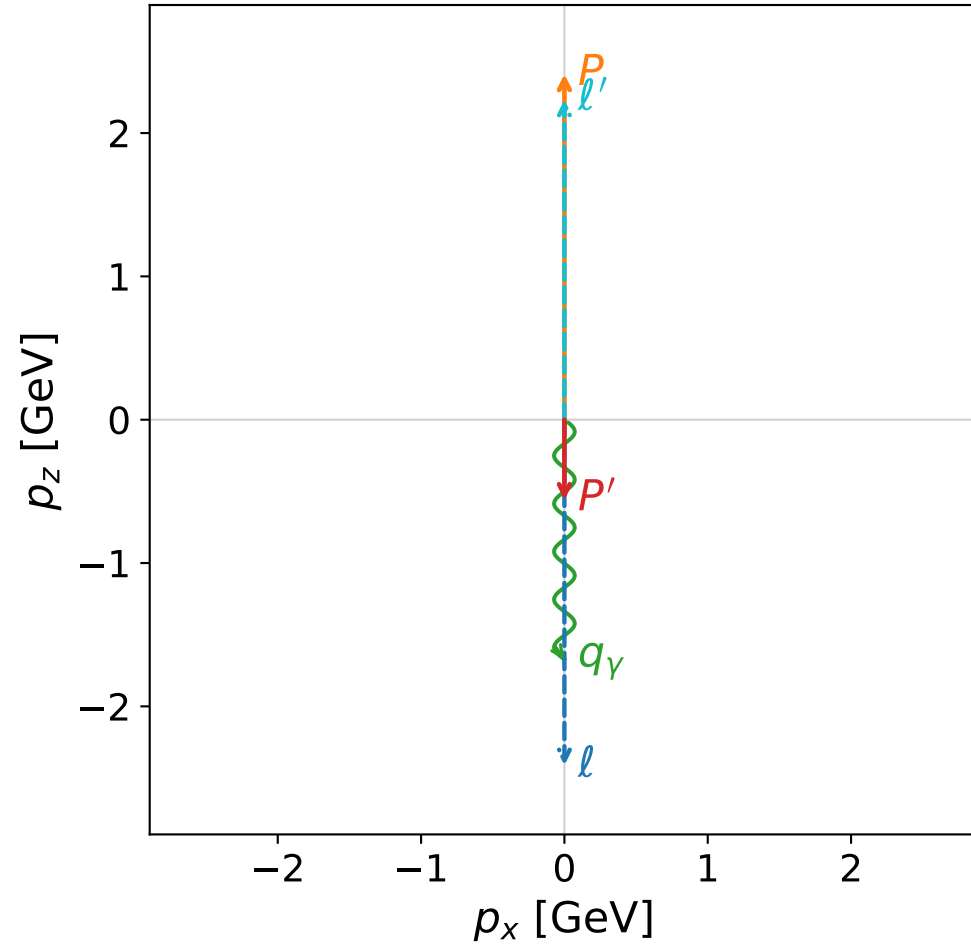
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_04\combined\
helicity_flipping\canonical_dghz_polarization_cluster_04_helicity_flipping_configurati
ons_electron.csv

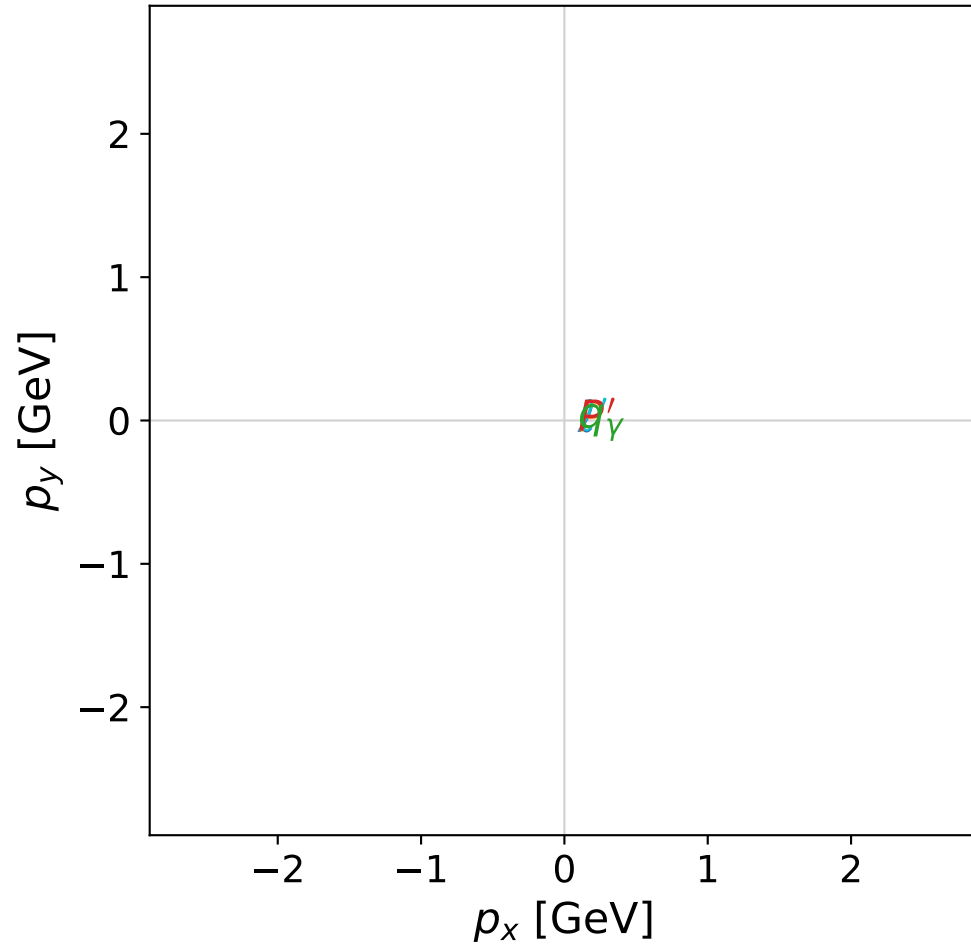
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_8

$d_{\text{GHZ}}=2.4276\text{e-}08$

region: polarization_cluster_04_local_minimum_8

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0008

lepton mixing angle: $\alpha_e=2.332037$ rad

incoming lepton: $-0.68982|+\rangle + 0.723981|-\rangle$

proton mixing angle: $\alpha_p=2.3320298$ rad

incoming proton: $-0.689815|+\rangle + 0.723986|-\rangle$

$s=24.9835$, $\sqrt{s}=4.99835$, $|\vec{P}|=2.41116$, $|\vec{P}'|=0.561095$

$\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=4.05473$, $\phi_\gamma=1.2574$

$E_\gamma=1.67212$, $\theta_{l'}=0$, $\phi_{l'}=4.56305$

$Q^2=21.5386$, $x_B=0.923711$, $t=-6.60174$, $W^2=2.6587$, $y=0.967382$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.4112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4112$

P $E= 2.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4112$

l' $E= 2.2332$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2332$

P' $E= 1.0930$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.5611$

q_γ $E= 1.6721$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6721$

GHZ mechanism: helicity flipping (full H G C)

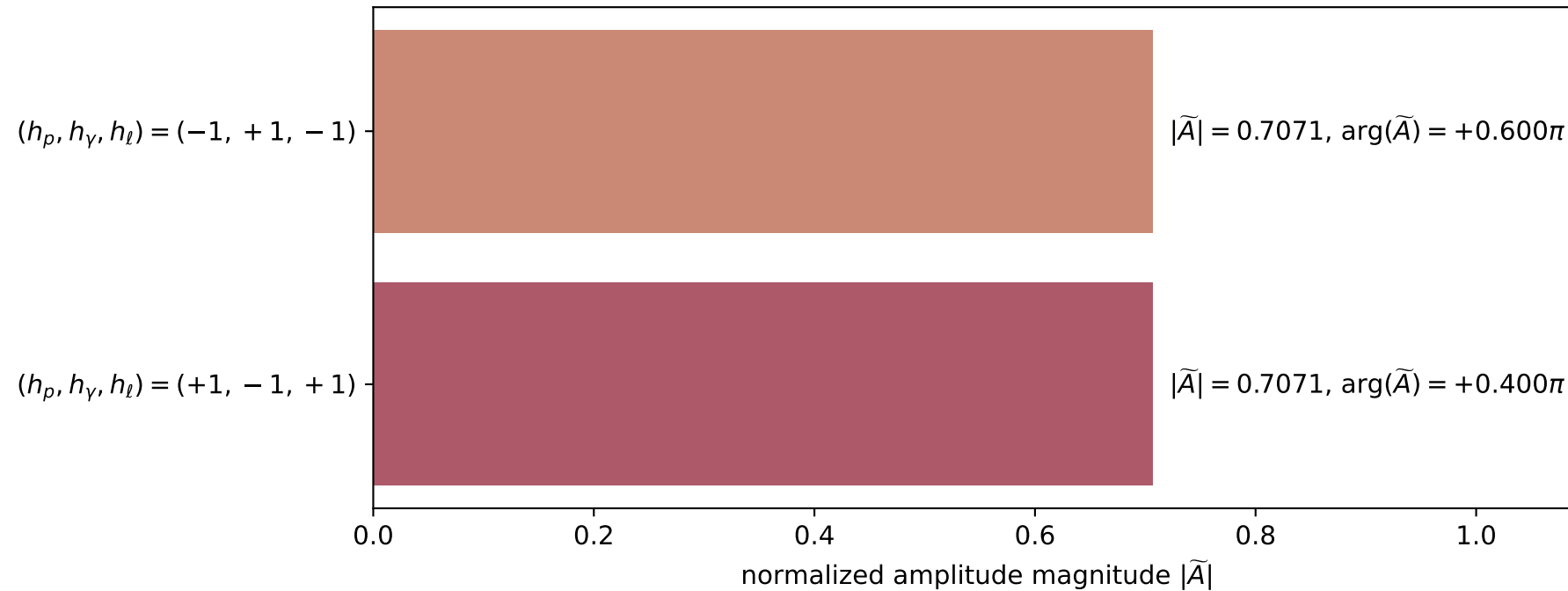
helicity powers: flip 100.000%, conserve 0.000%

R_flip/conserve=1.310e+24; I_h=-7.616e-15

virtual-photon powers: T 100.000%, L 0.000%

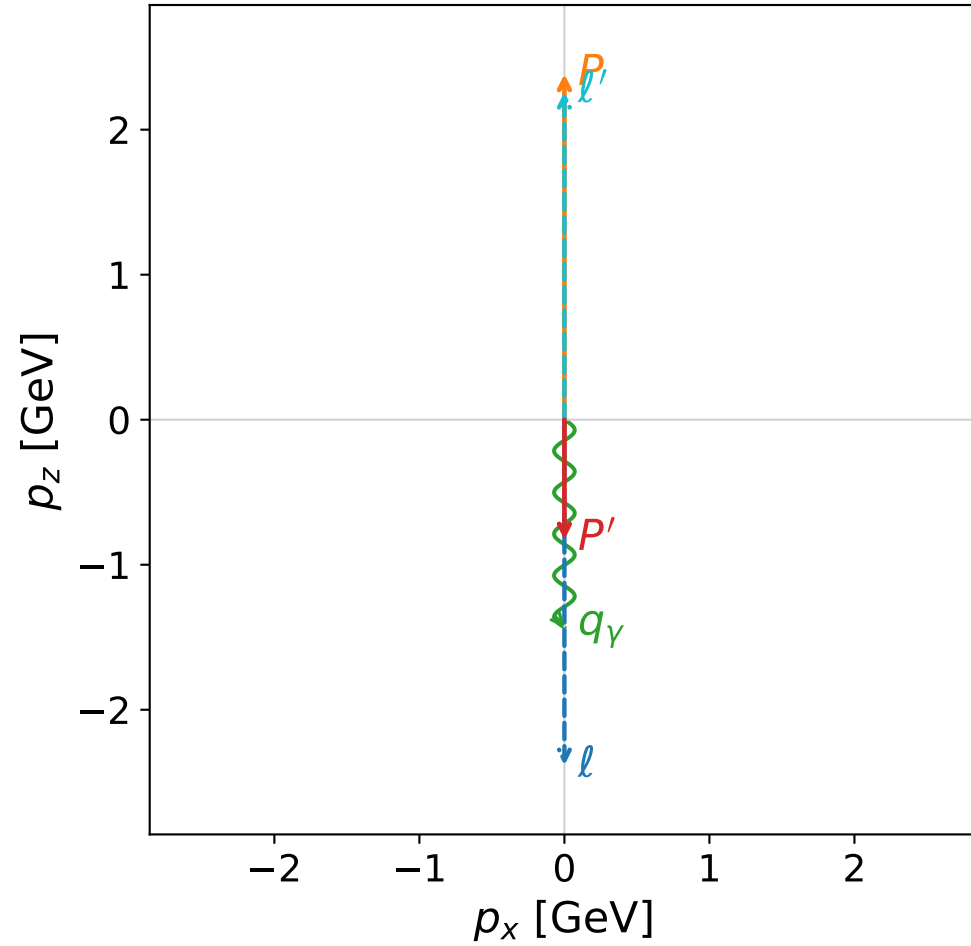
R_L/T=2.327e-29; I_TL=+2.303e-19

Leading normalized final-state amplitudes (N=2, retained=100.0%)

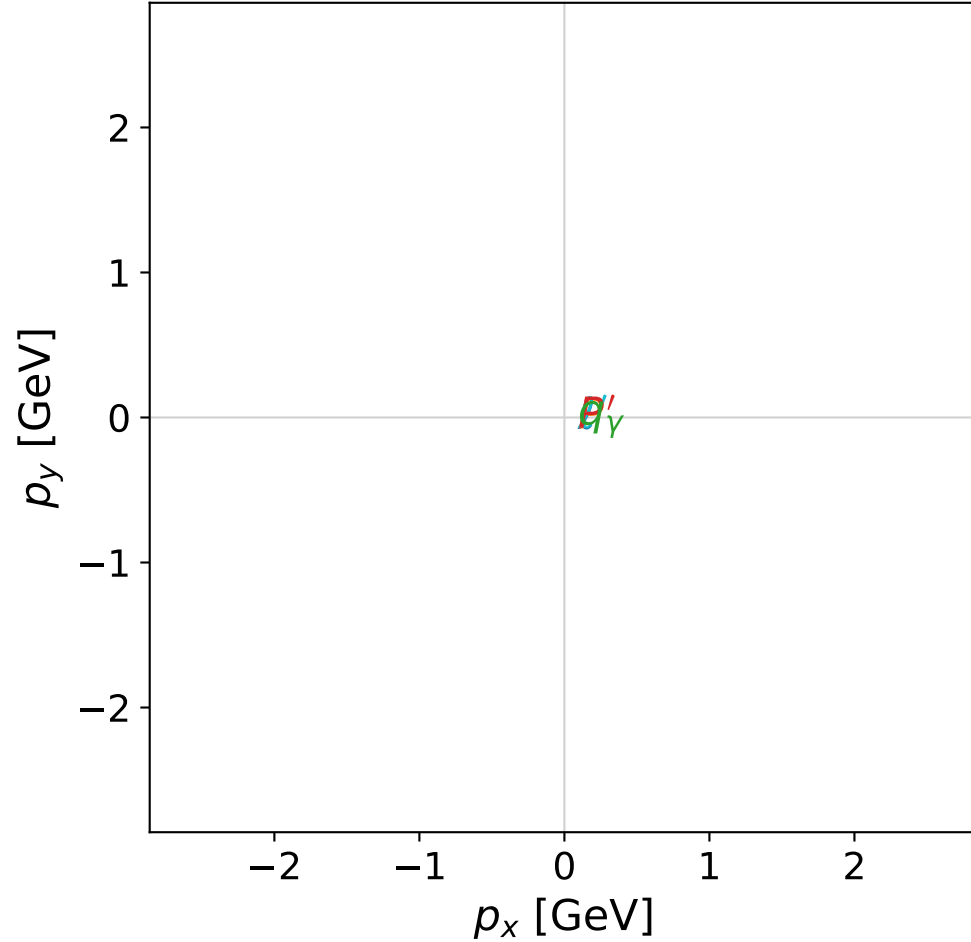


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_9

$d_{\text{GHZ}}=2.43084\text{e-}08$

region: polarization_cluster_04_local_minimum_9

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0009

lepton mixing angle: $\alpha_e=2.380064$ rad

incoming lepton: $-0.723782|+\rangle +0.690029|-\rangle$

proton mixing angle: $\alpha_p=2.3800831$ rad

incoming proton: $-0.723795|+\rangle +0.690015|-\rangle$

$s=24.4448$, $\sqrt{s}=4.94417$, $|\vec{P}|=2.38311$, $|\vec{P}'|=0.827879$

$\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14928$, $\phi_\gamma=4.99248$

$E_\gamma=1.4326$, $\theta_{l'}=0$, $\phi_{l'}=1.26818$

$Q^2=21.5479$, $x_B=0.946724$, $t=-8.5944$, $W^2=2.09243$, $y=0.965859$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.3831$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3831$

P $E= 2.5611$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3831$

l' $E= 2.2605$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2605$

P' $E= 1.2511$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.8279$

q_γ $E= 1.4326$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4326$

GHZ mechanism: helicity flipping (full H G C)

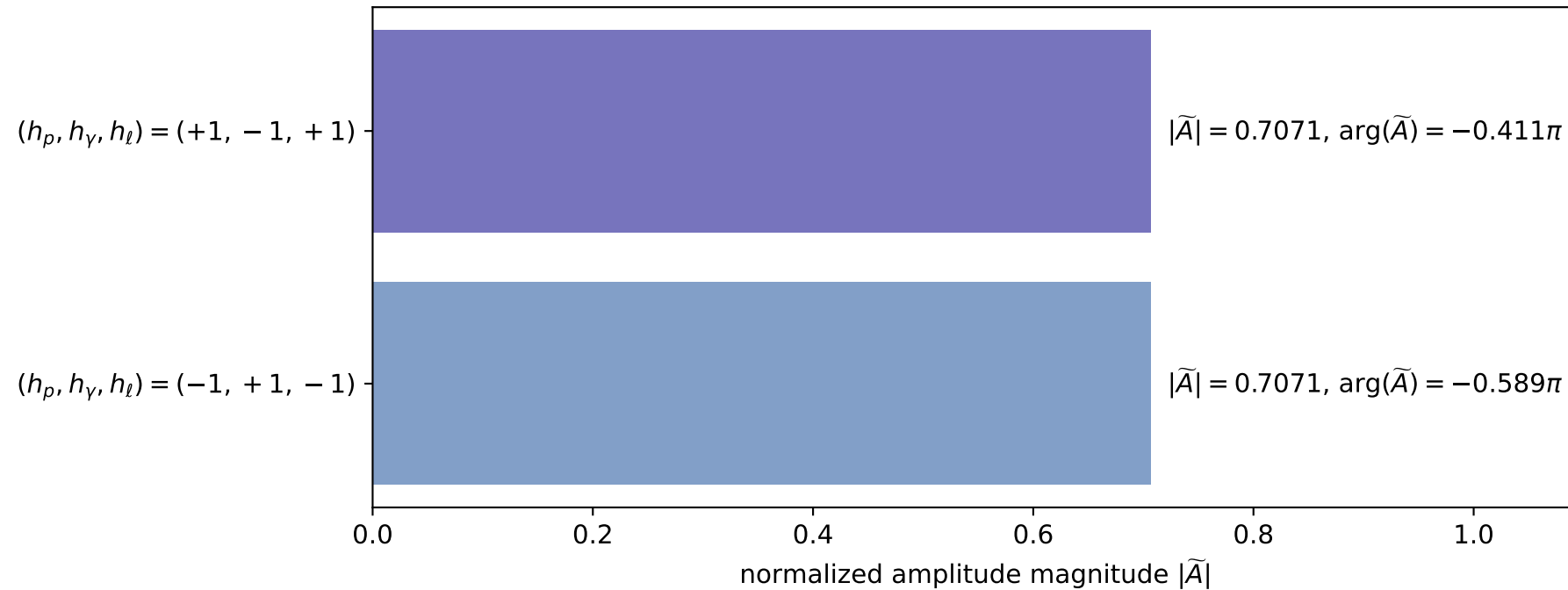
helicity powers: flip 100.000%, conserve 0.000%

R_flip/conserve=9.516e+23; I_h=+1.002e-14

virtual-photon powers: T 100.000%, L 0.000%

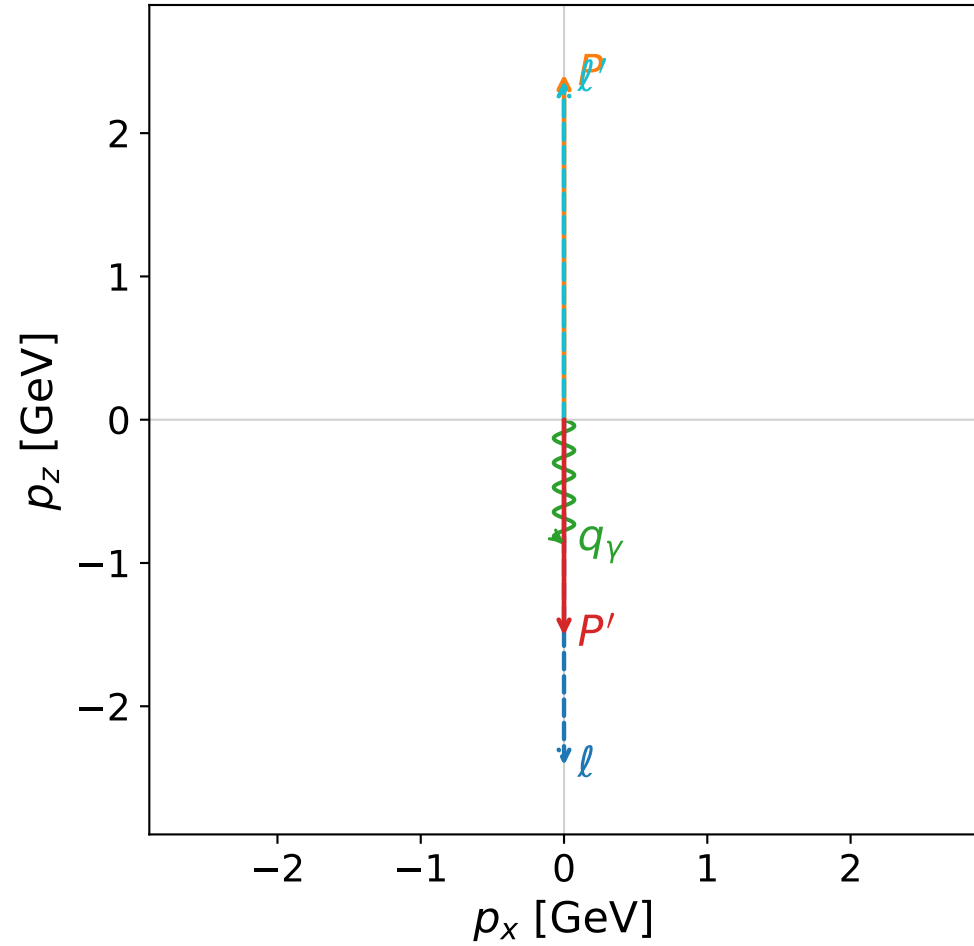
R_L/T=6.682e-29; I_TL=-1.383e-19

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

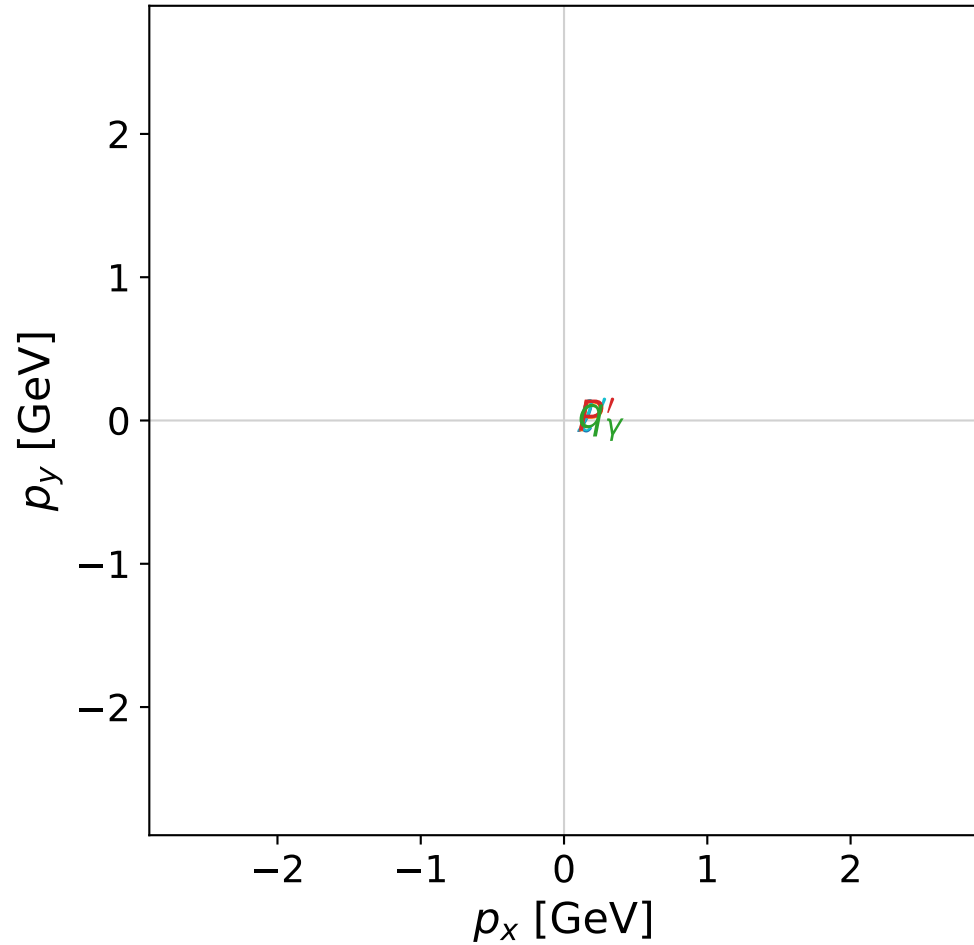


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

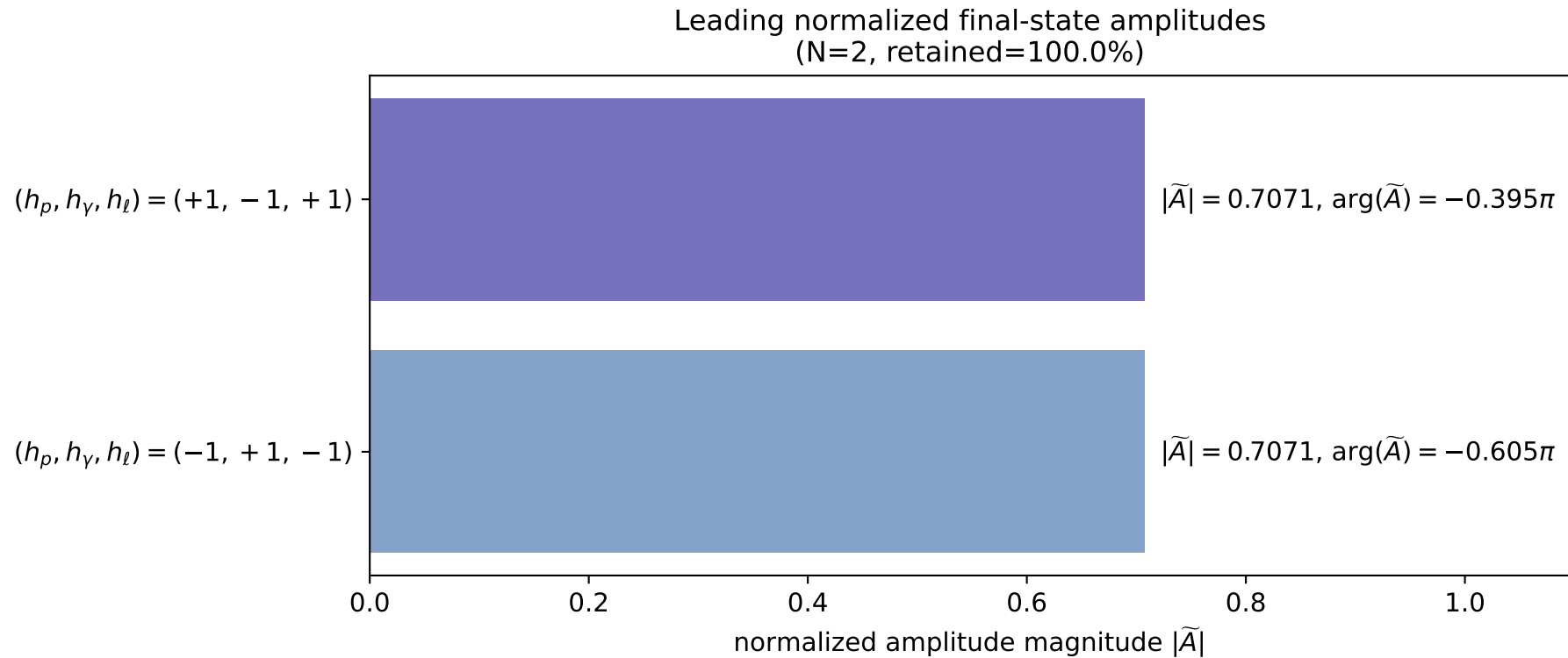


dghz_mixing_angles_polarization_04_local_minimum_18
 $d_{\text{GHZ}}=2.51612\text{e-}08$
 region: polarization_cluster_04_local_minimum_18
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0018
 lepton mixing angle: $\alpha_e=2.5703729$ rad
 incoming lepton: $-0.841242|+\rangle +0.540659|-\rangle$
 proton mixing angle: $\alpha_p=2.5703847$ rad
 incoming proton: $-0.841248|+\rangle +0.540649|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.50685$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.529804$, $\phi_\gamma=5.04178$
 $E_\gamma=0.859098$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.10903$
 $Q^2=22.8268$, $x_B=0.980219$, $t=-14.6965$, $W^2=1.34049$, $y=0.965478$

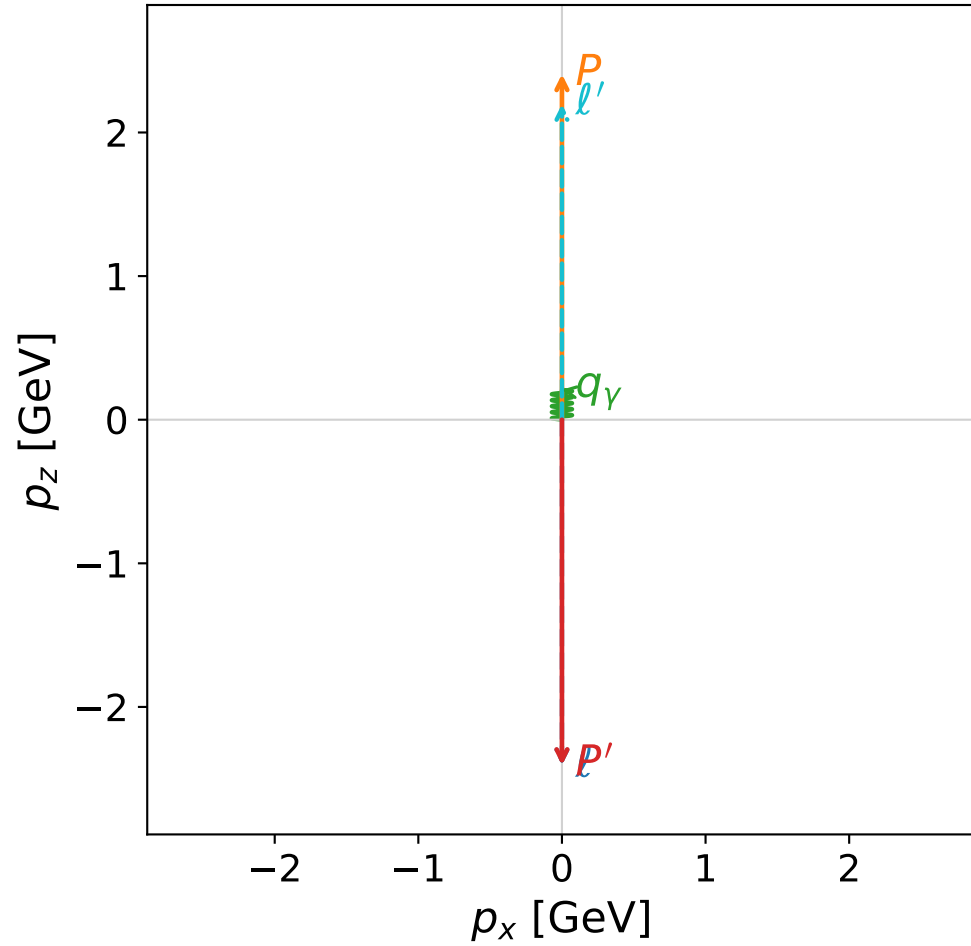
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3660$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3660$
 P' $E= 1.7750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5069$
 q_γ $E= 0.8591$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.8591$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.895\text{e}+23$; $I_h=+3.087\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.996\text{e-}29$; $I_{TL}=+9.567\text{e-}19$

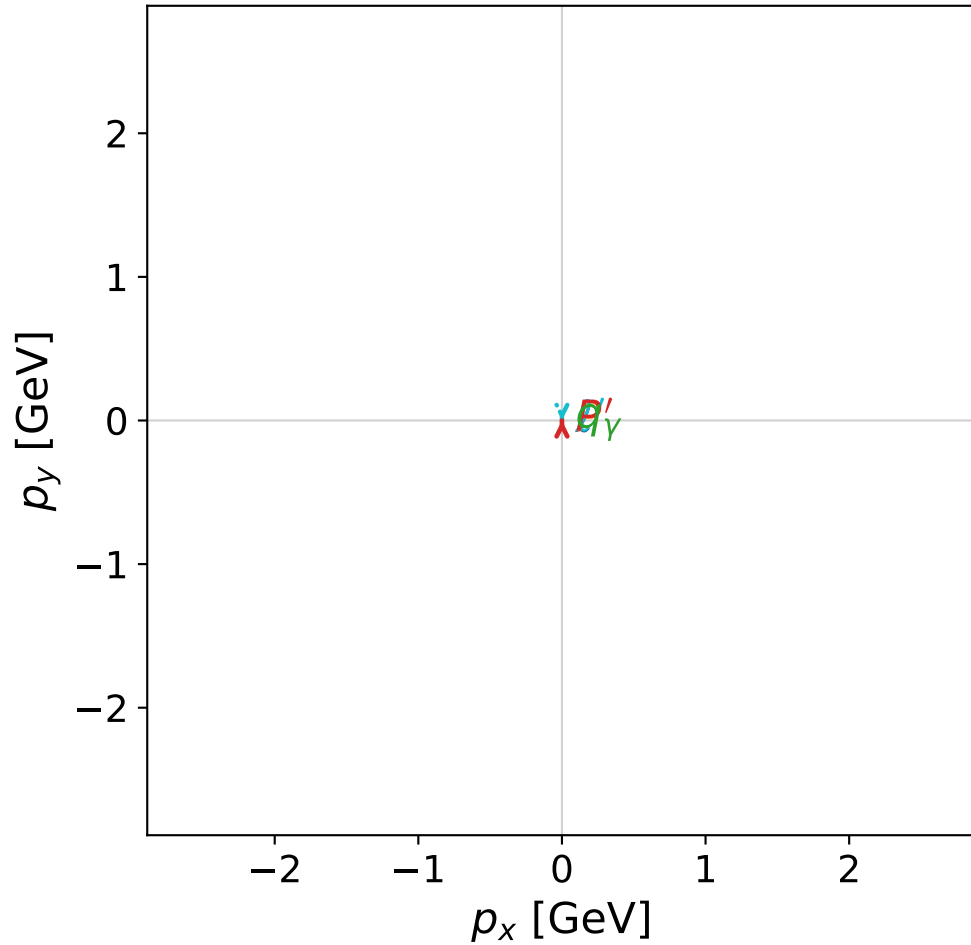


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

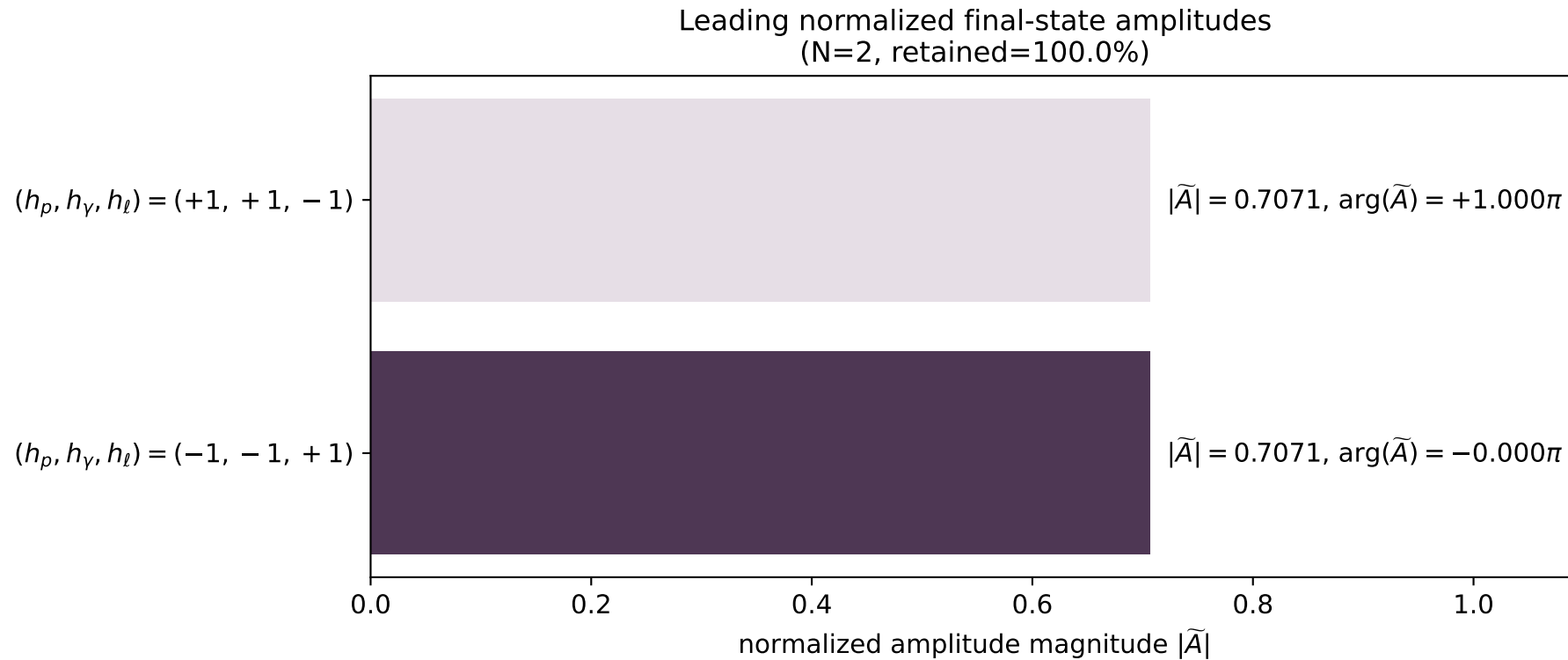


dghz_mixing_angles_polarization_04_local_minimum_22
 $d_{\text{GHZ}}=2.56605\text{e-}08$
 region: polarization_cluster_04_local_minimum_22
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0022
 lepton mixing angle: $\alpha_e=2.4937434$ rad
 incoming lepton: $-0.797384|+\rangle +0.603473|-\rangle$
 proton mixing angle: $\alpha_p=2.218653$ rad
 incoming proton: $-0.603479|+\rangle +0.797379|-\rangle$

$s=24.8915$, $\sqrt{s}=4.98914$, $|\vec{P}|=2.4064$, $|\vec{P}'|=2.4064$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.87557$, $\phi_\gamma=0.0606964$
 $E_\gamma=0.208957$, $\theta_{l'}=0$, $\phi_{l'}=5.01716$
 $Q^2=21.1516$, $x_B=0.91027$, $t=-23.163$, $W^2=2.96488$, $y=0.967722$

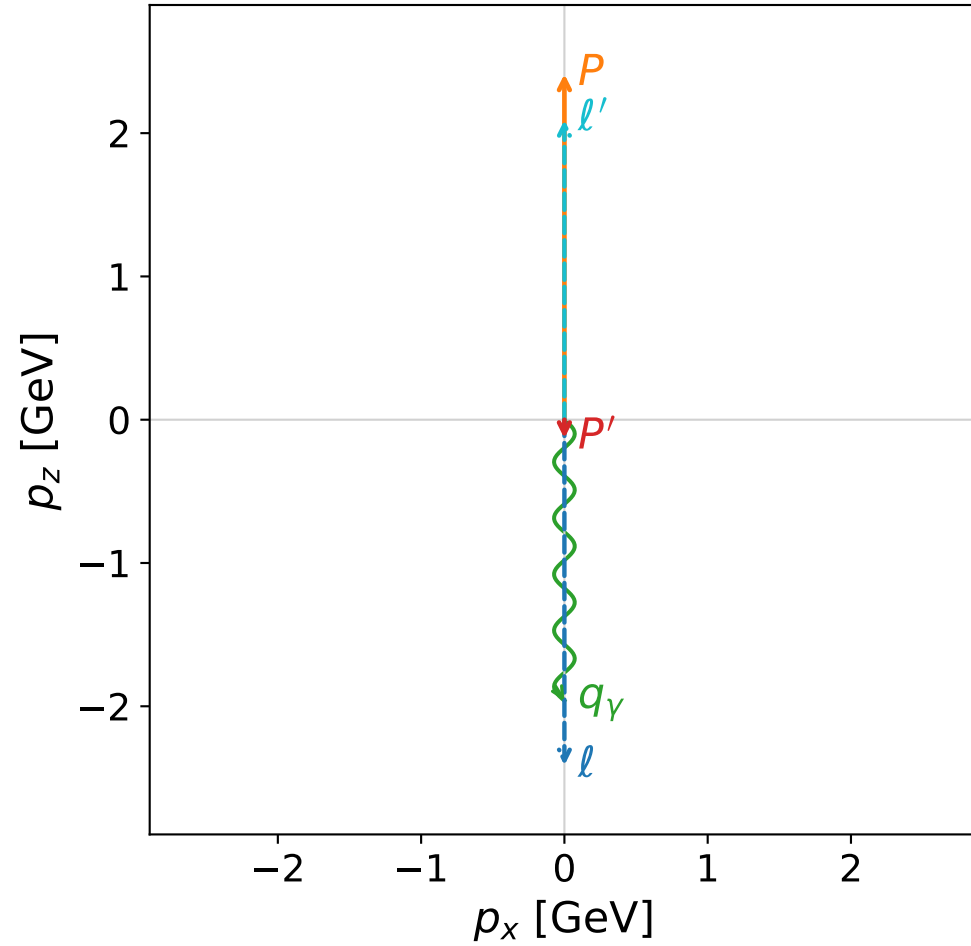
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 P $E= 2.5827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4064$
 ℓ' $E= 2.1974$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.1974$
 P' $E= 2.5827$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 q_γ $E= 0.2090$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2090$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.230\text{e}+22$; $I_h=-5.055\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.379\text{e-}25$; $I_{TL}=-5.055\text{e-}13$

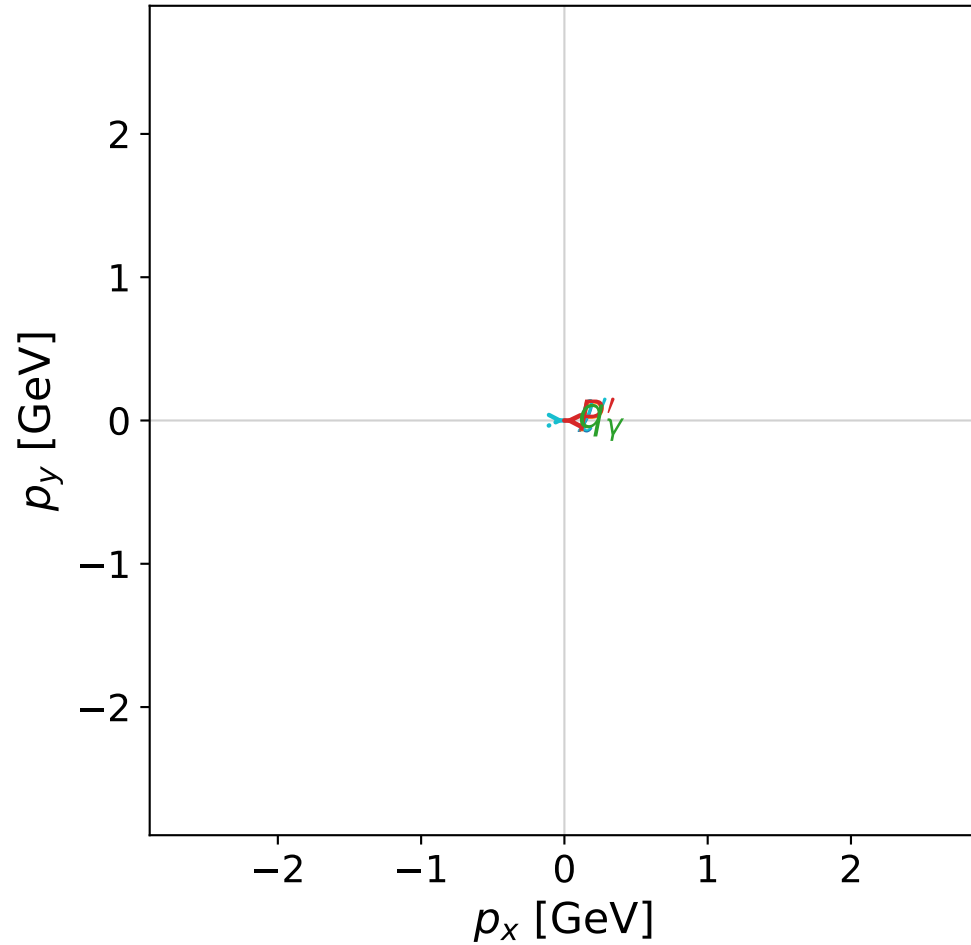


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

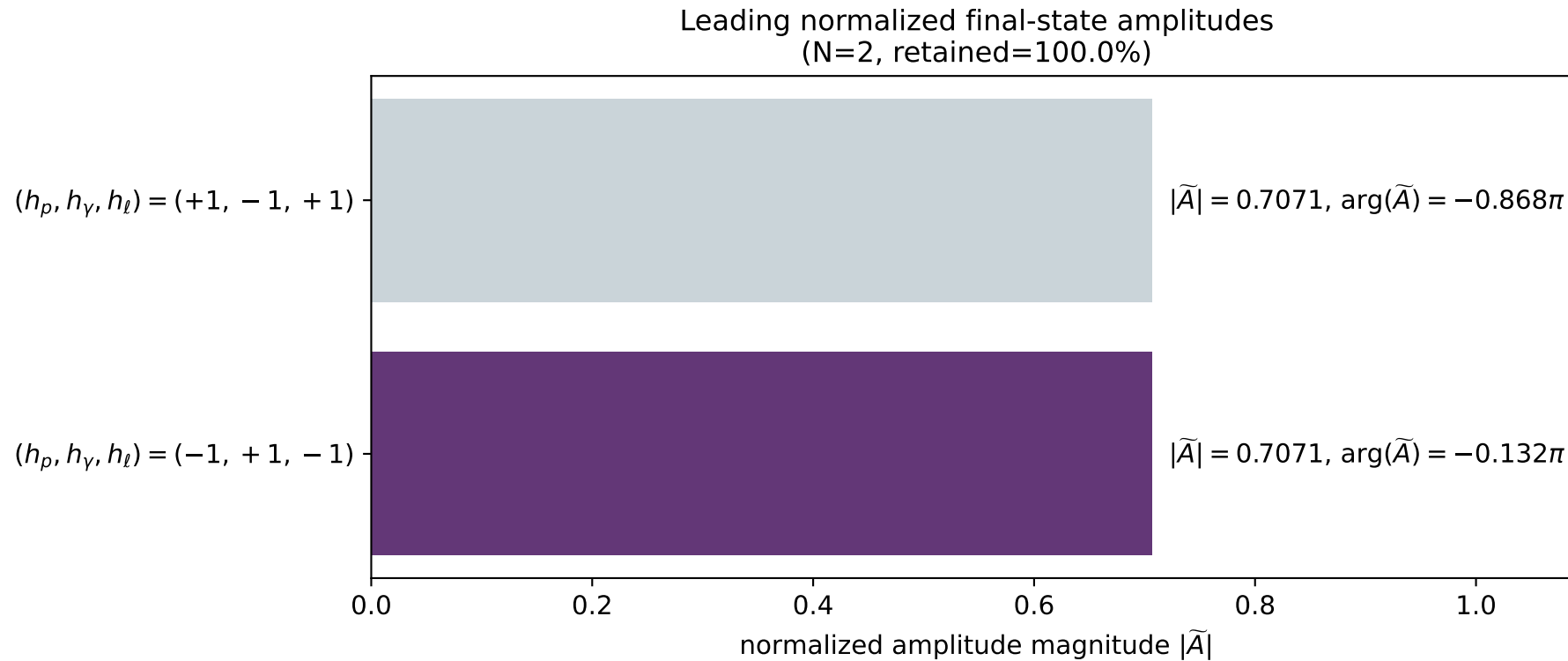


dghz_mixing_angles_polarization_04_local_minimum_29
 $d_{\text{GHZ}}=2.60368\text{e-}08$
 region: polarization_cluster_04_local_minimum_29
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0029
 lepton mixing angle: $\alpha_e=2.3616845$ rad
 incoming lepton: $-0.710978|+\rangle +0.703214|-\rangle$
 proton mixing angle: $\alpha_p=2.3617041$ rad
 incoming proton: $-0.710992|+\rangle +0.7032|-\rangle$

$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.13022$
 $\theta_{P'}=3.14095$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.1261$, $\phi_\gamma=0.397731$
 $E_\gamma=1.96136$, $\theta_{l'}=3.98045\text{e-}05$, $\phi_{l'}=6.2677$
 $Q^2=20.1795$, $x_B=0.86298$, $t=-3.77005$, $W^2=4.08383$, $y=0.969481$

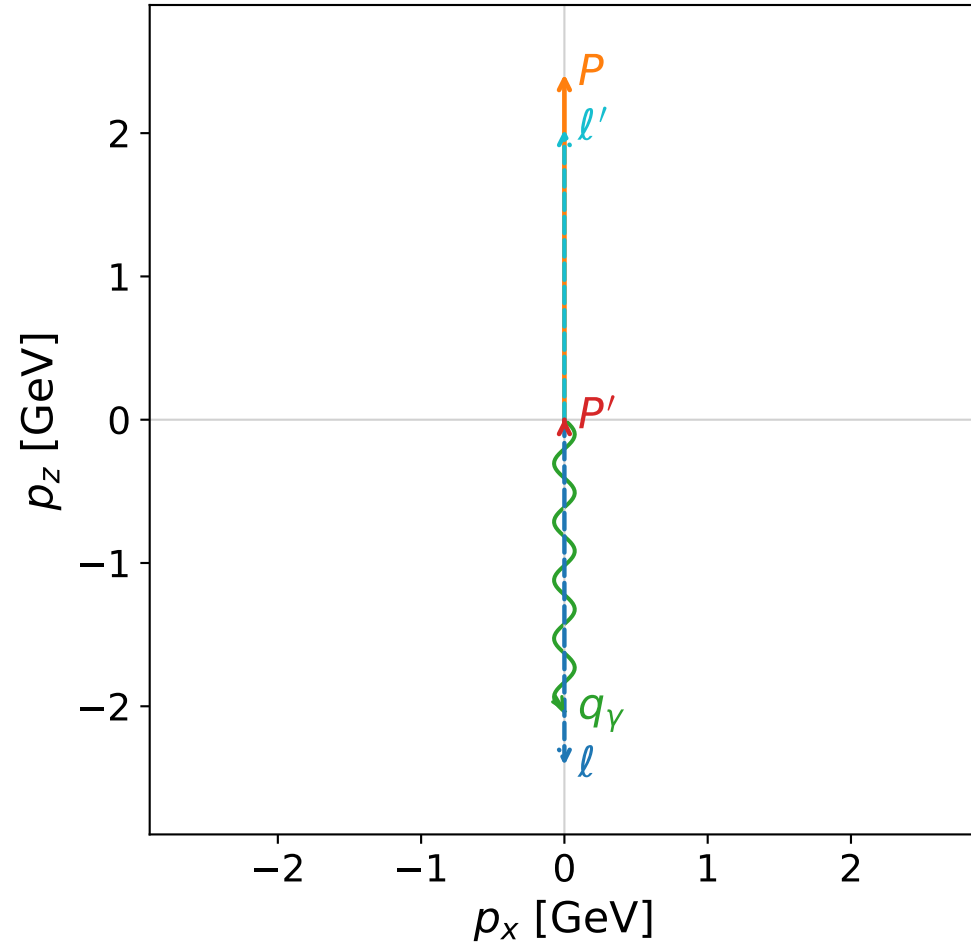
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0916$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0916$
 P' $E= 0.9470$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= -0.1302$
 q_γ $E= 1.9614$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9614$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.165\text{e+}16$; $I_h=+5.705\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.174\text{e-}12$; $I_{TL}=-3.080\text{e-}08$

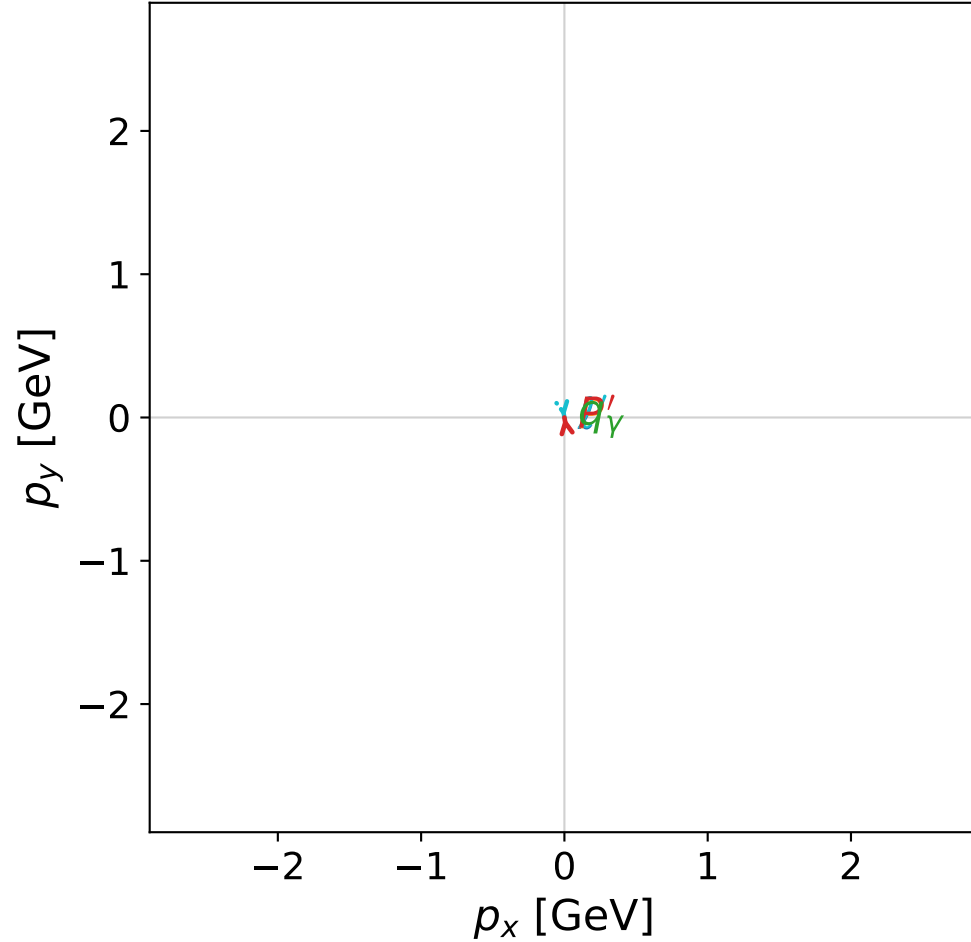


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

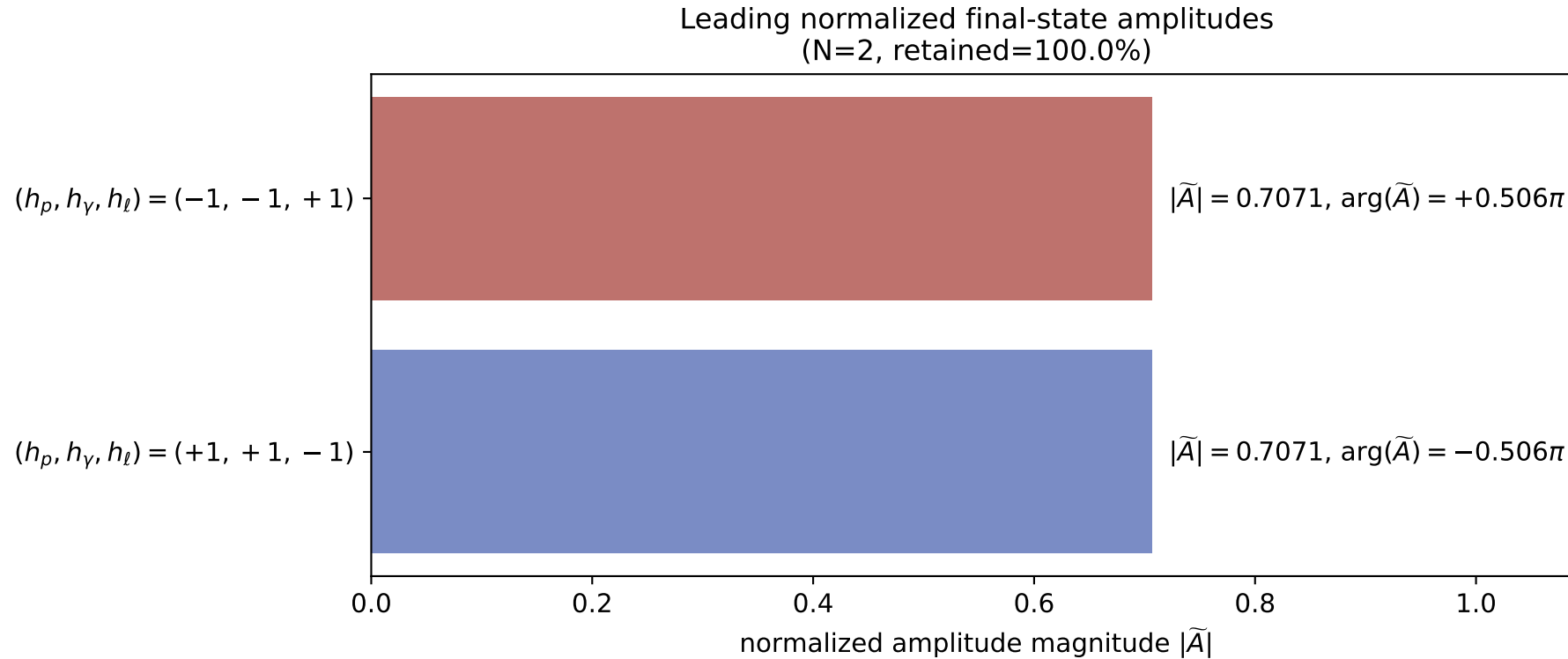


dghz_mixing_angles_polarization_04_local_minimum_30
 $d_{\text{GHZ}}=2.62358\text{e-}08$
 region: polarization_cluster_04_local_minimum_30
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0030
 lepton mixing angle: $\alpha_e=2.3547844$ rad
 incoming lepton: $-0.706109|+\rangle +0.708103|-\rangle$
 proton mixing angle: $\alpha_p=2.3547913$ rad
 incoming proton: $-0.706114|+\rangle +0.708098|-\rangle$

$s=24.9986$, $\sqrt{s}=4.99986$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0112972$
 $\theta_{P'}=0.0085095$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.74098$, $\phi_\gamma=1.5897$
 $E_\gamma=2.03655$, $\theta_{\ell'}=4.7467\text{e-}05$, $\phi_{\ell'}=4.88257$
 $Q^2=19.5391$, $x_B=0.834792$, $t=-3.0411$, $W^2=4.74669$, $y=0.970447$

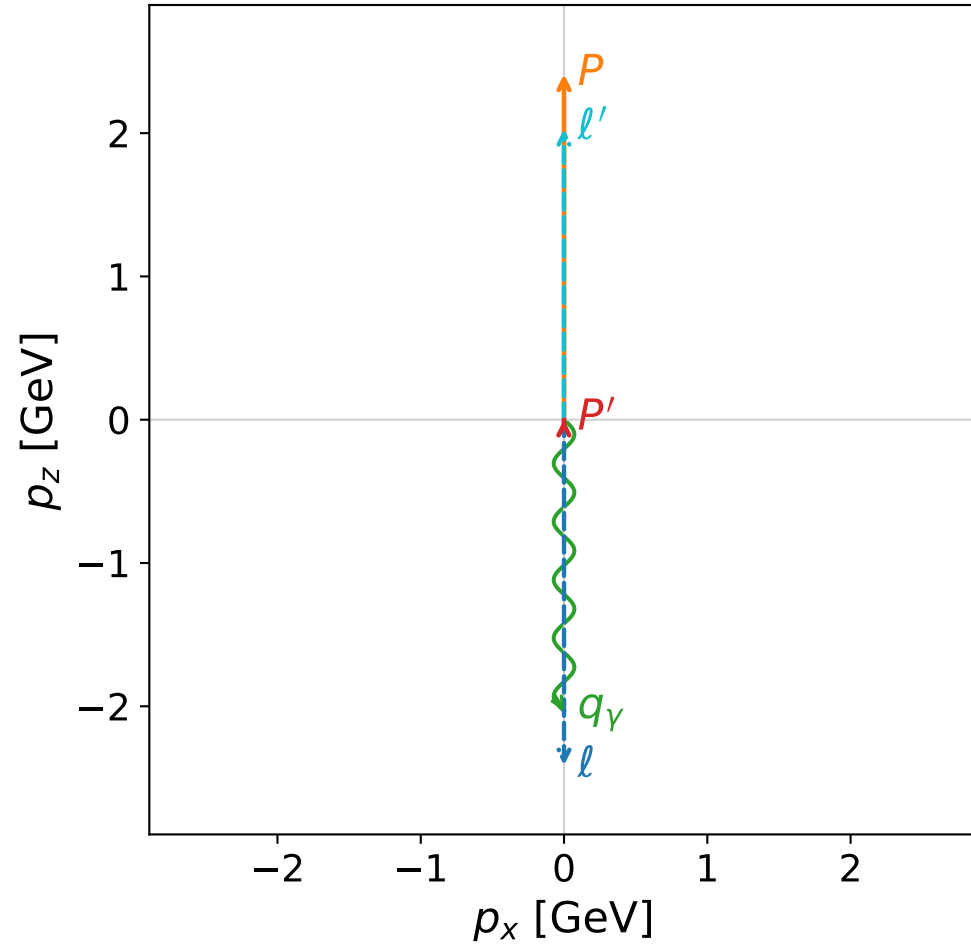
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0252$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 2.0252$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 0.0113$
 q_γ $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.746\text{e+}16$; $I_h=-2.834\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.651\text{e-}12$; $I_{TL}=+3.363\text{e-}09$

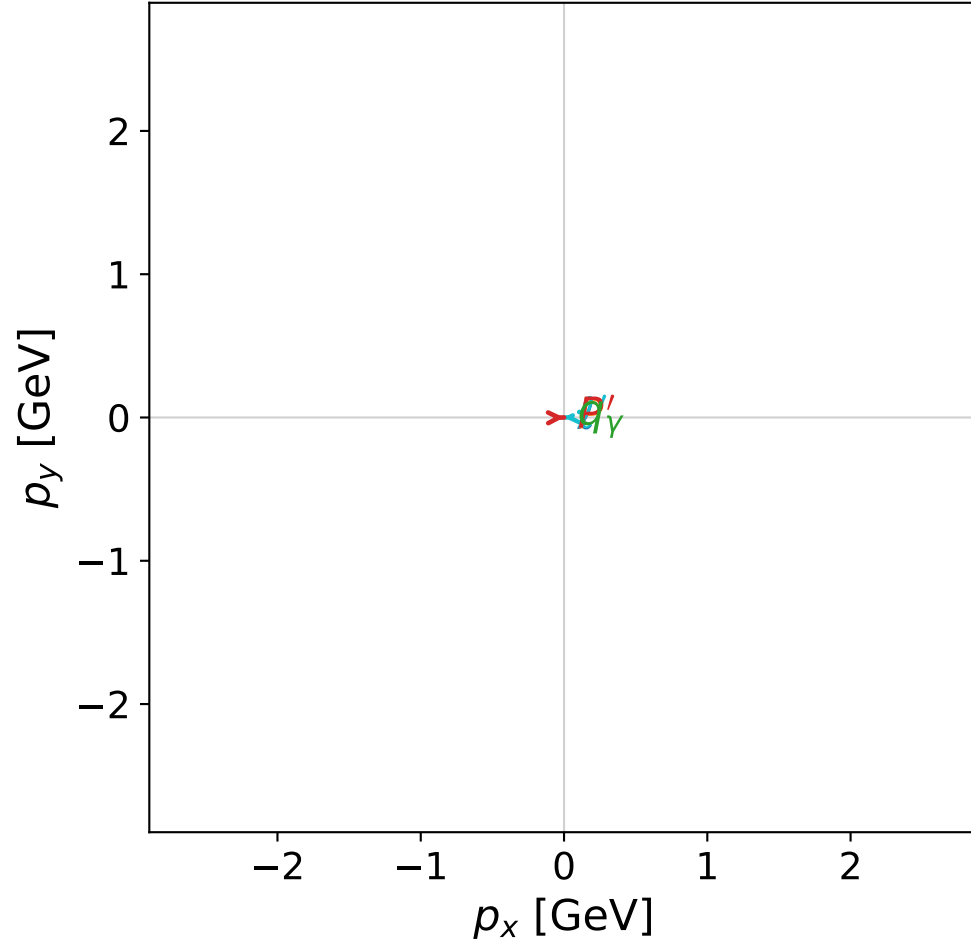


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

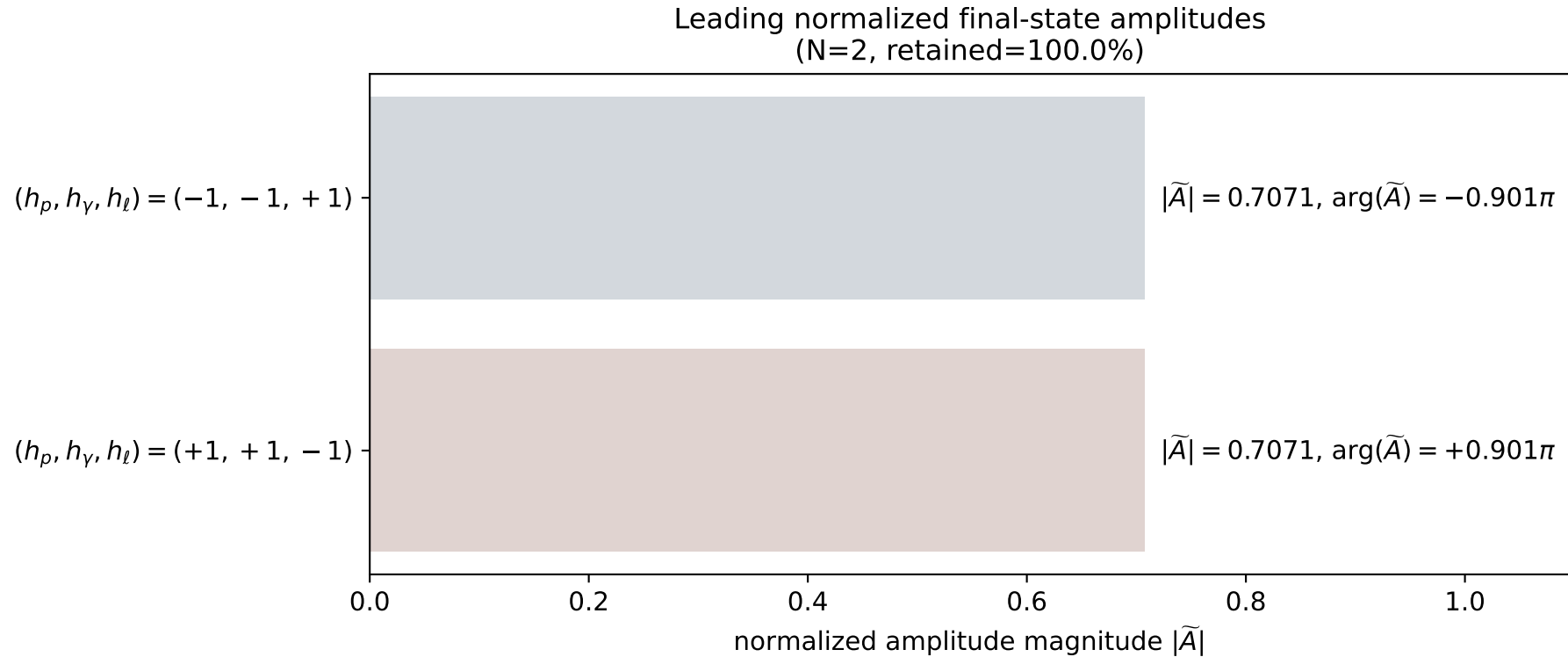


dghz_mixing_angles_polarization_04_local_minimum_39
 $d_{\text{GHZ}}=2.64421\text{e-}08$
 region: polarization_cluster_04_local_minimum_39
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0039
 lepton mixing angle: $\alpha_e=2.4732875$ rad
 incoming lepton: $-0.784873|+\rangle +0.619657|-\rangle$
 proton mixing angle: $\alpha_p=2.4732933$ rad
 incoming proton: $-0.784877|+\rangle +0.619652|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00214662$
 $\theta_{P'}=0.00981748$, $\theta_Y=3.14159$, $\phi_{P'}=0.019635$, $\phi_Y=3.45408$
 $E_Y=2.03207$, $\theta_{l'}=1.03817\text{e-}05$, $\phi_{l'}=3.16123$
 $Q^2=19.5848$, $x_B=0.836754$, $t=-3.08503$, $W^2=4.70074$, $y=0.970381$

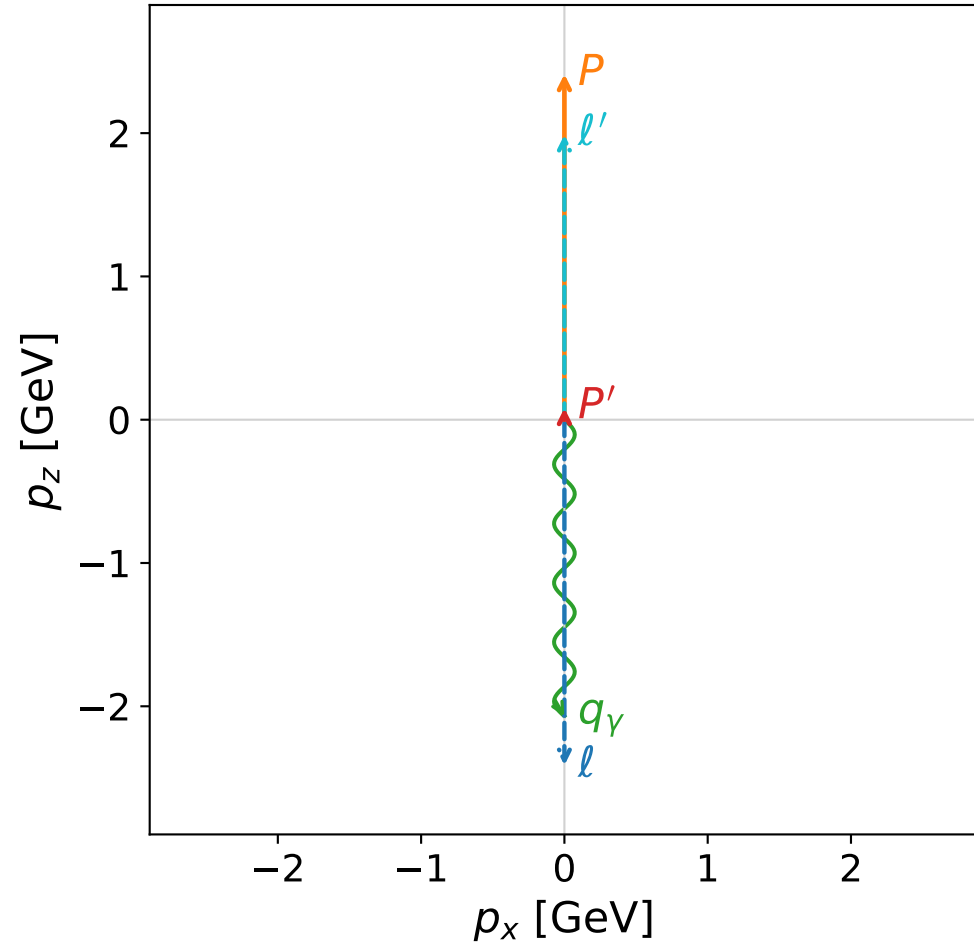
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0299$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0299$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0021$
 q_Y $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.263\text{e+}17$; $I_h=-3.114\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.274\text{e-}13$; $I_{TL}=+2.146\text{e-}07$

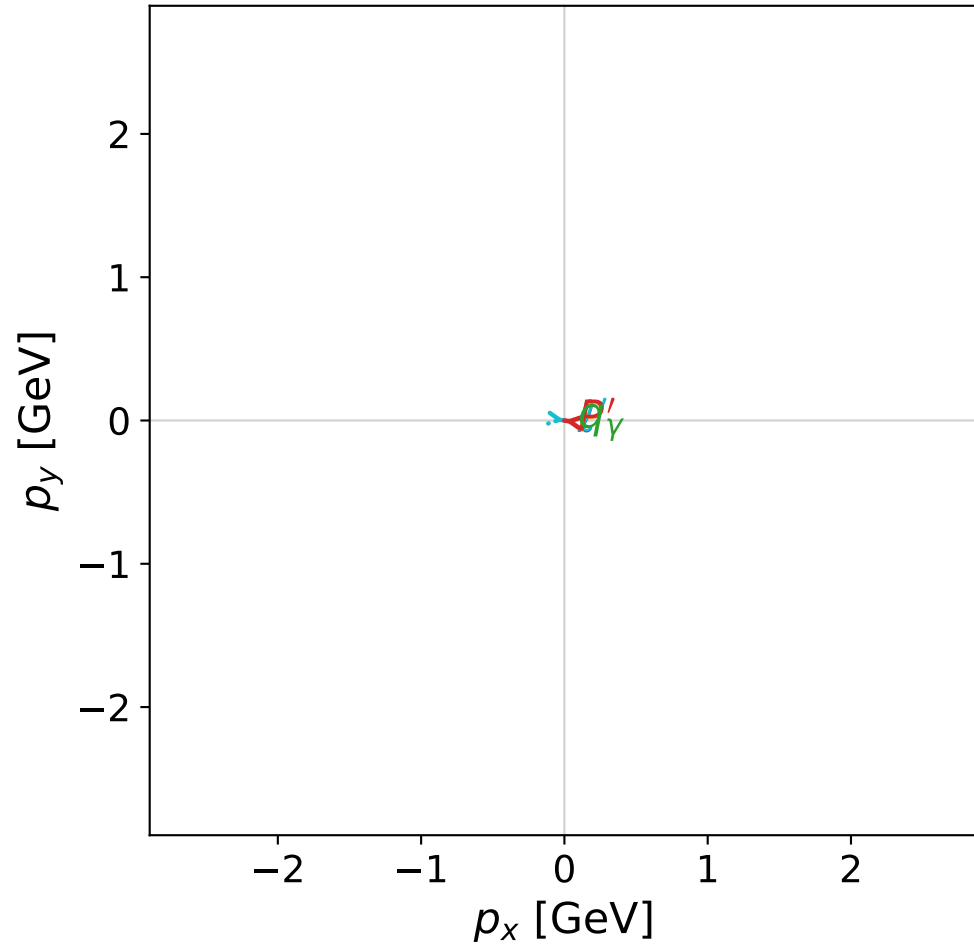


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

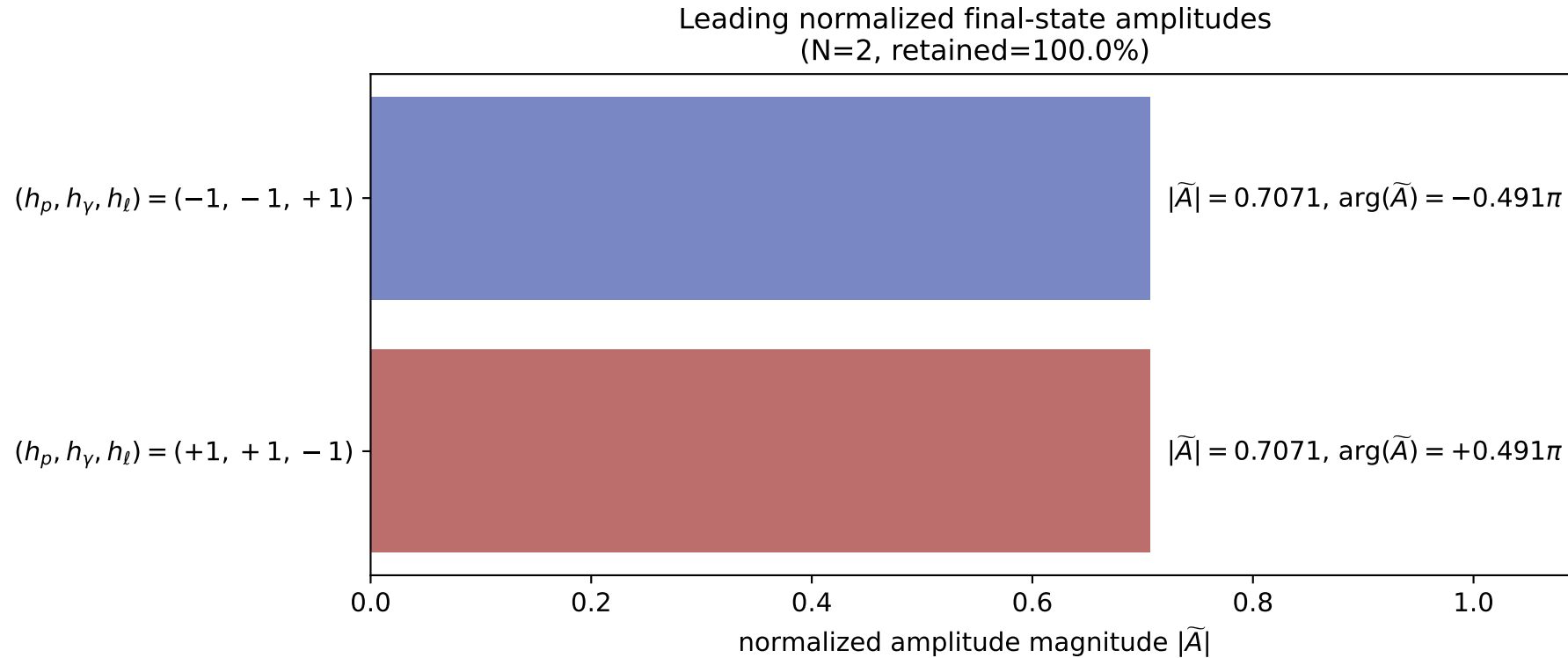


dghz_mixing_angles_polarization_04_local_minimum_50
 $d_{\text{GHZ}}=2.68327\text{e-}08$
 region: polarization_cluster_04_local_minimum_50
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0050
 lepton mixing angle: $\alpha_e=2.2455924$ rad
 incoming lepton: $-0.624738|+\rangle +0.780834|-\rangle$
 proton mixing angle: $\alpha_p=2.2455942$ rad
 incoming proton: $-0.624739|+\rangle +0.780833|-\rangle$

$s=24.9974$, $\sqrt{s}=4.99974$, $|\vec{P}|=2.41188$, $|\vec{P}'|=0.0814227$
 $\theta_{P'}=7.88291\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.99389$, $\phi_\gamma=4.74182$
 $E_\gamma=2.06982$, $\theta_{\ell'}=3.23049\text{e-}07$, $\phi_{\ell'}=6.13548$
 $Q^2=19.183$, $x_B=0.819169$, $t=-2.72062$, $W^2=5.11448$, $y=0.970983$

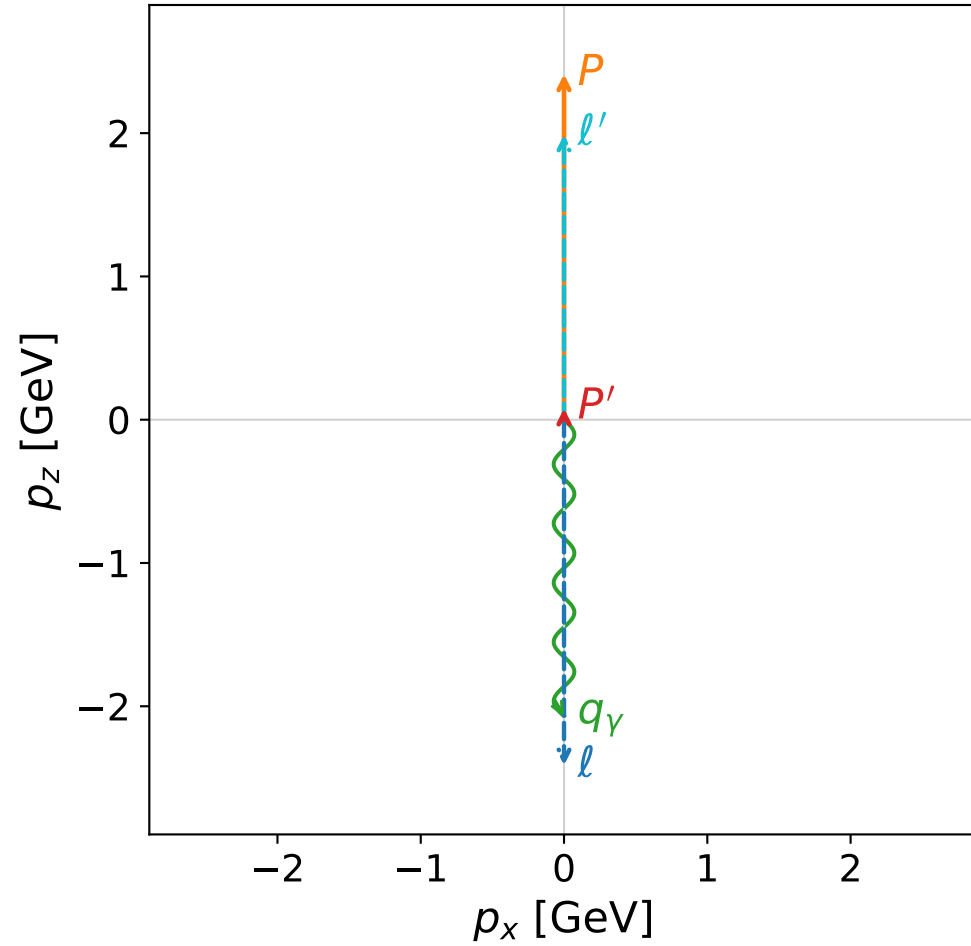
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 1.9884$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9884$
 P' $E= 0.9415$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0814$
 q_γ $E= 2.0698$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0698$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.629\text{e+}20$; $I_h=-8.864\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.977\text{e-}16$; $I_{TL}=+6.847\text{e-}09$

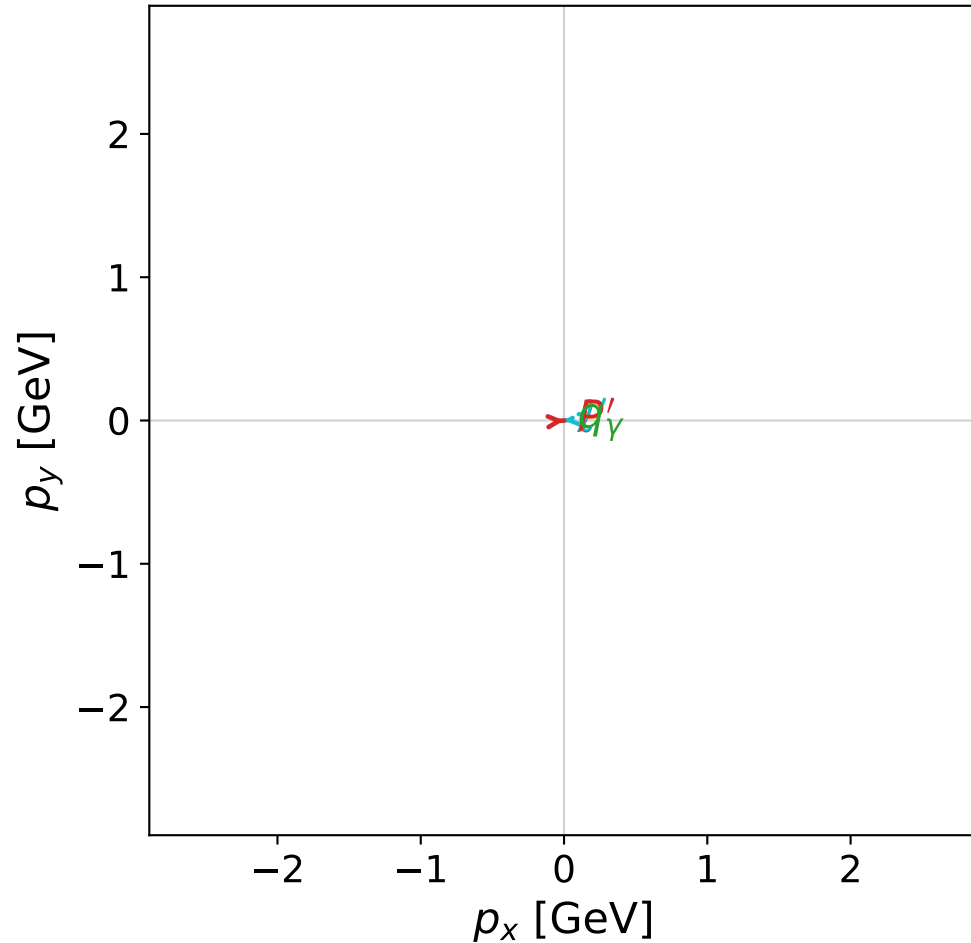


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

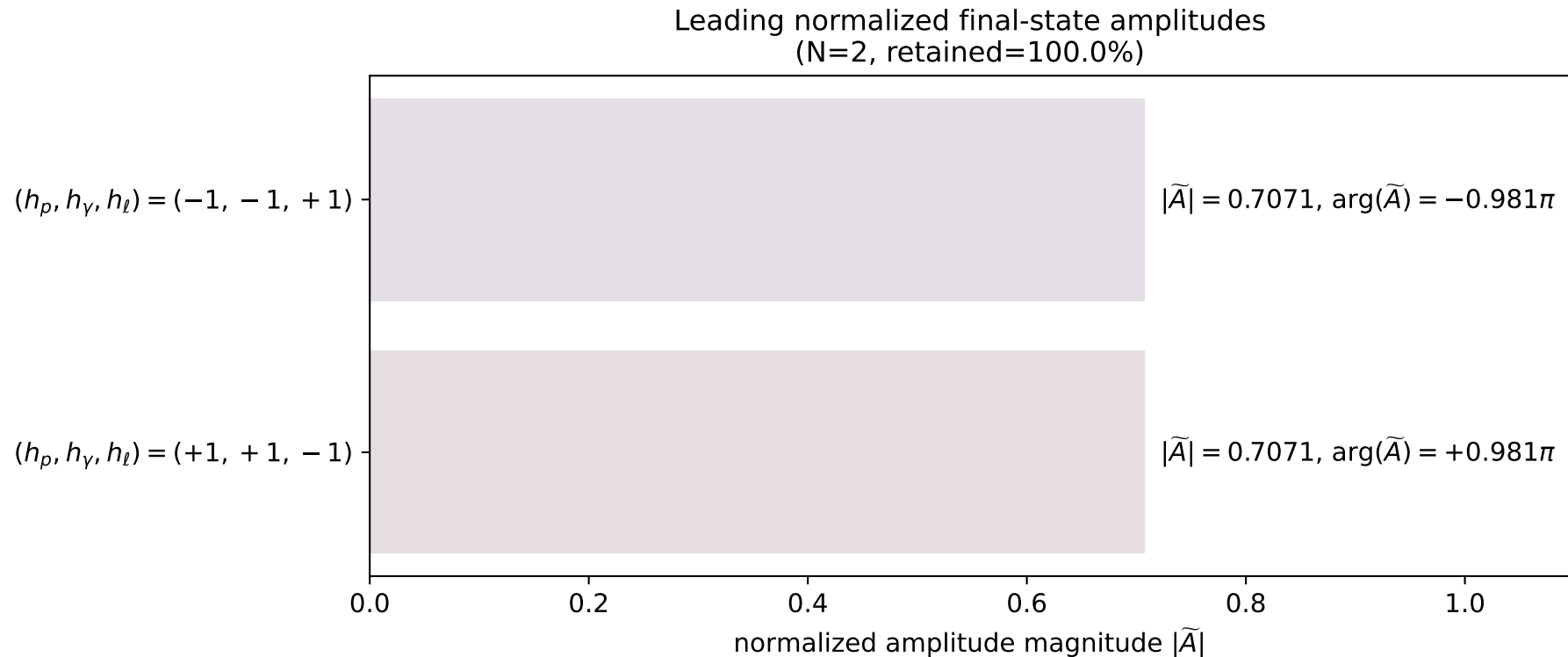


dghz_mixing_angles_polarization_04_local_minimum_59
 $d_{\text{GHZ}}=2.72599\text{e-}08$
 region: polarization_cluster_04_local_minimum_59
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0059
 lepton mixing angle: $\alpha_e=2.4105855$ rad
 incoming lepton: $-0.744502|+\rangle +0.66762|-\rangle$
 proton mixing angle: $\alpha_p=2.4105852$ rad
 incoming proton: $-0.744502|+\rangle +0.66762|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0792932$
 $\theta_{P'}=0.00083419$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0772728$, $\phi_\gamma=3.20129$
 $E_\gamma=2.06897$, $\theta_{\ell'}=3.32443\text{e-}05$, $\phi_{\ell'}=3.21887$
 $Q^2=19.1966$, $x_B=0.819668$, $t=-2.73017$, $W^2=5.10319$, $y=0.970969$

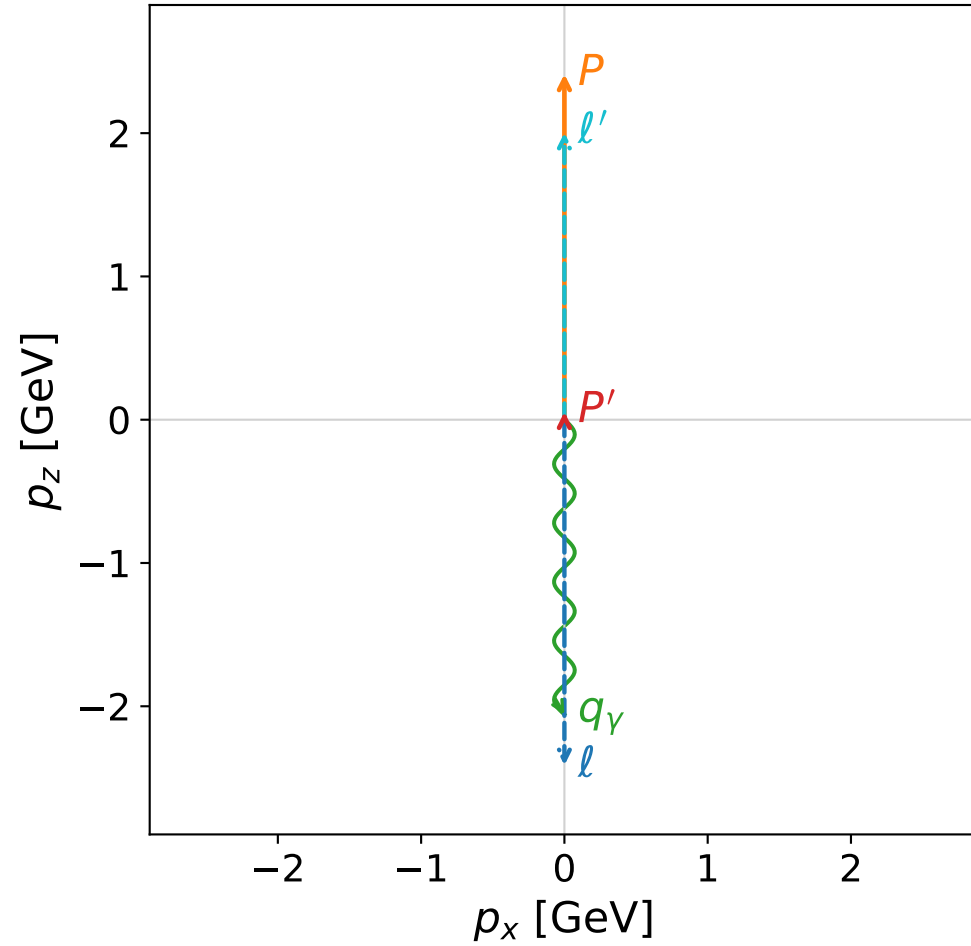
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9897$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.9897$
 P' $E= 0.9413$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0793$
 q_γ $E= 2.0690$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0690$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.583\text{e+}16$; $I_h=-4.471\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.059\text{e-}12$; $I_{TL}=+3.450\text{e-}07$

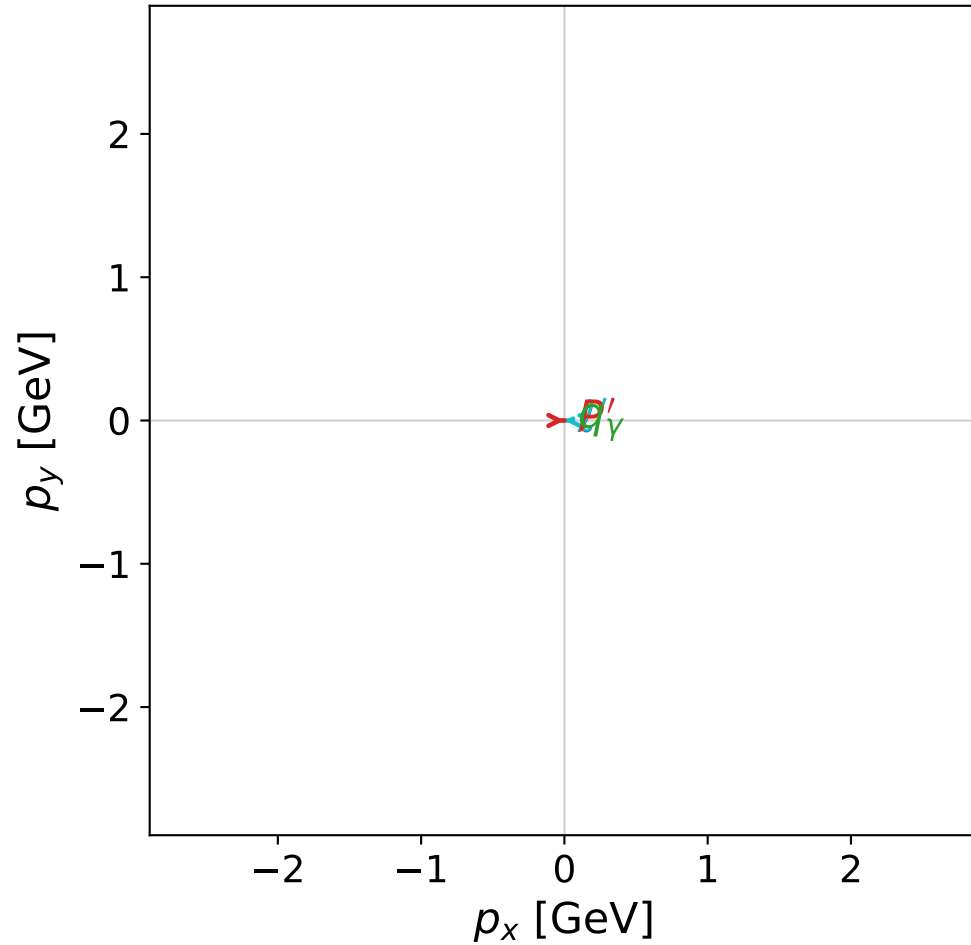


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



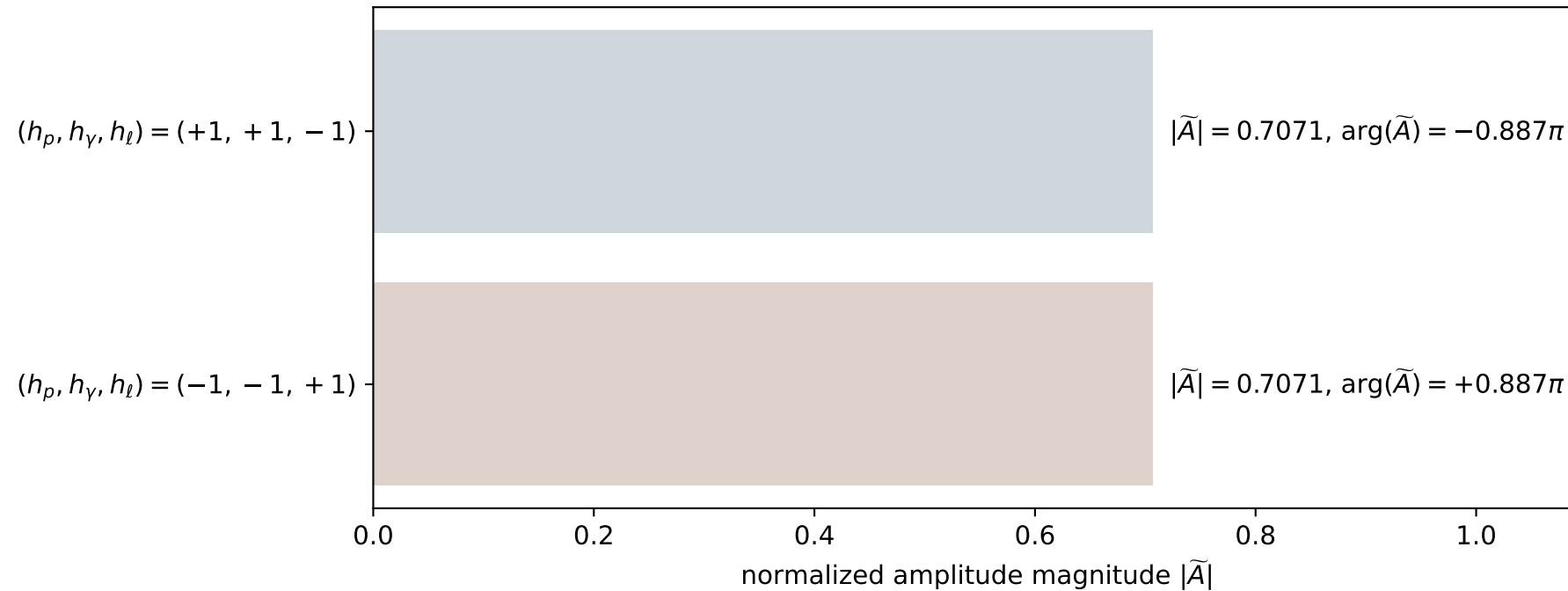
dghz_mixing_angles_polarization_04_local_minimum_63
 $d_{\text{GHZ}}=2.75718\text{e-}08$
 region: polarization_cluster_04_local_minimum_63
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0063
 lepton mixing angle: $\alpha_e=2.4612083$ rad
 incoming lepton: $-0.777331|+\rangle +0.629092|-\rangle$
 proton mixing angle: $\alpha_p=2.461193$ rad
 incoming proton: $-0.777321|+\rangle +0.629104|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0542678$
 $\theta_{P'}=0.00132352$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28313$, $\phi_\gamma=2.78563$
 $E_\gamma=2.05735$, $\theta_{\ell'}=3.58569\text{e-}05$, $\phi_{\ell'}=3.14154$
 $Q^2=19.3259$, $x_B=0.825355$, $t=-2.8417$, $W^2=4.96918$, $y=0.970773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0031$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0031$
 P' $E= 0.9396$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0543$
 q_γ $E= 2.0573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

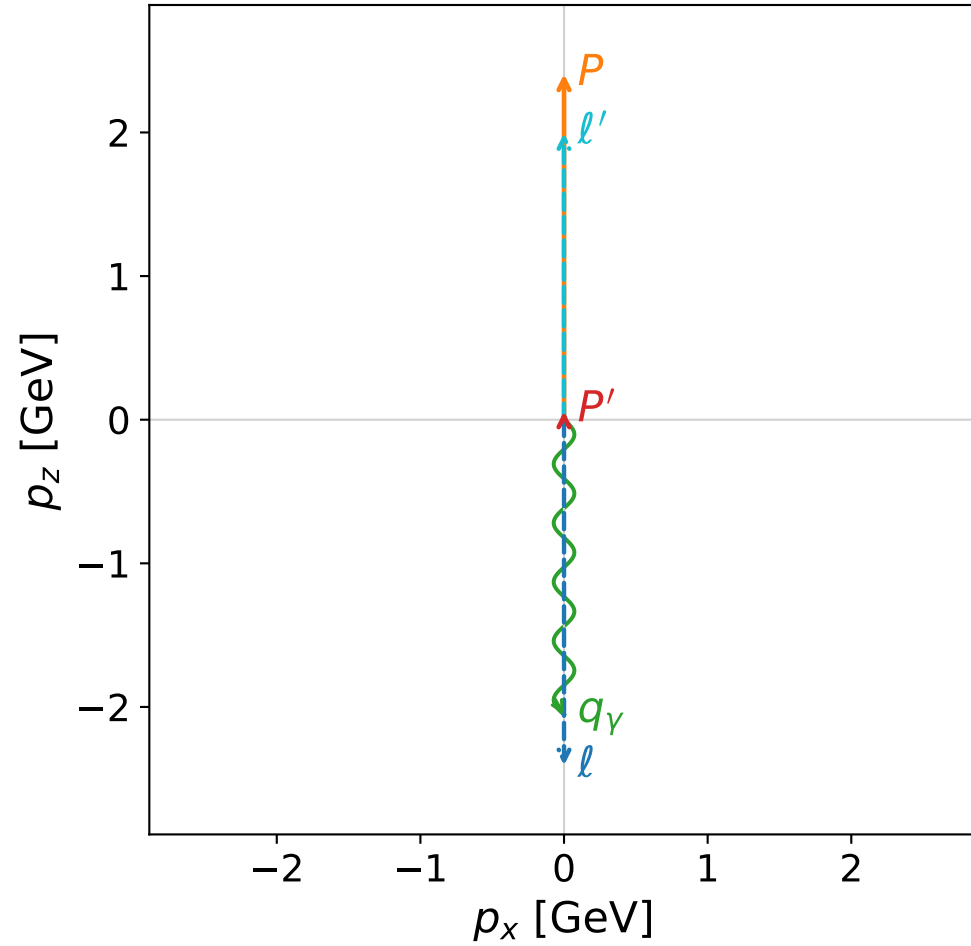
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.231\text{e+}16$; $I_h=-9.492\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.731\text{e-}12$; $I_{TL}=+7.066\text{e-}07$

Leading normalized final-state amplitudes (N=2, retained=100.0%)

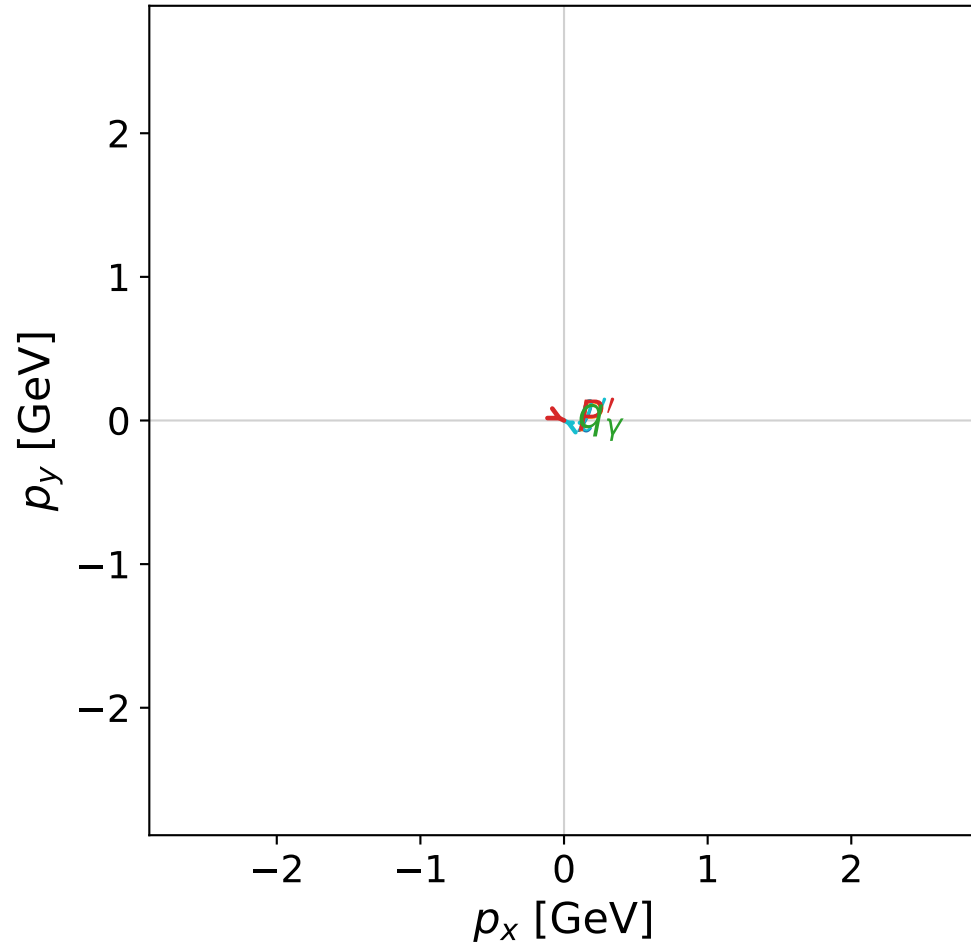


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

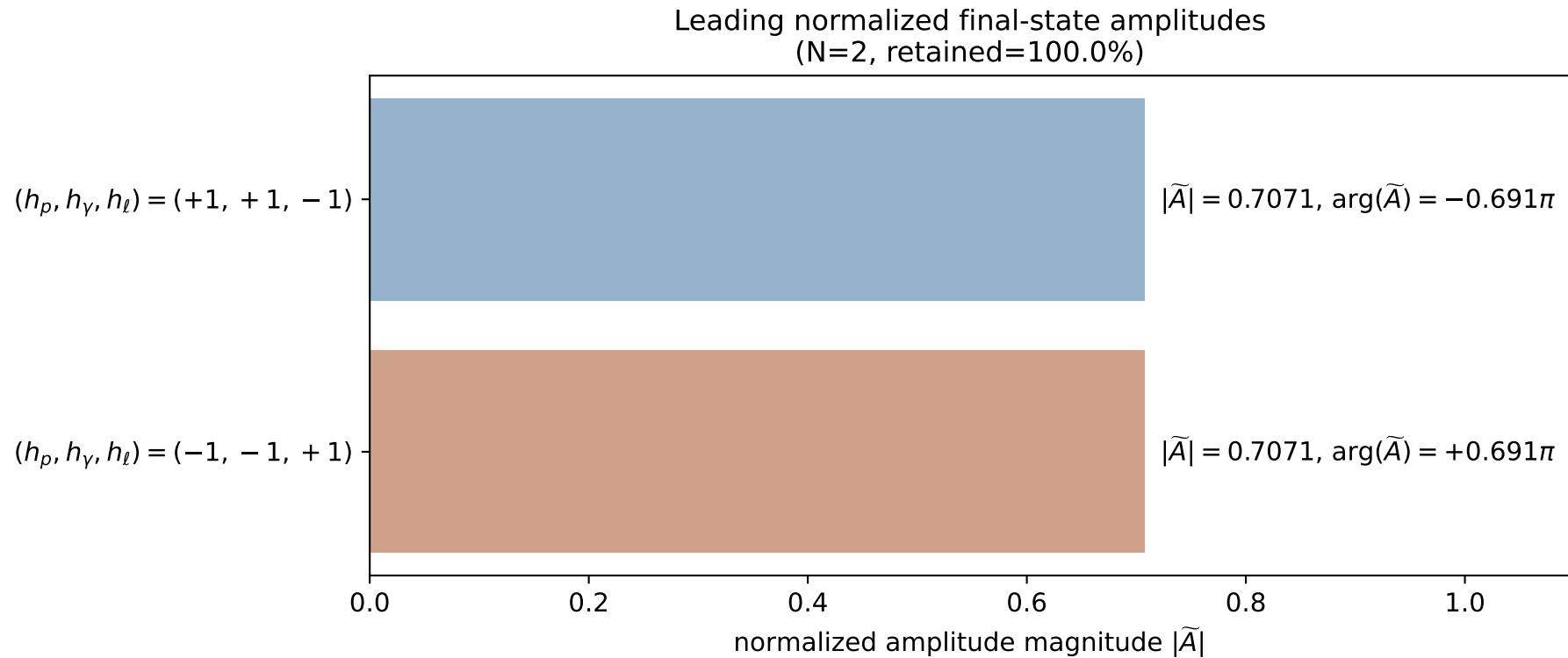


dghz_mixing_angles_polarization_04_local_minimum_64
 $d_{\text{GHZ}}=2.76858\text{e-}08$
 region: polarization_cluster_04_local_minimum_64
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0064
 lepton mixing angle: $\alpha_e=2.2443831$ rad
 incoming lepton: $-0.623793|+\rangle +0.781589|-\rangle$
 proton mixing angle: $\alpha_p=2.2443657$ rad
 incoming proton: $-0.62378|+\rangle +0.7816|-\rangle$

$s=24.887$, $\sqrt{s}=4.98869$, $|\vec{P}|=2.40616$, $|\vec{P}'|=0.0592441$
 $\theta_{P'}=7.76676\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.81389$, $\phi_\gamma=2.17126$
 $E_\gamma=2.05403$, $\theta_{\ell'}=2.30665\text{e-}06$, $\phi_{\ell'}=2.67229$
 $Q^2=19.1991$, $x_B=0.823871$, $t=-2.80969$, $W^2=4.98427$, $y=0.970691$

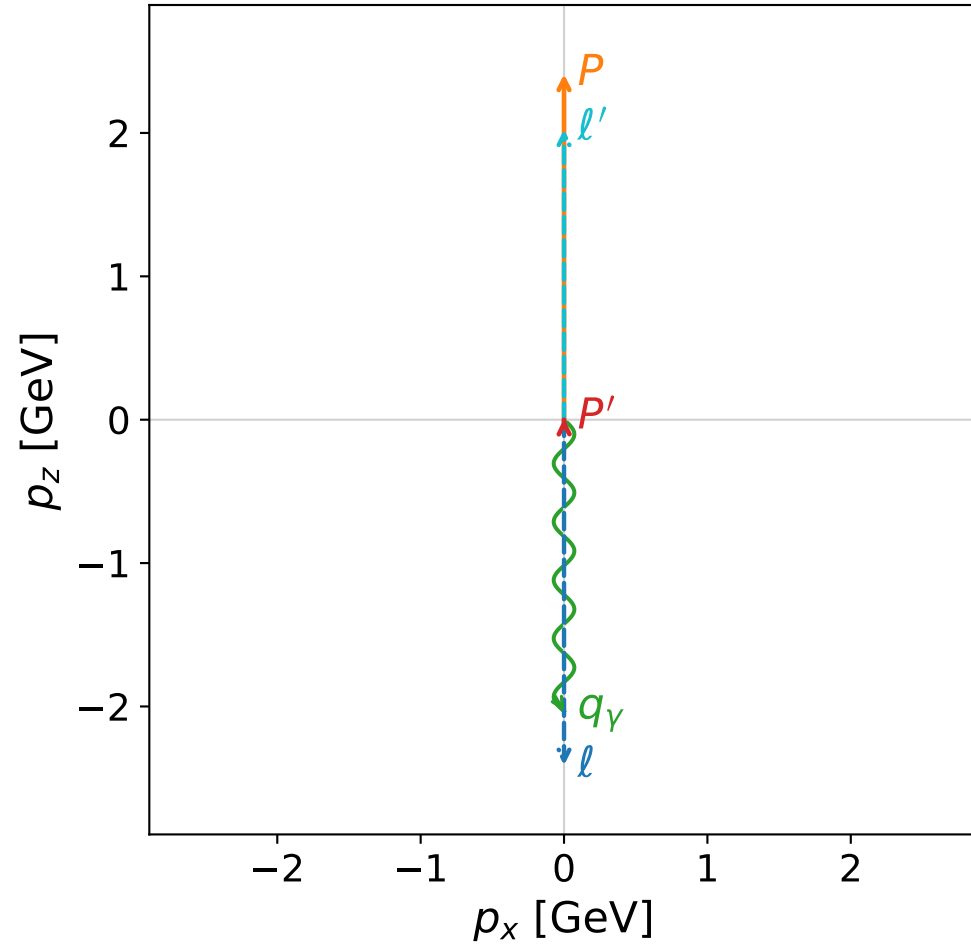
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4062$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4062$
 P $E= 2.5825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4062$
 ℓ' $E= 1.9948$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9948$
 P' $E= 0.9399$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0592$
 q_γ $E= 2.0540$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0540$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.482\text{e+}19$; $I_h=+5.823\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.434\text{e-}14$; $I_{TL}=-4.364\text{e-}08$

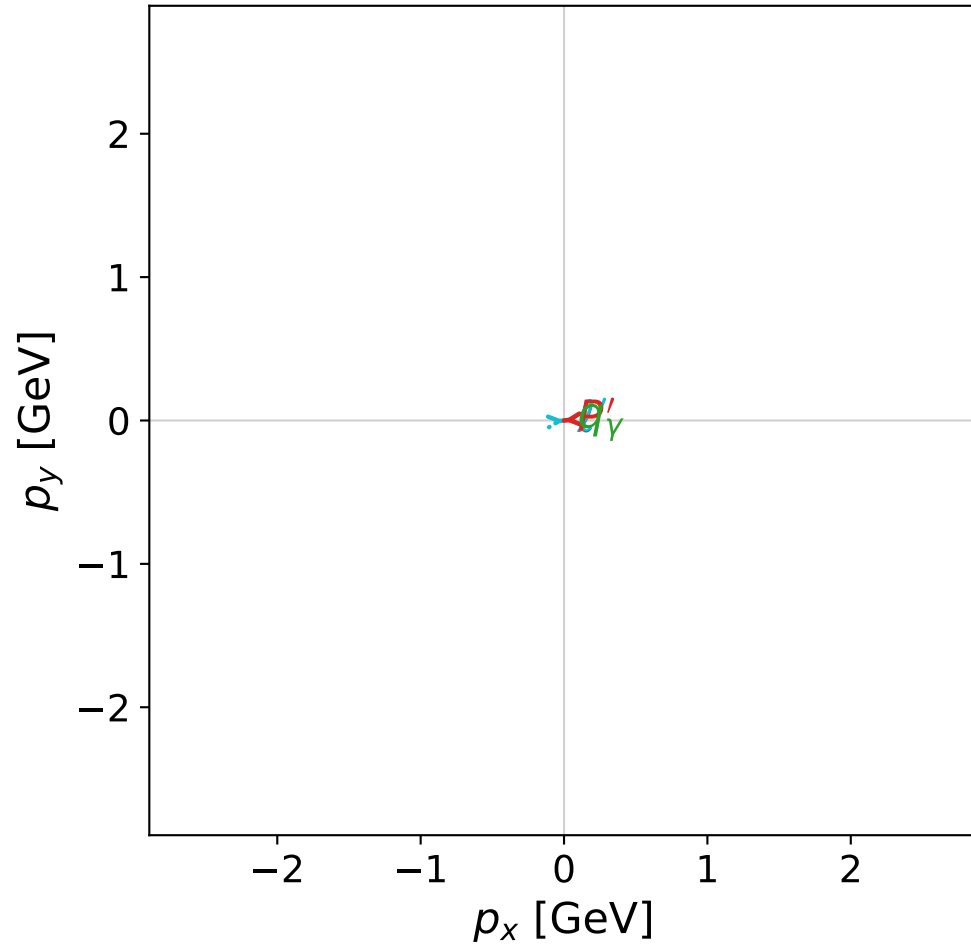


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

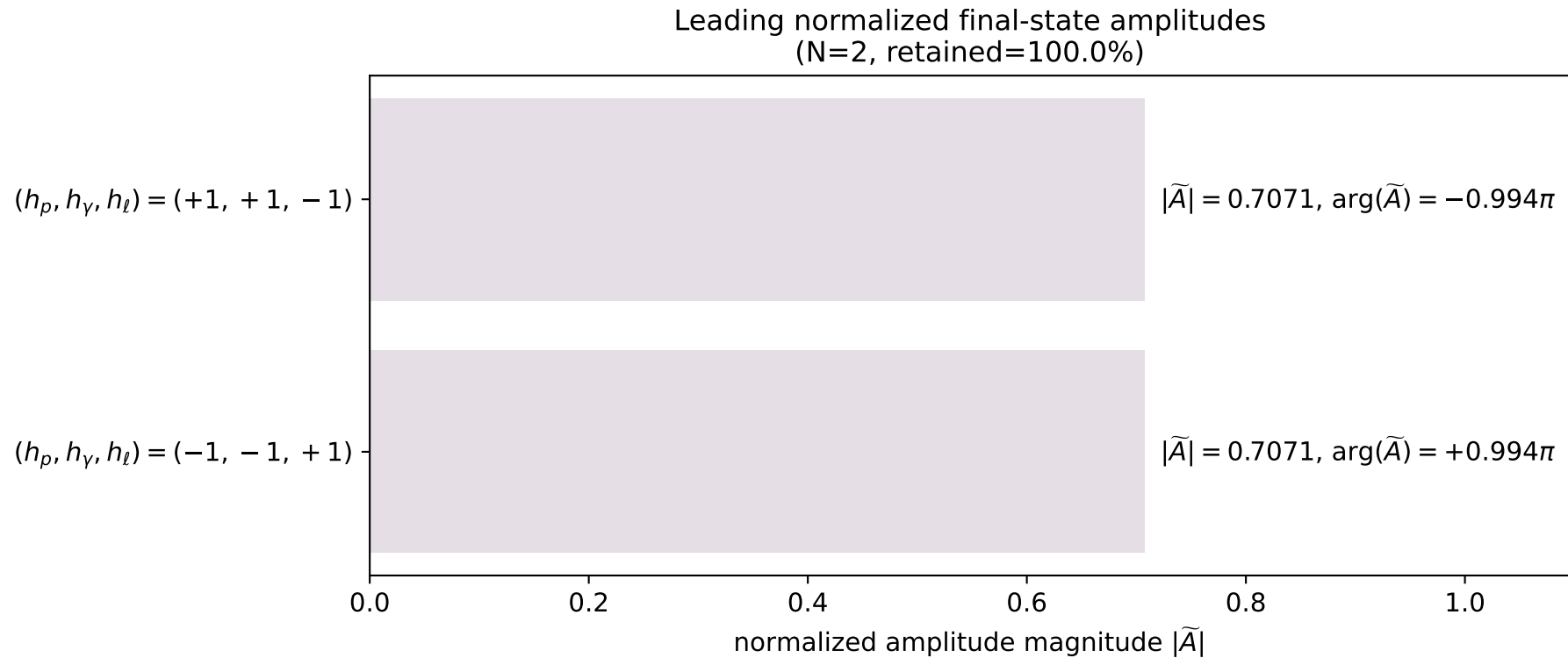


dghz_mixing_angles_polarization_04_local_minimum_74
 $d_{\text{GHZ}}=2.93202\text{e-}08$
 region: polarization_cluster_04_local_minimum_74
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0074
 lepton mixing angle: $\alpha_e=2.1227555$ rad
 incoming lepton: $-0.524356|+\rangle +0.851499|-\rangle$
 proton mixing angle: $\alpha_p=2.12275$ rad
 incoming proton: $-0.524352|+\rangle +0.851502|-\rangle$

$s=24.9688$, $\sqrt{s}=4.99688$, $|\vec{P}|=2.4104$, $|\vec{P}'|=0.00957823$
 $\theta_{P'}=0.00965936$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.2353$, $\phi_\gamma=3.12295$
 $E_\gamma=2.0342$, $\theta_{\ell'}=4.56964\text{e-}05$, $\phi_{\ell'}=0.0937025$
 $Q^2=19.5206$, $x_B=0.835073$, $t=-3.04662$, $W^2=4.73517$, $y=0.970402$

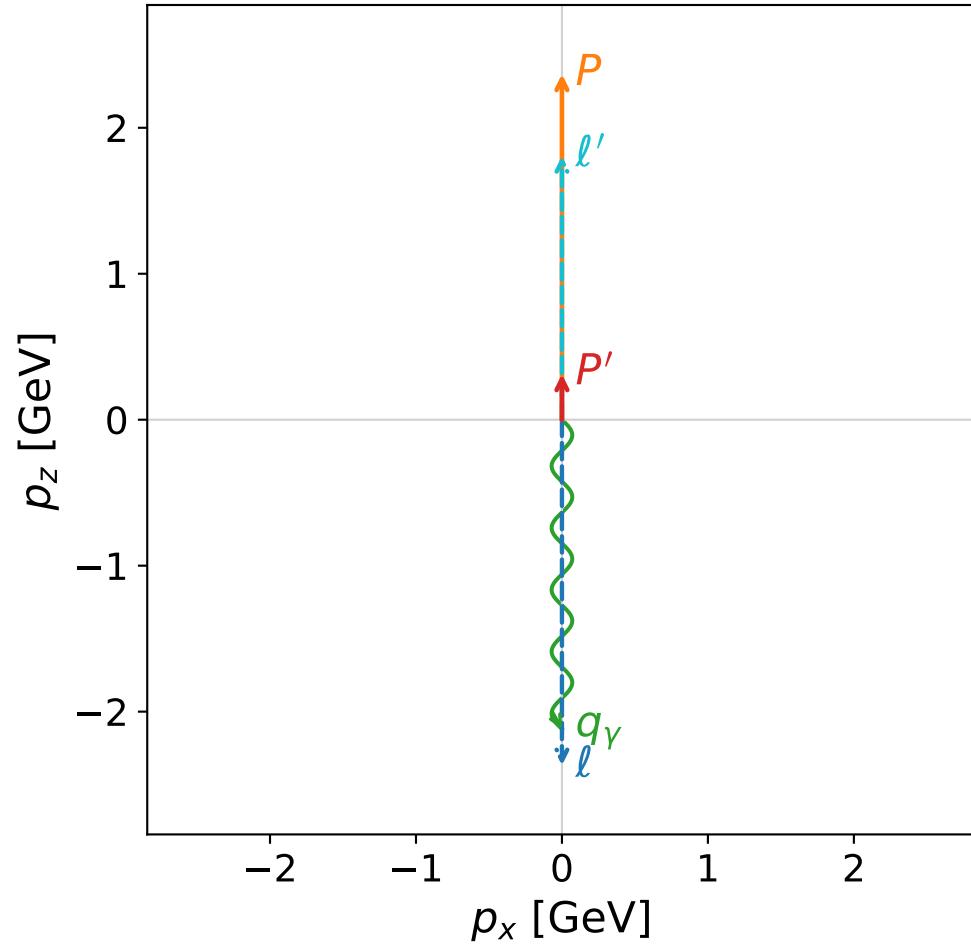
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 ℓ' $E= 2.0246$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.0246$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0096$
 q_γ $E= 2.0342$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0342$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.017\text{e+}16$; $I_h=-2.880\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.003\text{e-}11$; $I_{TL}=+2.009\text{e-}06$

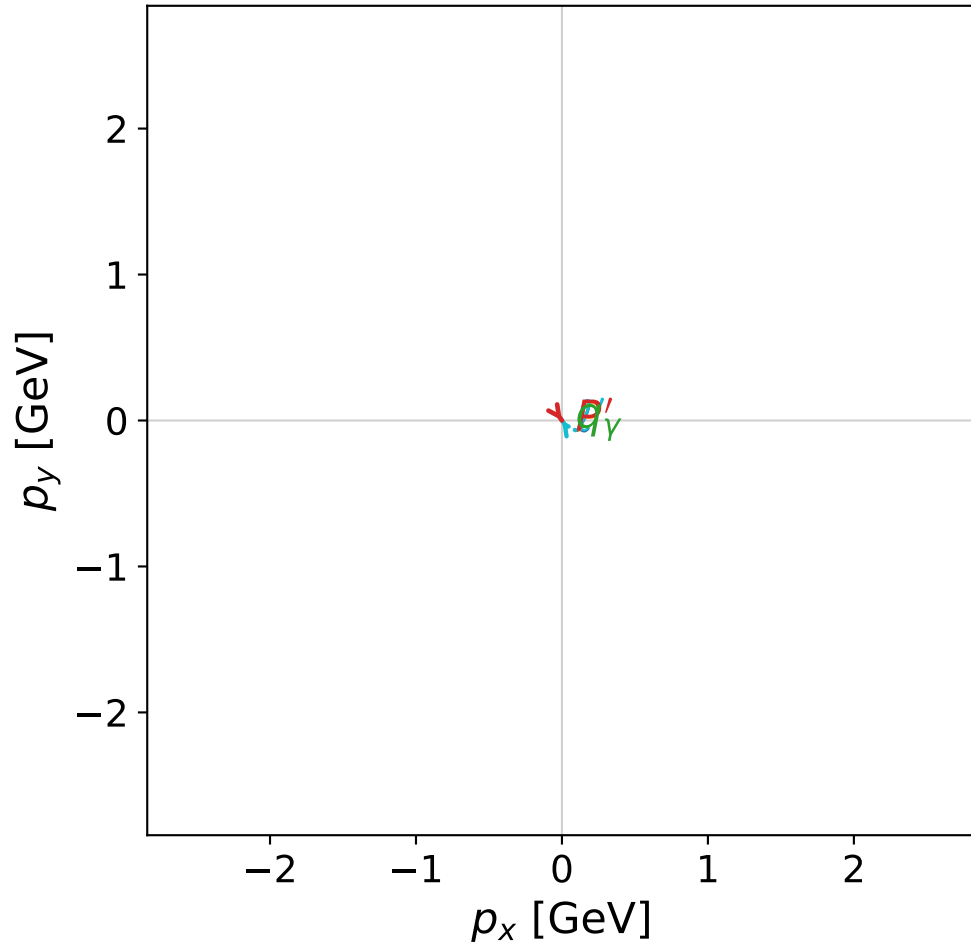


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

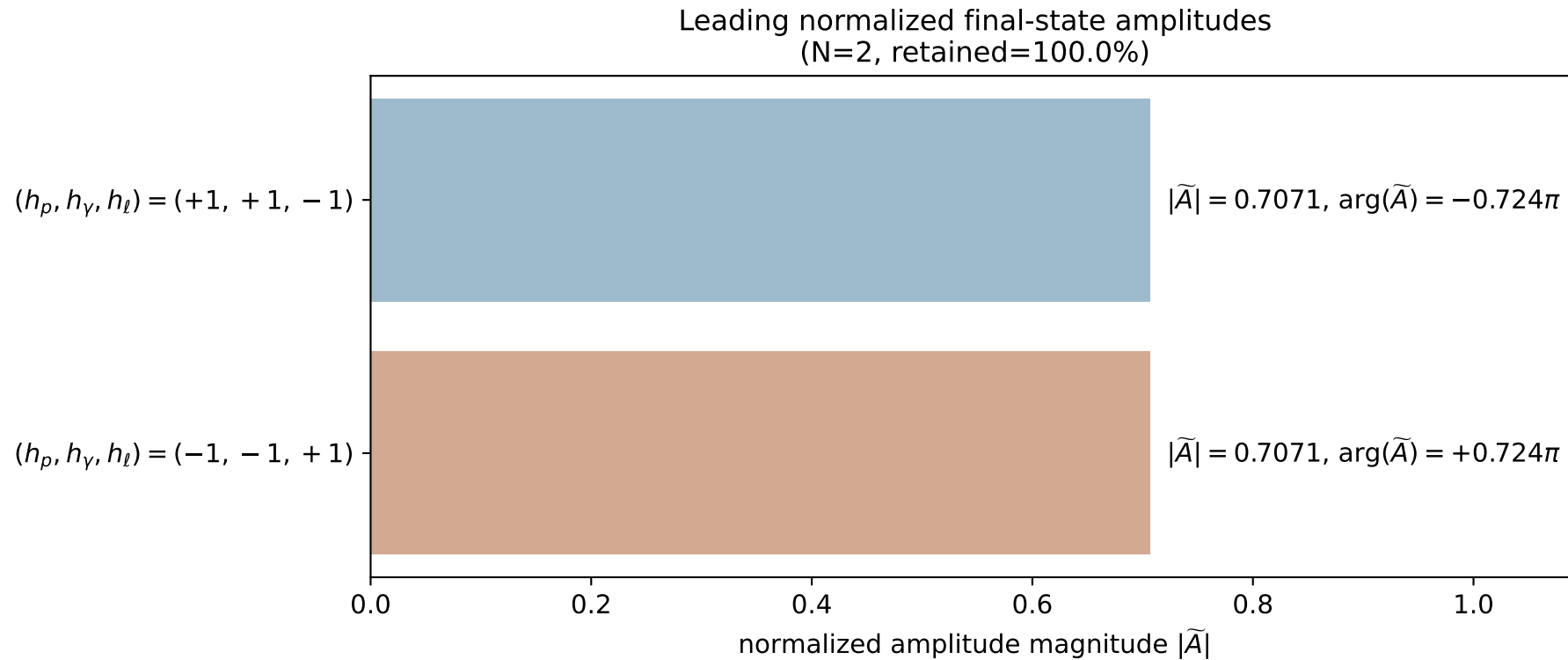


dghz_mixing_angles_polarization_04_local_minimum_80
 $d_{\text{GHZ}}=3.02153\text{e-}08$
 region: polarization_cluster_04_local_minimum_80
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0080
 lepton mixing angle: $\alpha_e=2.3404957$ rad
 incoming lepton: $-0.695919|+\rangle +0.71812|-\rangle$
 proton mixing angle: $\alpha_p=2.3404627$ rad
 incoming proton: $-0.695896|+\rangle +0.718143|-\rangle$

$s=24.1511$, $\sqrt{s}=4.91438$, $|\vec{P}|=2.36767$, $|\vec{P}'|=0.31225$
 $\theta_{P'}=9.71376\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.3268$, $\phi_\gamma=2.27603$
 $E_\gamma=2.11901$, $\theta_{l'}=1.67875\text{e-}06$, $\phi_{l'}=2.18521$
 $Q^2=17.1113$, $x_B=0.756322$, $t=-1.79708$, $W^2=6.39291$, $y=0.9722$

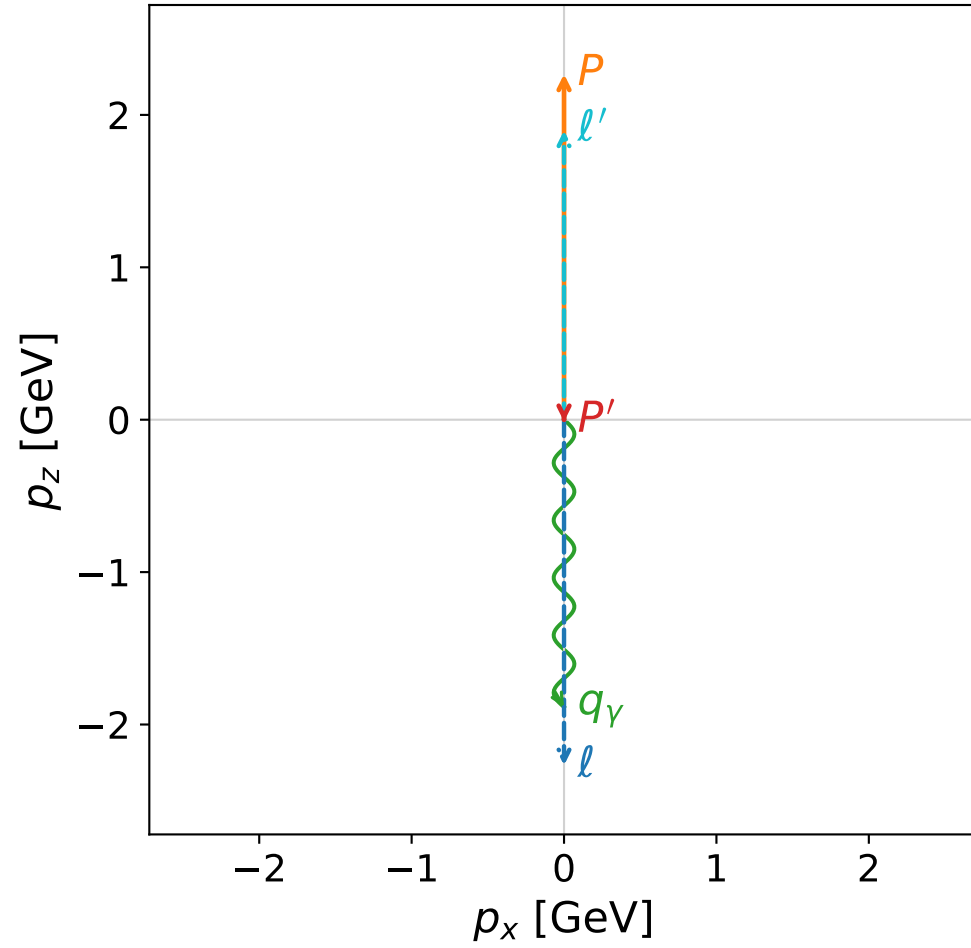
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3677$
 P $E= 2.5467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3677$
 l' $E= 1.8068$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8068$
 P' $E= 0.9886$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3123$
 q_γ $E= 2.1190$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1190$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.877\text{e+}19$; $I_h=+3.675\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.219\text{e-}14$; $I_{TL}=-3.811\text{e-}09$

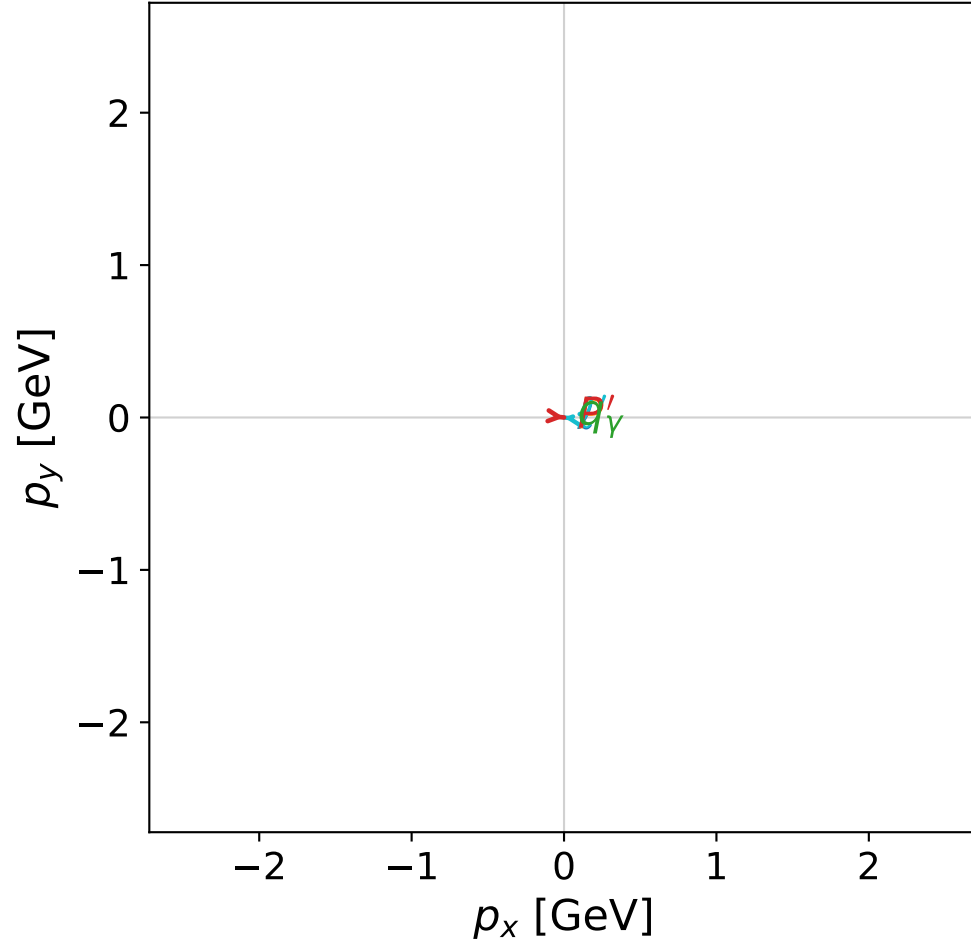


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

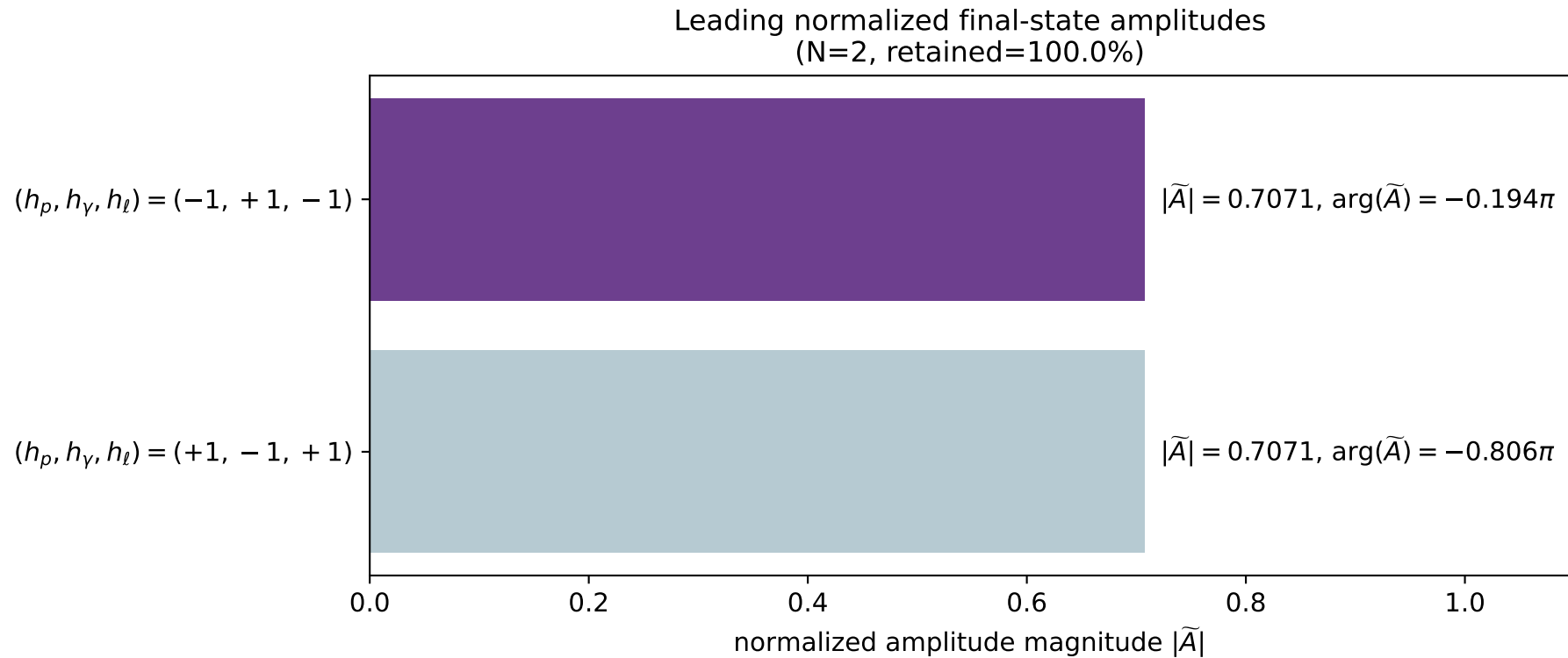


dghz_mixing_angles_polarization_04_local_minimum_81
 $d_{\text{GHZ}}=3.03827\text{e-}08$
 region: polarization_cluster_04_local_minimum_81
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0081
 lepton mixing angle: $\alpha_e=2.3663814$ rad
 incoming lepton: $-0.714273|+\rangle +0.699867|-\rangle$
 proton mixing angle: $\alpha_p=2.3663611$ rad
 incoming proton: $-0.714259|+\rangle +0.699881|-\rangle$

$s=22.294$, $\sqrt{s}=4.72165$, $|\vec{P}|=2.26765$, $|\vec{P}'|=0.0124921$
 $\theta_{P'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.17498$, $\phi_\gamma=3.64321$
 $E_\gamma=1.88554$, $\theta_{\ell'}=7.48922\text{e-}07$, $\phi_{\ell'}=3.03338$
 $Q^2=17.2163$, $x_B=0.831433$, $t=-2.90107$, $W^2=4.37032$, $y=0.966967$

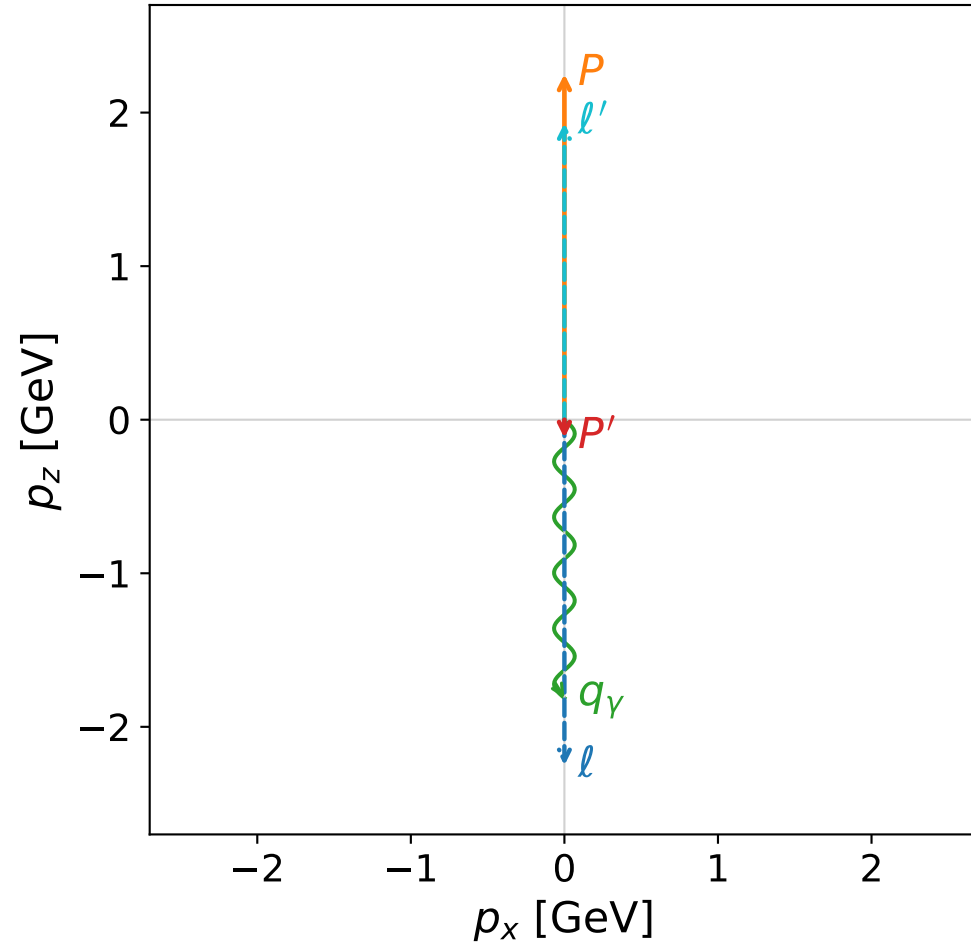
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2677$
 P $E= 2.4540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2677$
 ℓ' $E= 1.8980$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8980$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0125$
 q_γ $E= 1.8855$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8855$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.289\text{e+}20$; $I_h=-2.055\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.506\text{e-}15$; $I_{TL}=+1.434\text{e-}09$

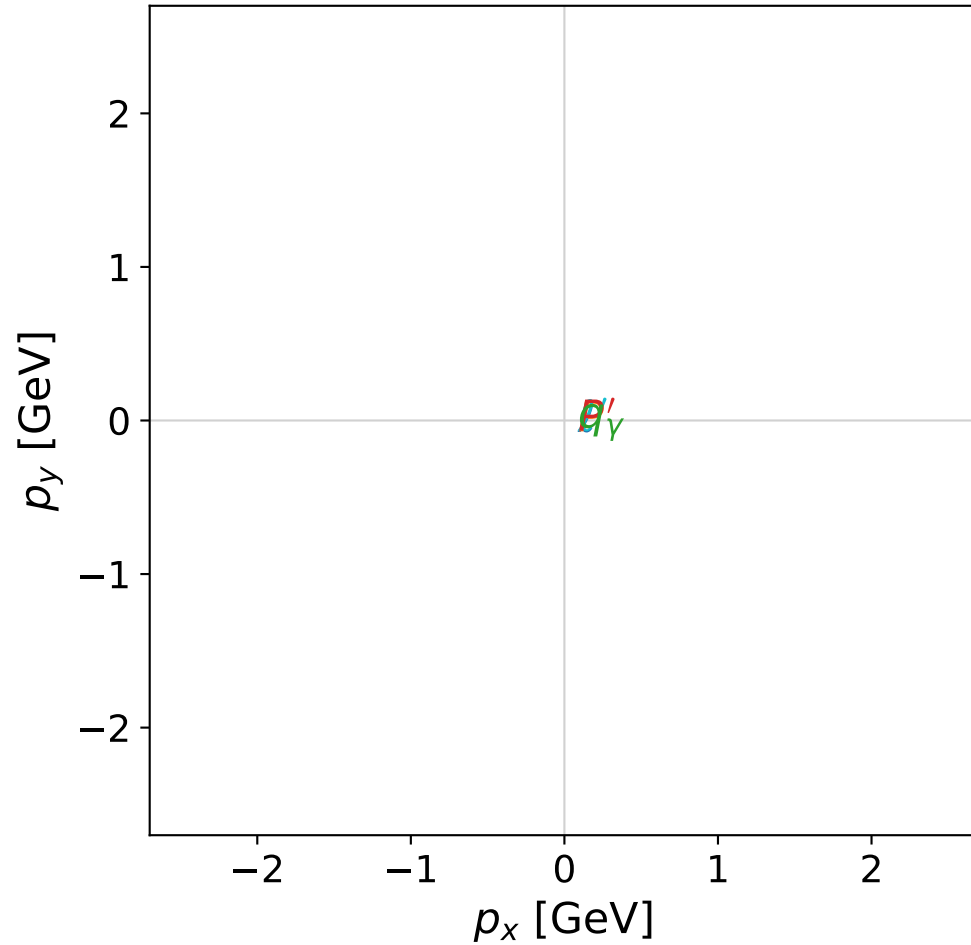


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

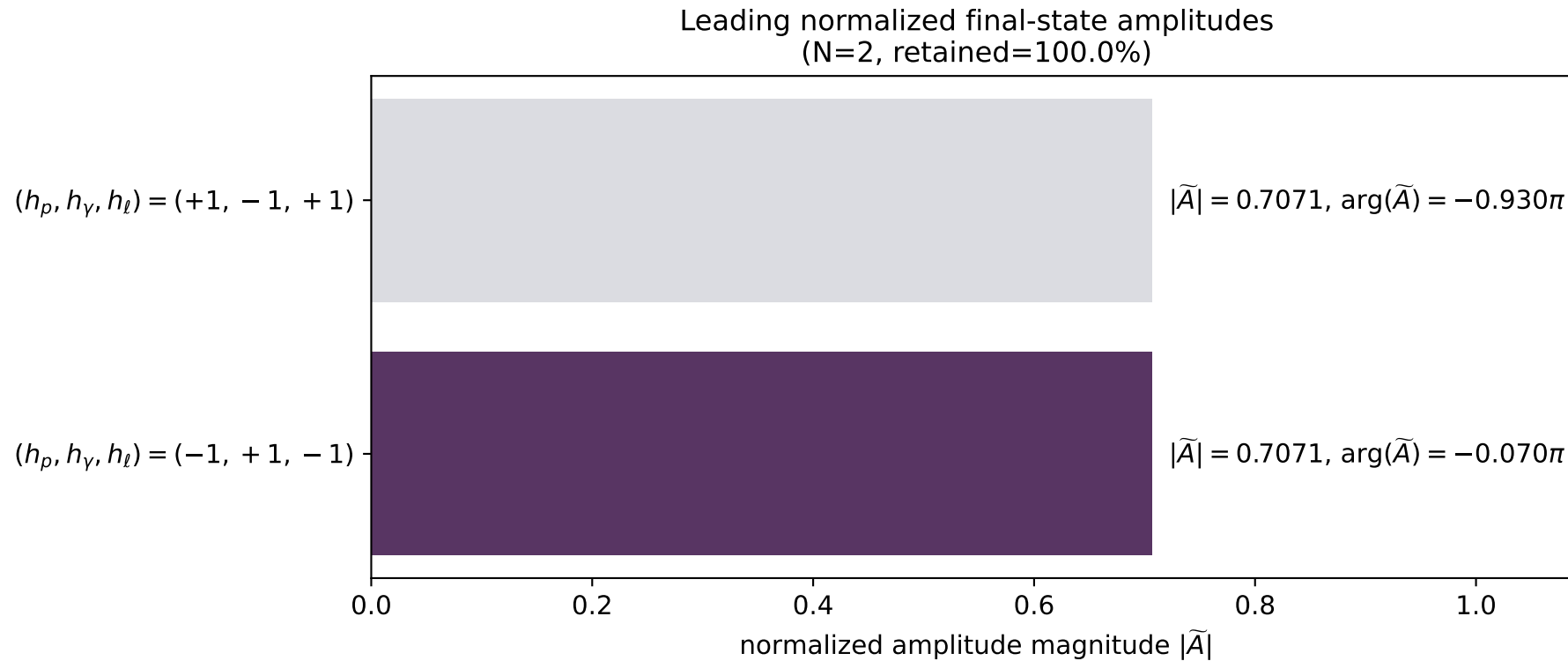


dghz_mixing_angles_polarization_04_local_minimum_82
 $d_{\text{GHZ}}=3.04333\text{e-}08$
 region: polarization_cluster_04_local_minimum_82
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0082
 lepton mixing angle: $\alpha_e=2.2763558$ rad
 incoming lepton: $-0.64846|+\rangle +0.761249|-\rangle$
 proton mixing angle: $\alpha_p=2.2763635$ rad
 incoming proton: $-0.648466|+\rangle +0.761244|-\rangle$

$s=21.9839$, $\sqrt{s}=4.68869$, $|\vec{P}|=2.25052$, $|\vec{P}'|=0.118469$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.11906$, $\phi_Y=3.36218$
 $E_Y=1.81239$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.205792$
 $Q^2=17.3817$, $x_B=0.852909$, $t=-3.3839$, $W^2=3.87747$, $y=0.965662$

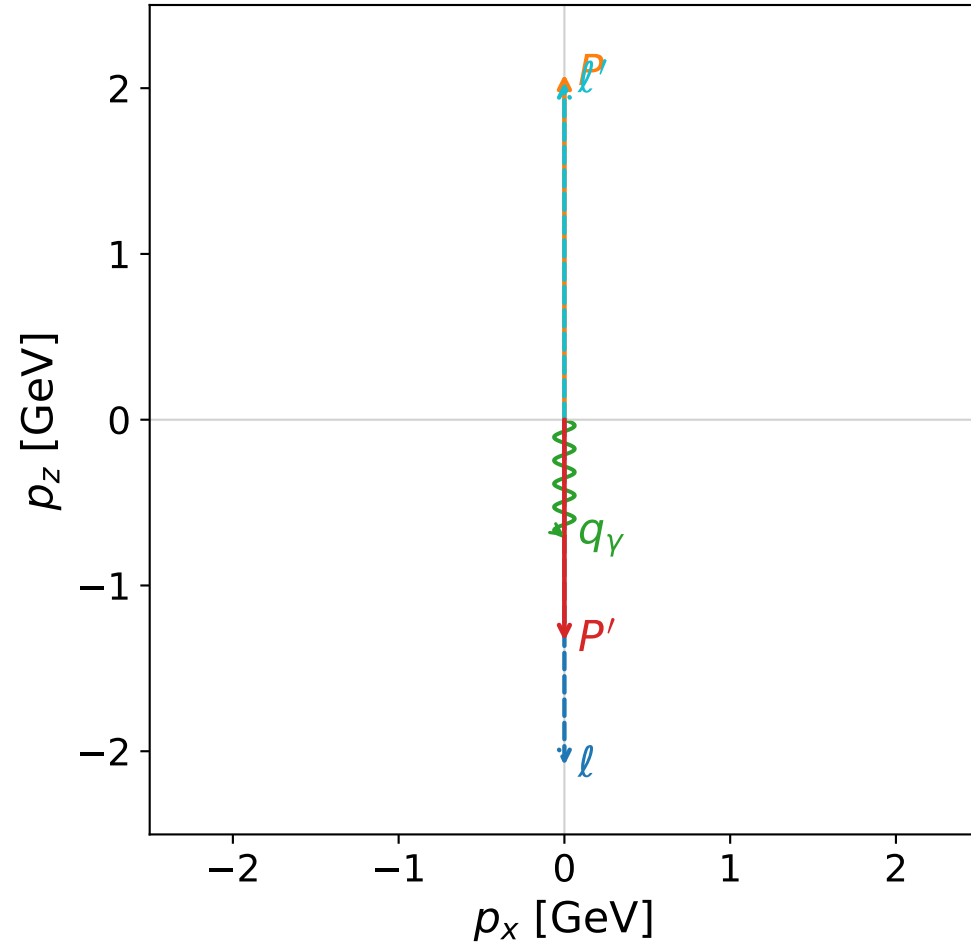
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2505$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2505$
 P $E= 2.4382$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2505$
 ℓ' $E= 1.9309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9309$
 P' $E= 0.9455$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1185$
 q_Y $E= 1.8124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8124$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.004\text{e+}24$; $I_h=+4.098\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.316\text{e-}27$; $I_{TL}=-4.324\text{e-}18$

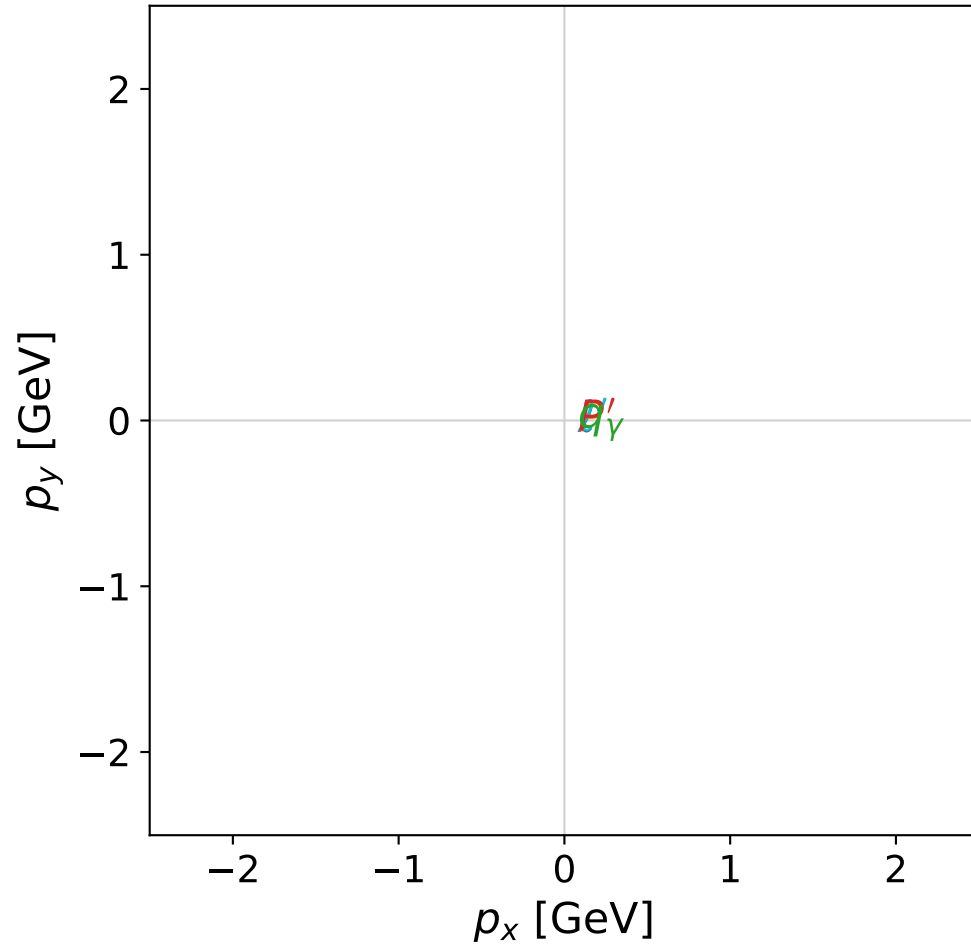


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

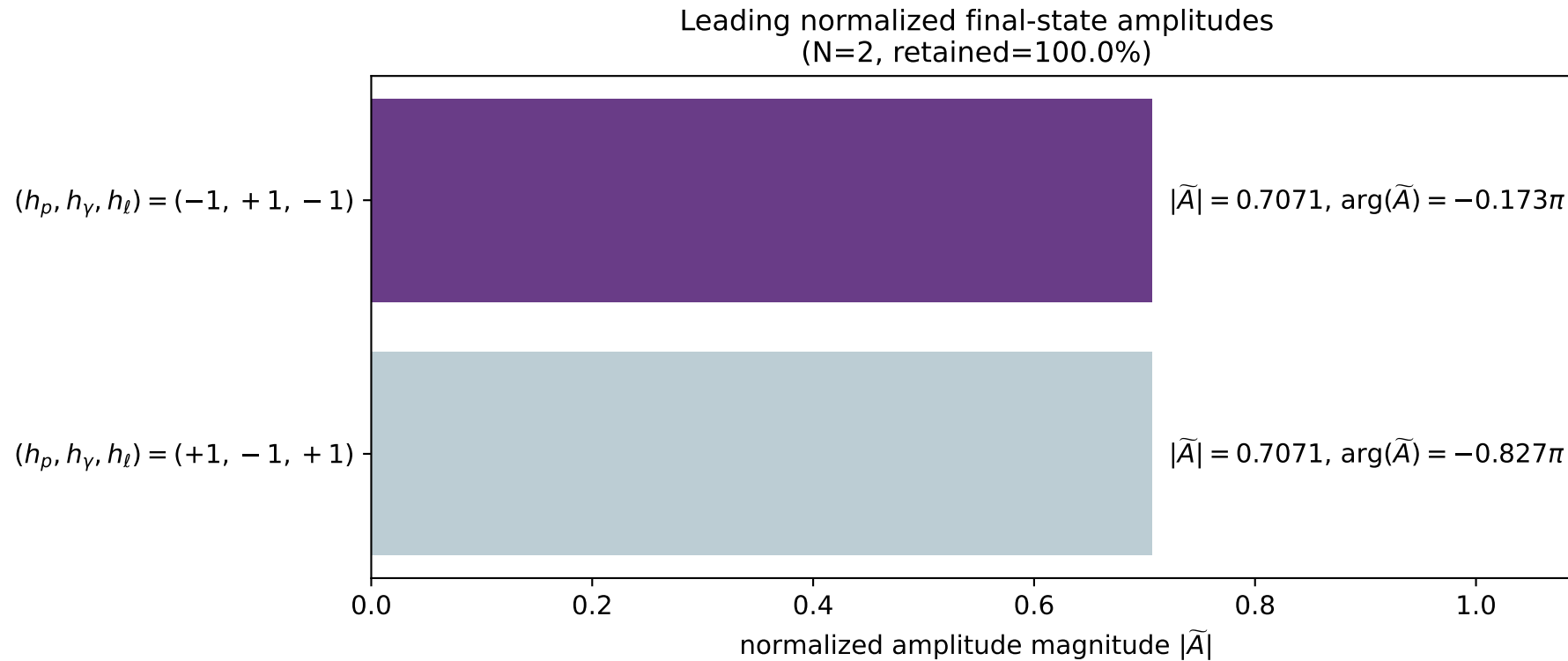


dghz_mixing_angles_polarization_04_local_minimum_84
 $d_{\text{GHZ}}=3.10329\text{e-}08$
 region: polarization_cluster_04_local_minimum_84
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0084
 lepton mixing angle: $\alpha_e=2.2903682$ rad
 incoming lepton: $-0.659063|+\rangle +0.752088|-\rangle$
 proton mixing angle: $\alpha_p=2.2903511$ rad
 incoming proton: $-0.65905|+\rangle +0.752099|-\rangle$

$s=19.1051$, $\sqrt{s}=4.37093$, $|\vec{P}|=2.08482$, $|\vec{P}'|=1.33443$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0100816$, $\phi_\gamma=3.68413$
 $E_\gamma=0.702689$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.69661$
 $Q^2=16.9881$, $x_B=0.976044$, $t=-11.2623$, $W^2=1.2968$, $y=0.955001$

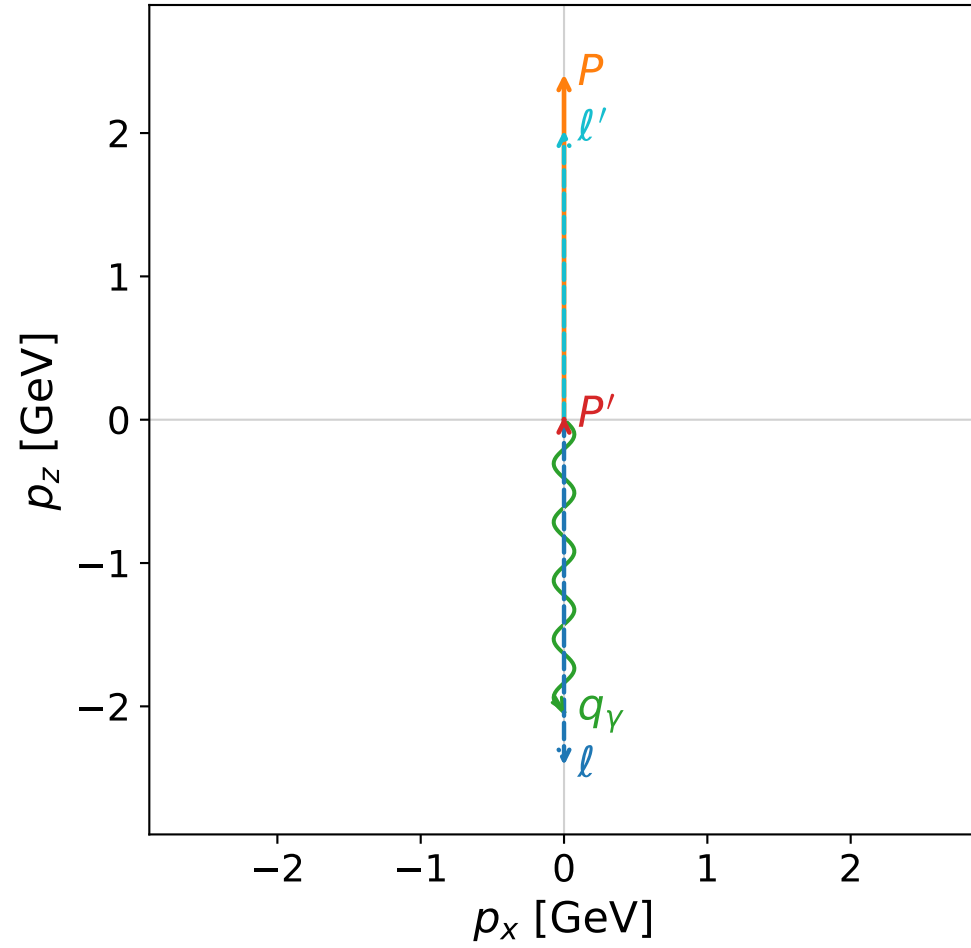
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0848$
 P $E= 2.2861$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0848$
 ℓ' $E= 2.0371$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0371$
 P' $E= 1.6311$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3344$
 q_γ $E= 0.7027$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.7027$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.054\text{e+}23$; $I_h=+2.230\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.719\text{e-}30$; $I_{TL}=-1.289\text{e-}19$

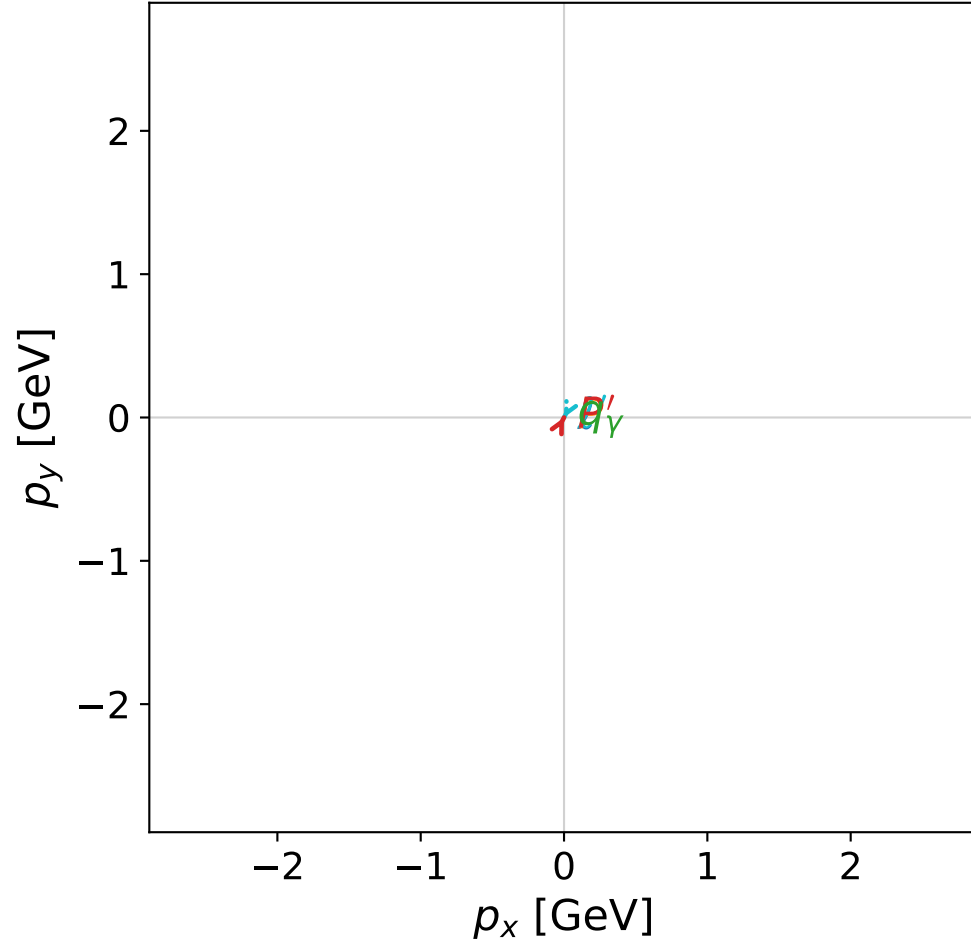


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

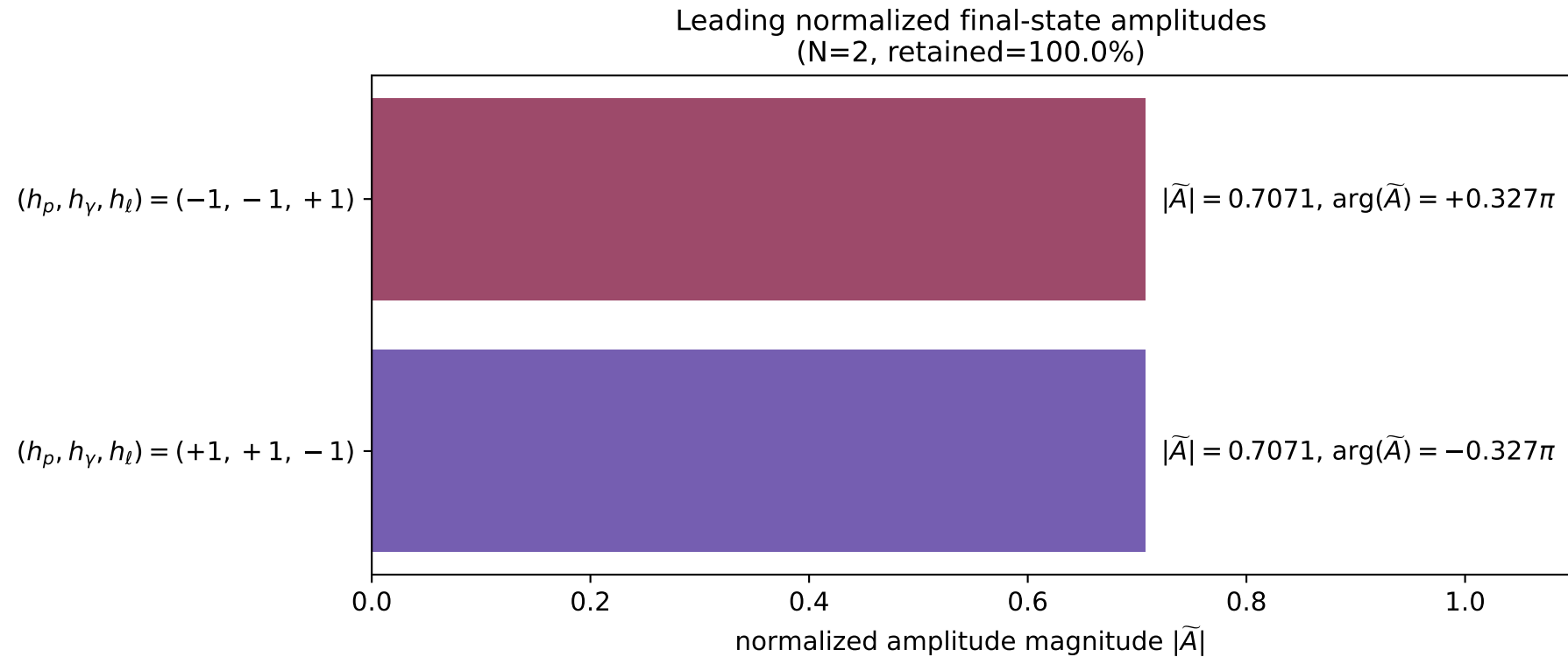


dghz_mixing_angles_polarization_04_local_minimum_85
 $d_{\text{GHZ}}=3.10962\text{e-}08$
 region: polarization_cluster_04_local_minimum_85
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0085
 lepton mixing angle: $\alpha_e=2.1008763$ rad
 incoming lepton: $-0.505602|+\rangle +0.862767|-\rangle$
 proton mixing angle: $\alpha_p=2.1008833$ rad
 incoming proton: $-0.505608|+\rangle +0.862763|-\rangle$

$s=24.9853$, $\sqrt{s}=4.99853$, $|\vec{P}|=2.41125$, $|\vec{P}'|=0.0216116$
 $\theta_{P'}=0.00498868$, $\theta_Y=3.14159$, $\phi_{P'}=1.09327$, $\phi_Y=1.0281$
 $E_Y=2.04095$, $\theta_{\ell'}=5.33904\text{e-}05$, $\phi_{\ell'}=4.23487$
 $Q^2=19.4765$, $x_B=0.832523$, $t=-2.99111$, $W^2=4.79789$, $y=0.970509$

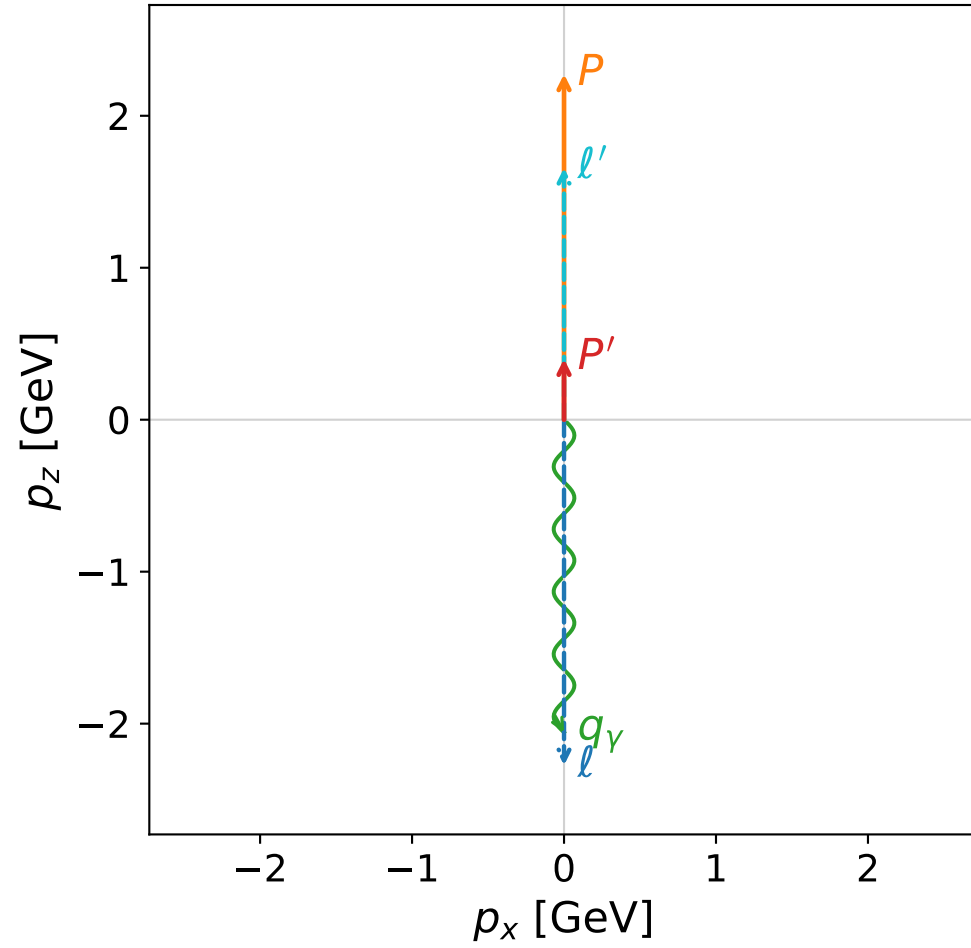
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0193$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0193$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0216$
 q_γ $E= 2.0409$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0409$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.094\text{e+}16$; $I_h=+1.721\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.477\text{e-}11$; $I_{TL}=-1.219\text{e-}06$

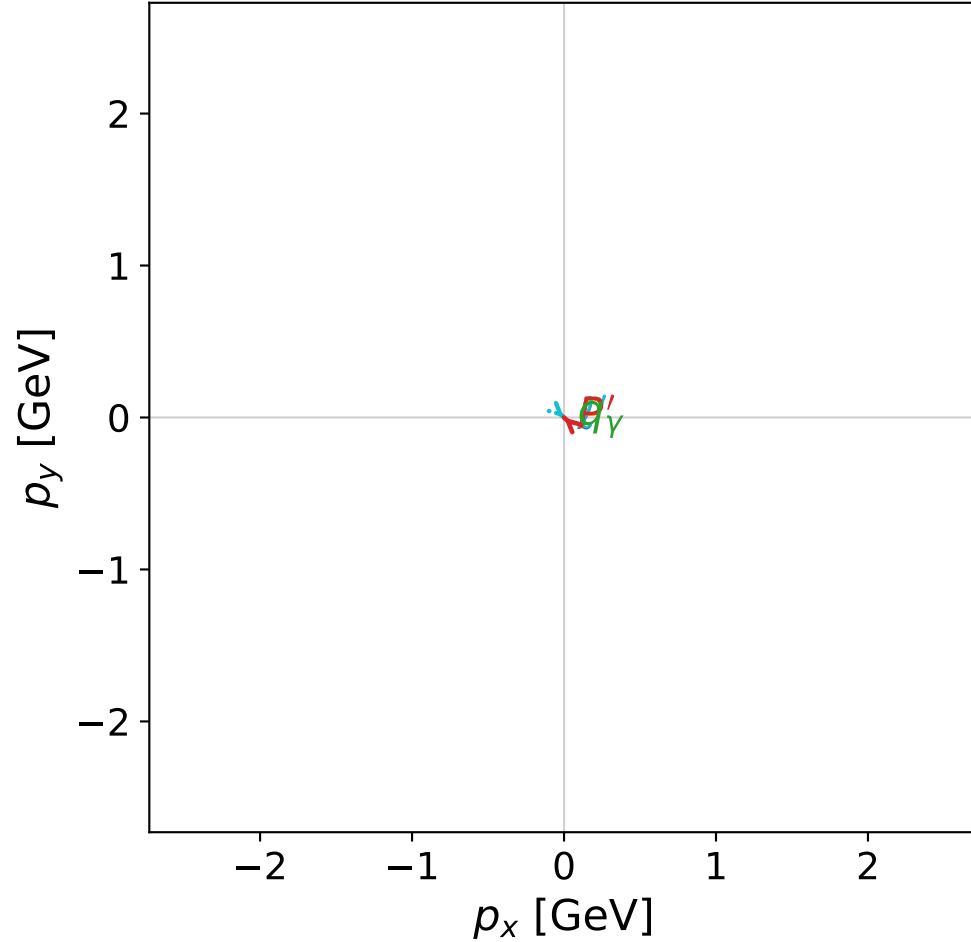


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

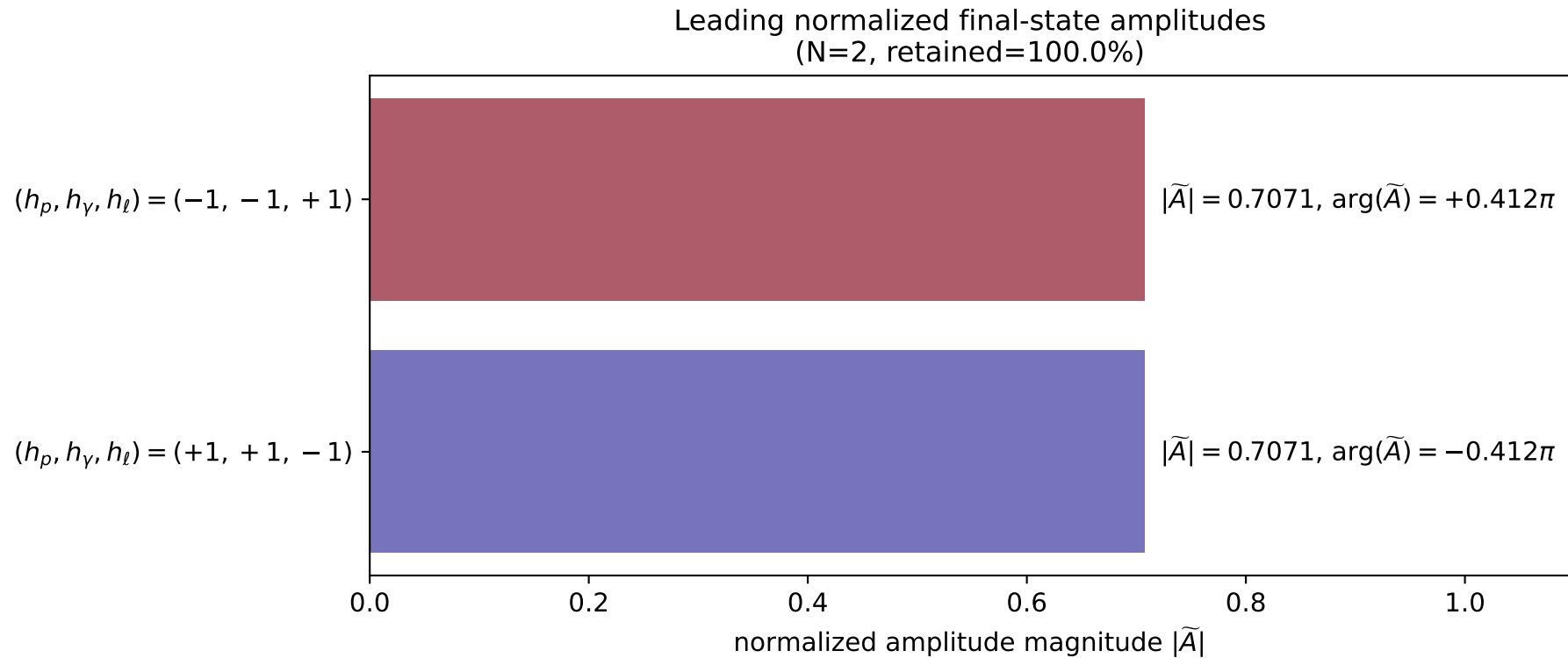


dghz_mixing_angles_polarization_04_local_minimum_88
 $d_{\text{GHZ}}=3.20878\text{e-}08$
 region: polarization_cluster_04_local_minimum_88
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0088
 lepton mixing angle: $\alpha_e=2.2717872$ rad
 incoming lepton: $-0.644975|+\rangle +0.764203|-\rangle$
 proton mixing angle: $\alpha_p=2.2717877$ rad
 incoming proton: $-0.644976|+\rangle +0.764203|-\rangle$

$s=22.4118$, $\sqrt{s}=4.73411$, $|\vec{P}|=2.27413$, $|\vec{P}'|=0.400283$
 $\theta_{P'}=1.00102\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.40749$, $\phi_\gamma=1.29355$
 $E_\gamma=2.05728$, $\theta_{\ell'}=2.41656\text{e-}07$, $\phi_{\ell'}=5.54909$
 $Q^2=15.0729$, $x_B=0.720637$, $t=-1.43729$, $W^2=6.72301$, $y=0.971395$

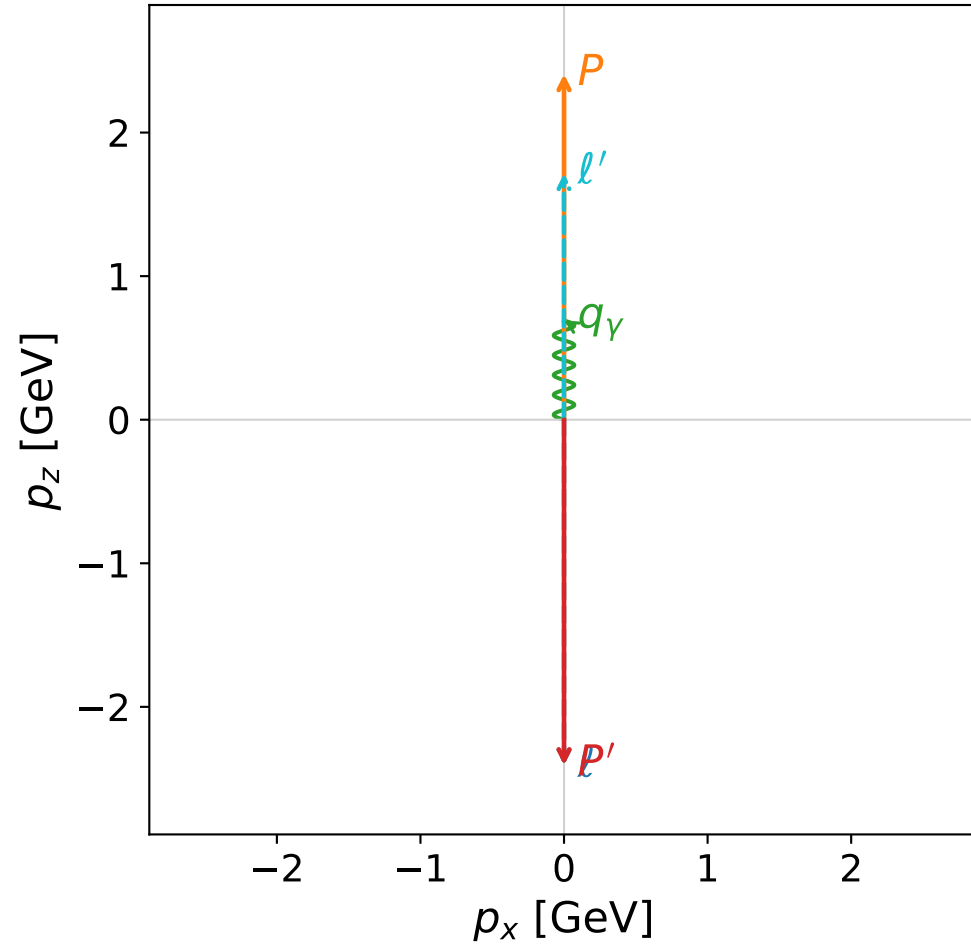
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2741$
 P $E= 2.4600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2741$
 ℓ' $E= 1.6570$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6570$
 P' $E= 1.0198$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.4003$
 q_γ $E= 2.0573$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.226\text{e+}21$; $I_h=-3.806\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.349\text{e-}16$; $I_{TL}=+4.445\text{e-}09$

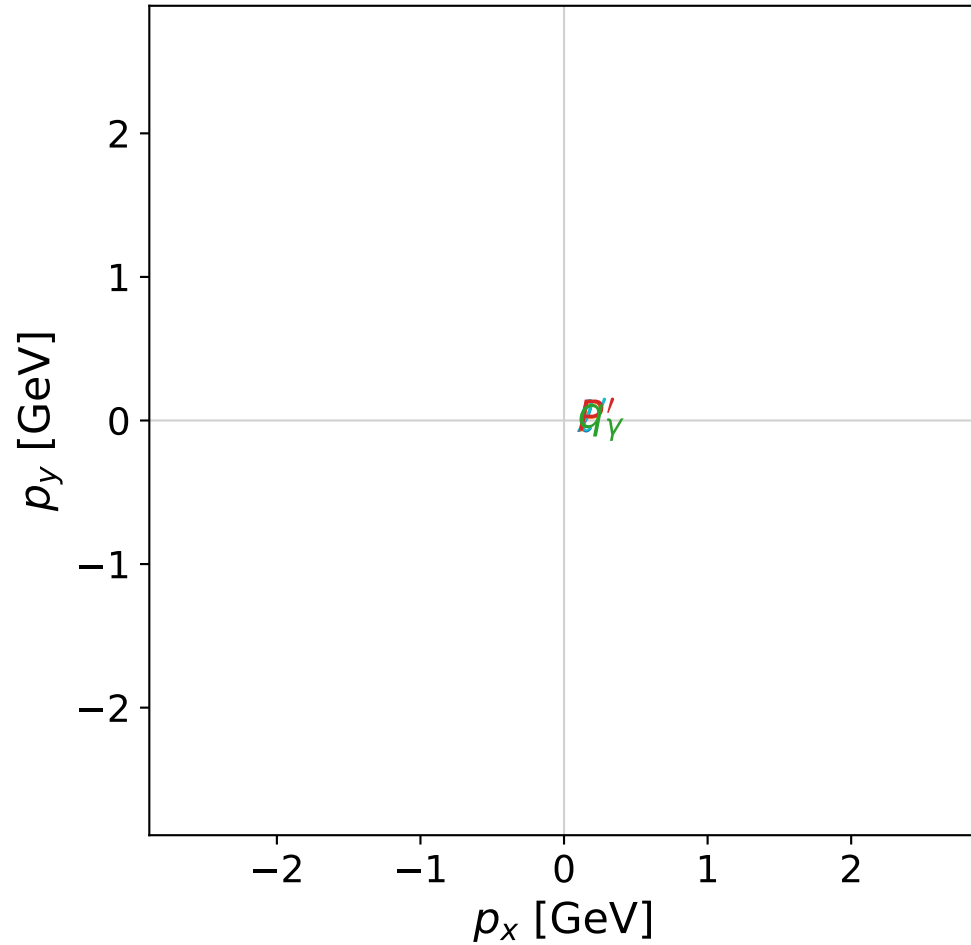


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

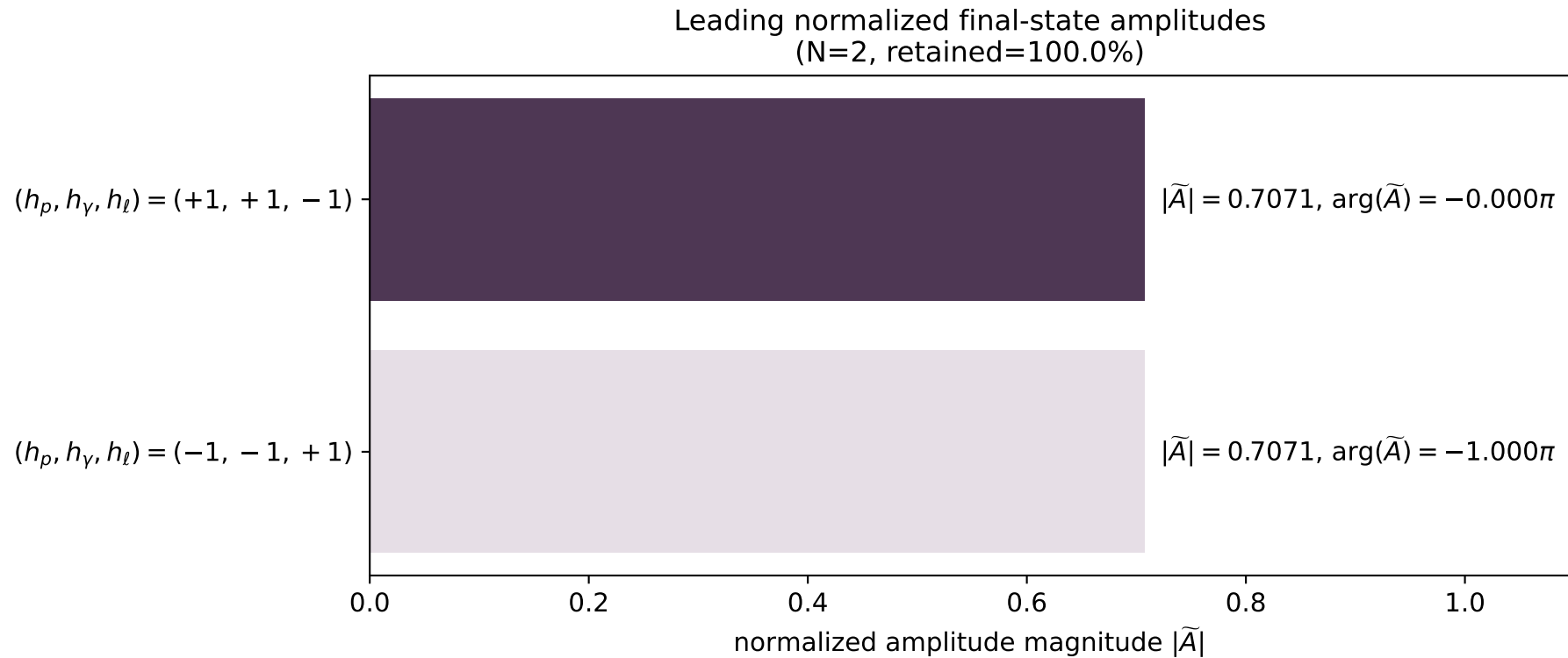


dghz_mixing_angles_polarization_04_local_minimum_91
 $d_{\text{GHZ}}=3.29073\text{e-}08$
 region: polarization_cluster_04_local_minimum_91
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0091
 lepton mixing angle: $\alpha_e=2.4970427$ rad
 incoming lepton: $-0.79937|+\rangle +0.600839|-\rangle$
 proton mixing angle: $\alpha_p=2.2153473$ rad
 incoming proton: $-0.60084|+\rangle +0.79937|-\rangle$

$s=24.9047$, $\sqrt{s}=4.99046$, $|\vec{P}|=2.40708$, $|\vec{P}'|=2.40708$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=4.19237$, $\phi_Y=1.989$
 $E_Y=0.691167$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.05077$
 $Q^2=16.5213$, $x_B=0.705442$, $t=-23.1761$, $W^2=7.77833$, $y=0.974816$

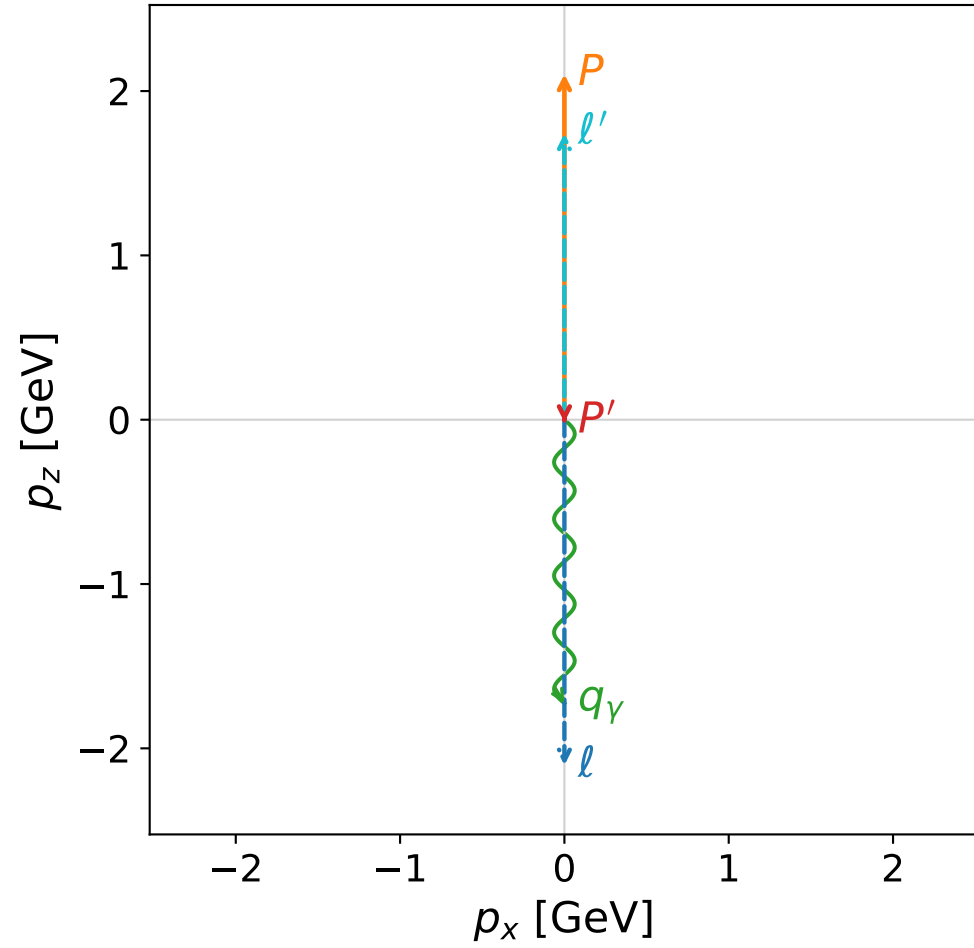
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4071$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4071$
 P $E= 2.5834$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4071$
 ℓ' $E= 1.7159$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7159$
 P' $E= 2.5834$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4071$
 q_Y $E= 0.6912$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6912$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.633\text{e+}23$; $I_h=-1.305\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.230\text{e-}27$; $I_{TL}=-1.305\text{e-}13$

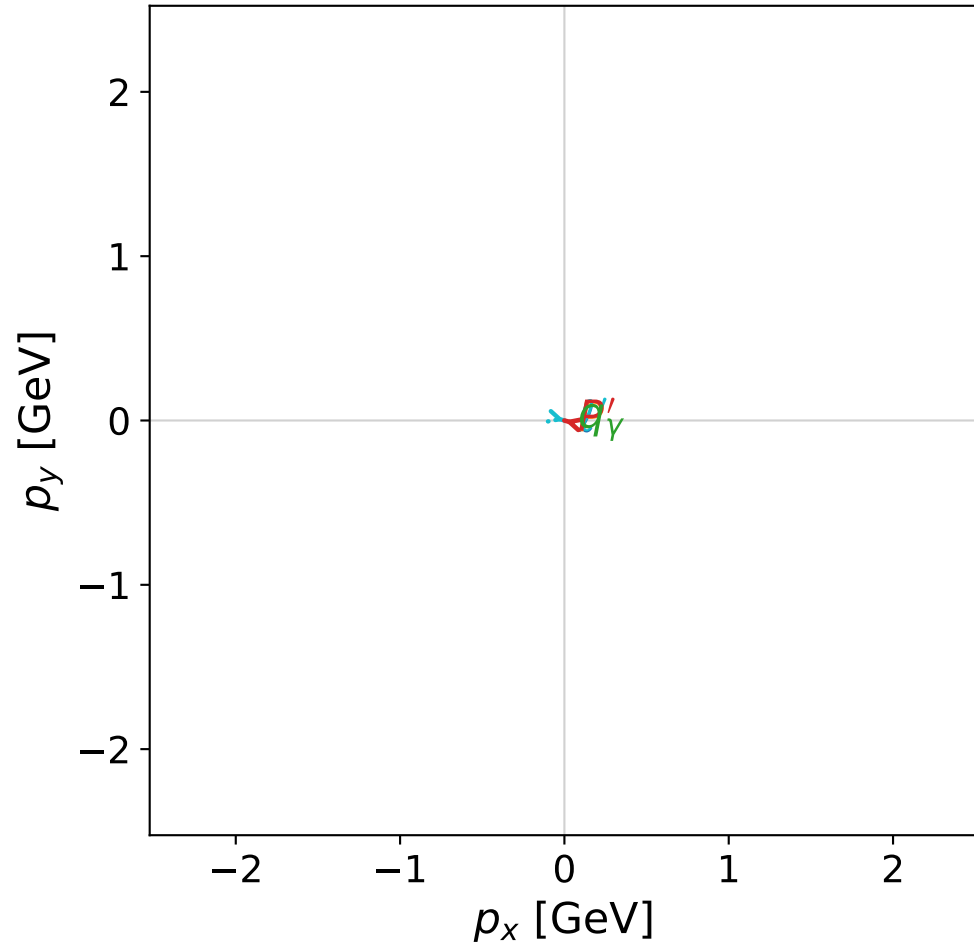


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

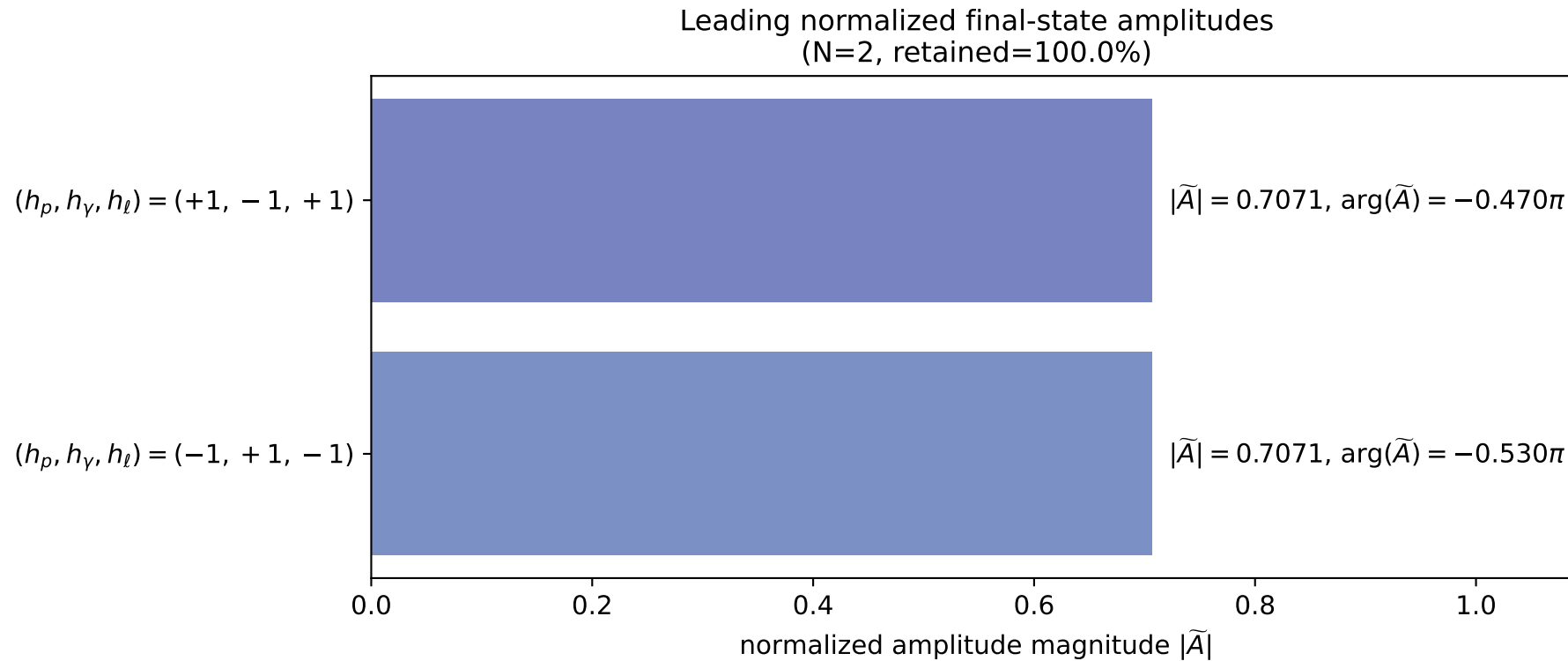


dghz_mixing_angles_polarization_04_local_minimum_98
 $d_{\text{GHZ}}=3.56394\text{e-}08$
 region: polarization_cluster_04_local_minimum_98
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0098
 lepton mixing angle: $\alpha_e=2.3614211$ rad
 incoming lepton: $-0.710793|+\rangle +0.703401|-\rangle$
 proton mixing angle: $\alpha_p=2.3614305$ rad
 incoming proton: $-0.7108|+\rangle +0.703395|-\rangle$

$s=19.4131$, $\sqrt{s}=4.40603$, $|\vec{P}|=2.10317$, $|\vec{P}'|=0.0160914$
 $\theta_{P'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.86856$, $\phi_\gamma=1.39128$
 $E_\gamma=1.7259$, $\theta_{\ell'}=1.00525\text{e-}06$, $\phi_{\ell'}=6.01016$
 $Q^2=14.6548$, $x_B=0.821572$, $t=-2.6288$, $W^2=4.06257$, $y=0.962461$

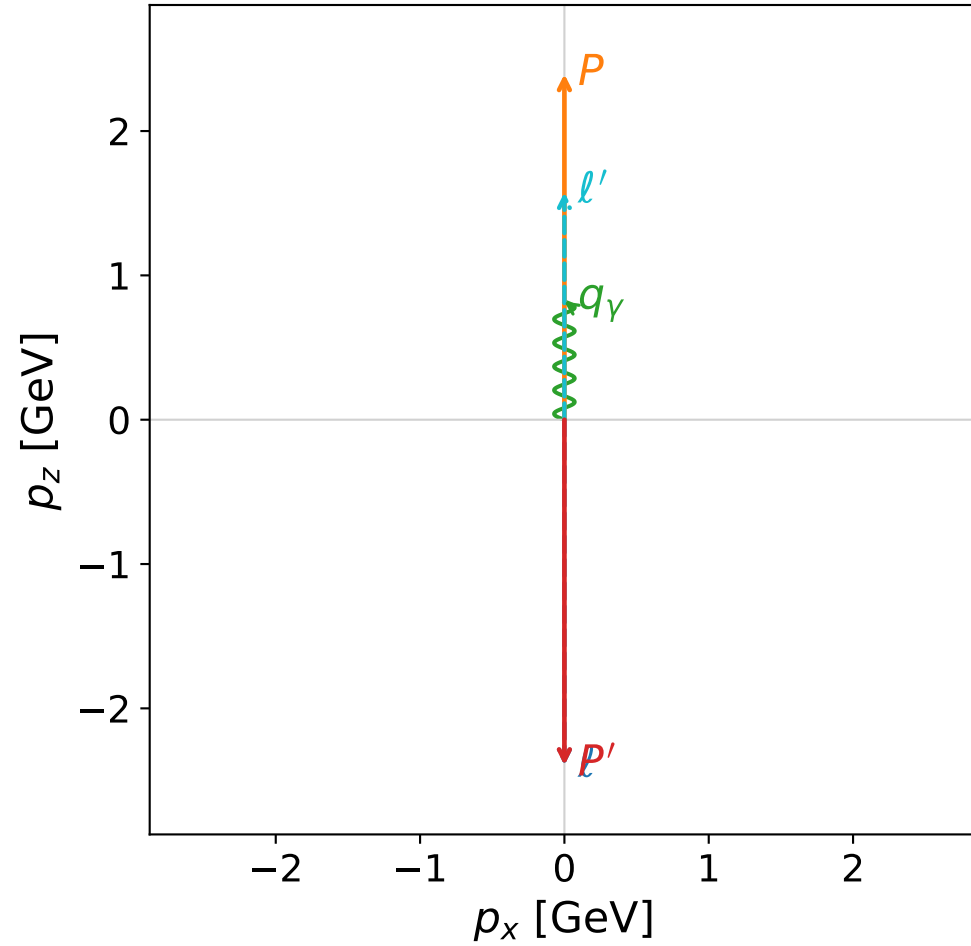
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1032$
 P $E= 2.3029$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1032$
 ℓ' $E= 1.7420$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7420$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0161$
 q_γ $E= 1.7259$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7259$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.060\text{e+}19$; $I_h=+1.501\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.731\text{e-}15$; $I_{TL}=-1.077\text{e-}09$

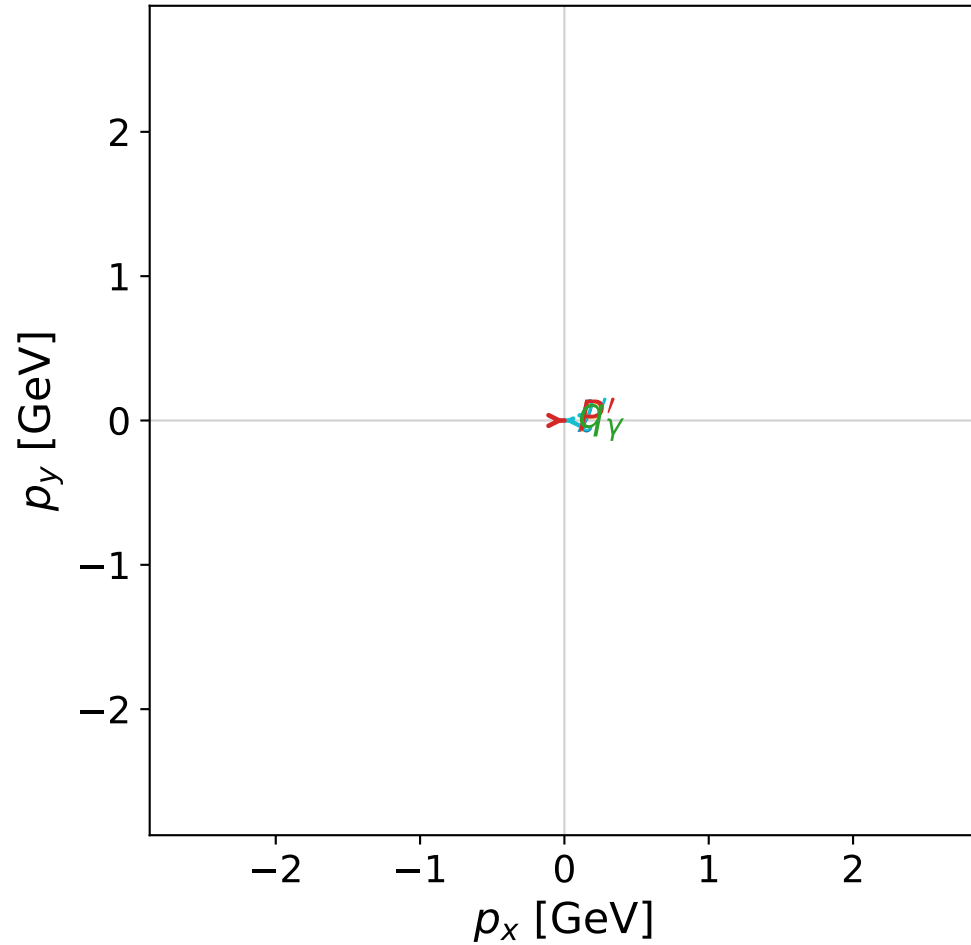


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

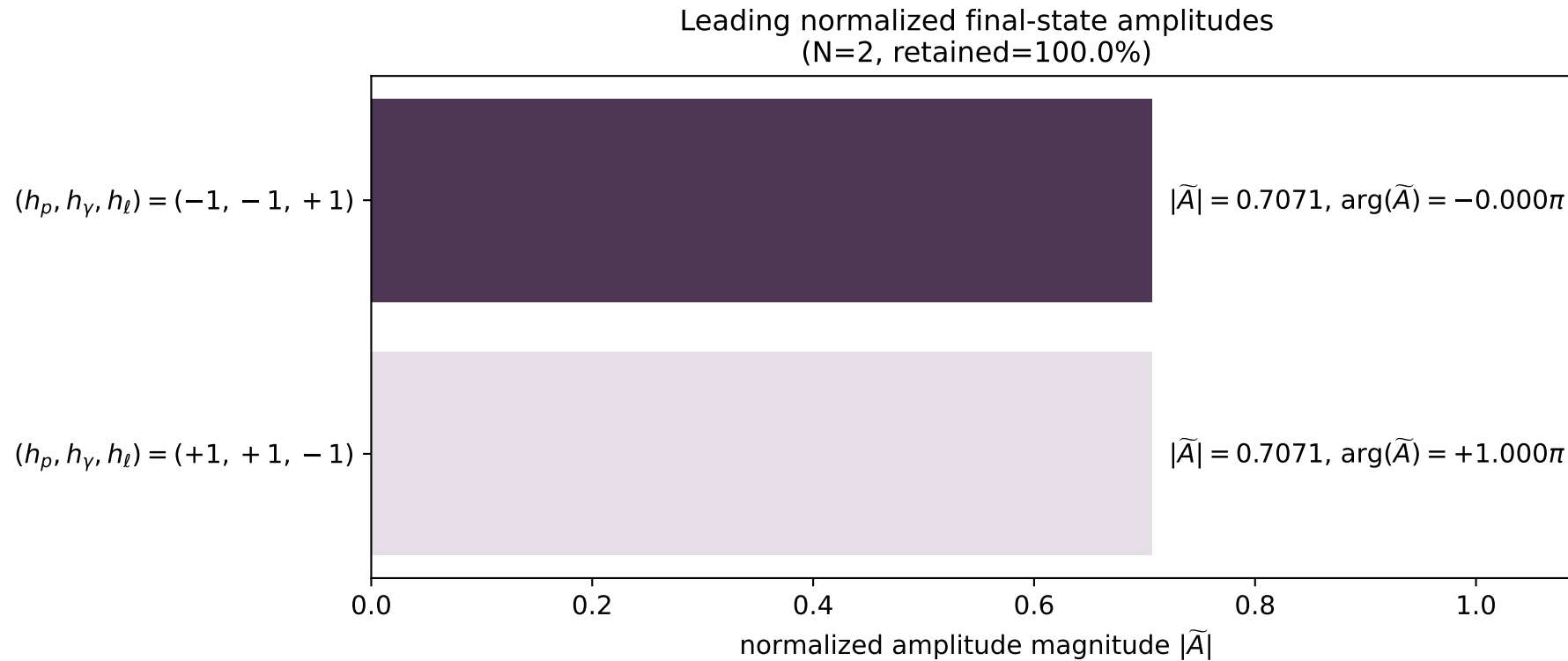


```
dghz_mixing_angles_polarization_04_local_minimum_101
d_{\rm GHZ}=3.64201e-08
region: polarization_cluster_04_local_minimum_101
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0101
lepton mixing angle: alpha_e=2.19789 rad
incoming lepton: -0.586794|+> +0.809736|->
proton mixing angle: alpha_p=2.514492 rad
incoming proton: -0.809732|+> +0.5868|->
```

```
s=24.6624, sqrt(s)=4.96613, |P|=2.39448, |P'|=2.39448
theta_{p'}=3.14159, theta_gamma=0, phi_{p'}=0.290949, phi_gamma=6.12241
E_gamma=0.819028, theta_{\ell'}=0, phi_{\ell'}=3.43254
Q^2=15.0896, x_B=0.64973, t=-22.9341, W^2=9.01465, y=0.976527
```

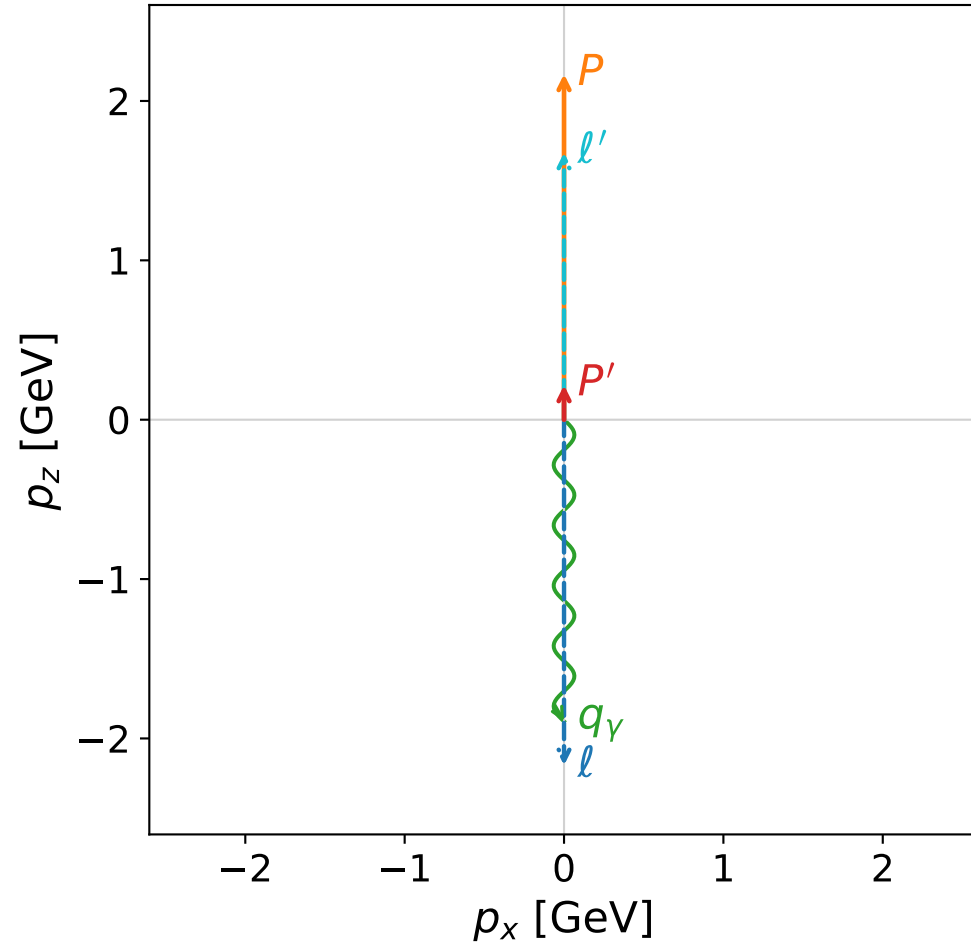
```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.3945 p_x=  0.0000 p_y=  0.0000 p_z= -2.3945
P      E=  2.5716 p_x=  0.0000 p_y=  0.0000 p_z=  2.3945
l'     E=  1.5755 p_x= -0.0000 p_y= -0.0000 p_z=  1.5755
P'     E=  2.5716 p_x=  0.0000 p_y=  0.0000 p_z= -2.3945
q_gamma E=  0.8190 p_x=  0.0000 p_y= -0.0000 p_z=  0.8190
```

```
GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=2.470e+23; I_h=-4.265e-14
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=1.007e-27; I_TL=-4.265e-14
```

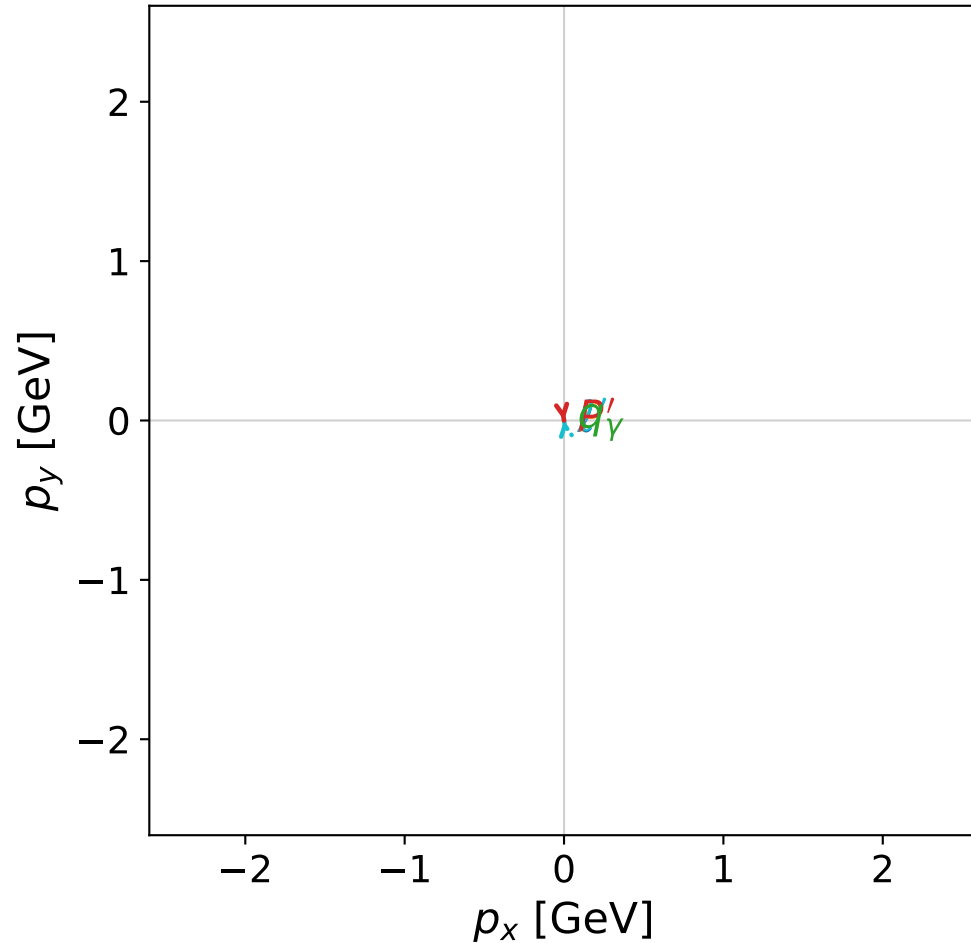


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_105

$d_{\text{GHZ}}=3.82625\text{e-}08$

region: polarization_cluster_04_local_minimum_105

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0105

lepton mixing angle: $\alpha_e=2.5411507$ rad

incoming lepton: $-0.825086|+\rangle +0.565007|-\rangle$

proton mixing angle: $\alpha_p=2.5411495$ rad

incoming proton: $-0.825085|+\rangle +0.565008|-\rangle$

$s=20.5321$, $\sqrt{s}=4.53124$, $|\vec{P}|=2.16853$, $|\vec{P}'|=0.216611$

$\theta_{P'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=4.86682$, $\phi_Y=5.92443$

$E_Y=1.89258$, $\theta_{\ell'}=9.91298\text{e-}06$, $\phi_{\ell'}=1.72522$

$Q^2=14.5376$, $x_B=0.765079$, $t=-1.84994$, $W^2=5.34368$, $y=0.966881$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.1685$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1685$

P $E= 2.3627$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1685$

ℓ' $E= 1.6760$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6760$

P' $E= 0.9627$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.2166$

q_Y $E= 1.8926$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8926$

GHZ mechanism: helicity flipping (full H G C)

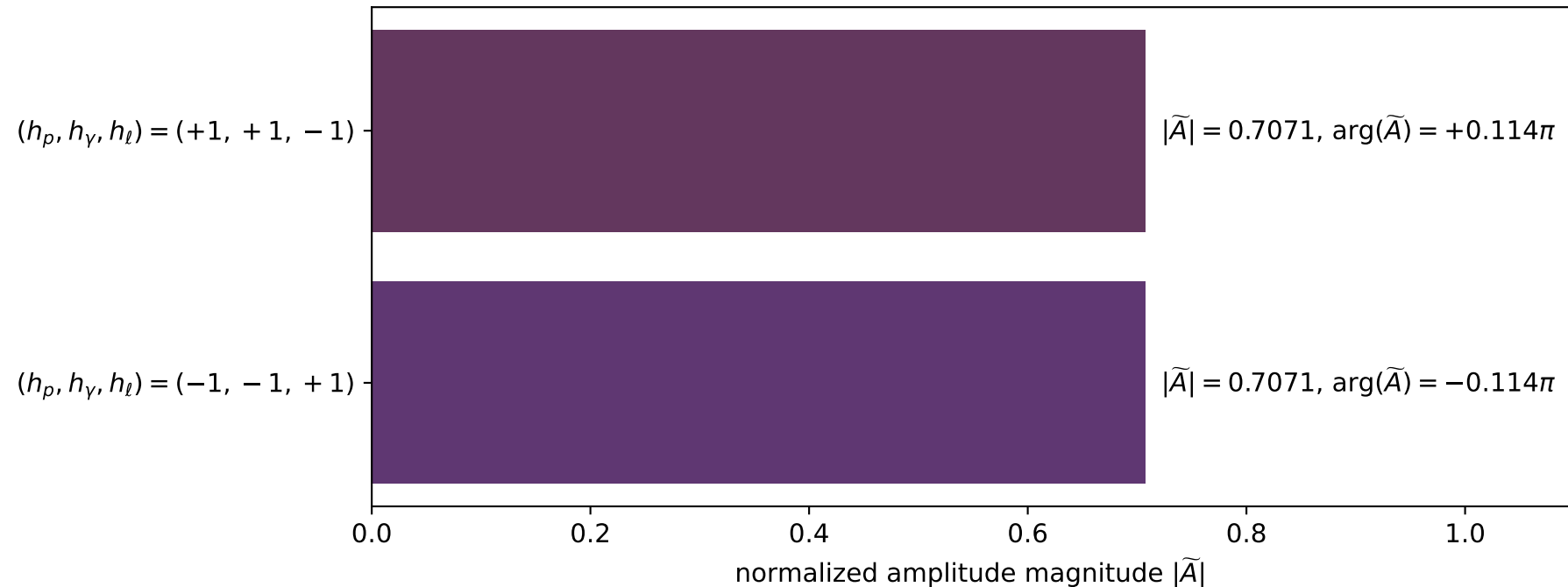
helicity powers: flip 100.000%, conserve 0.000%

$R_{\text{flip/conserve}}=5.763\text{e+}17$; $I_h=-7.976\text{e-}11$

virtual-photon powers: T 100.000%, L 0.000%

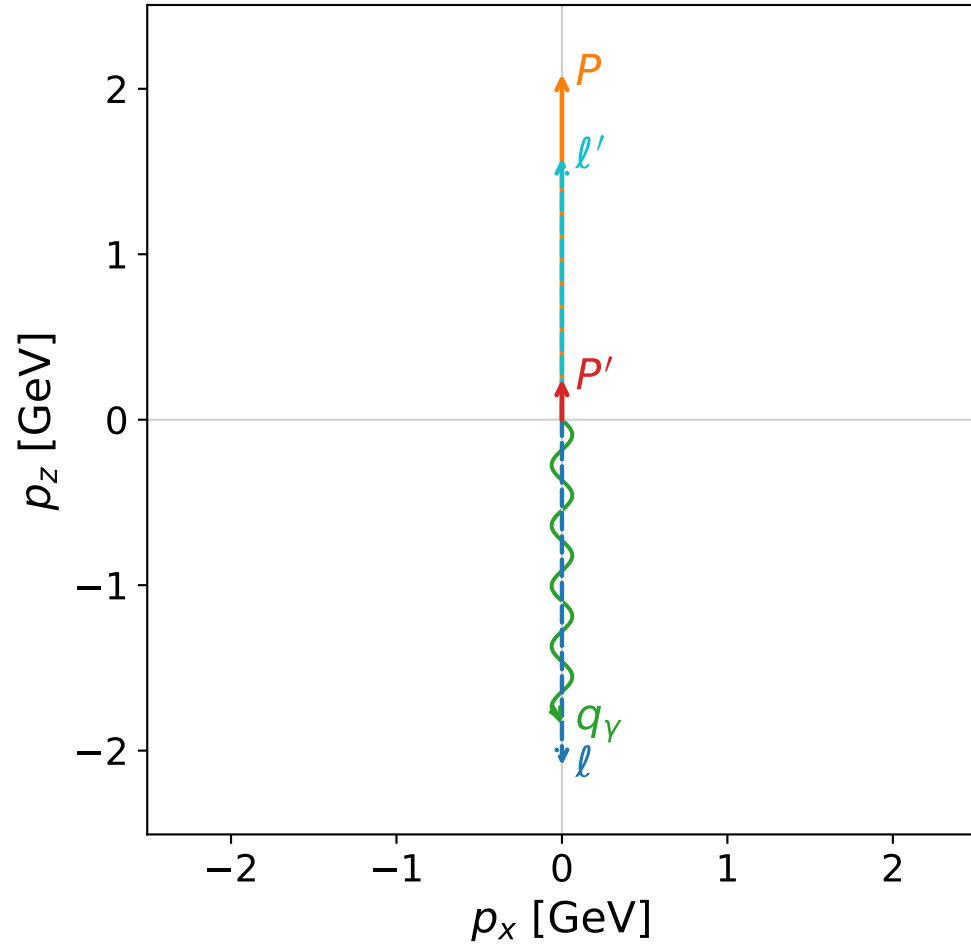
$R_L/T=9.702\text{e-}13$; $I_{TL}=+7.753\text{e-}08$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

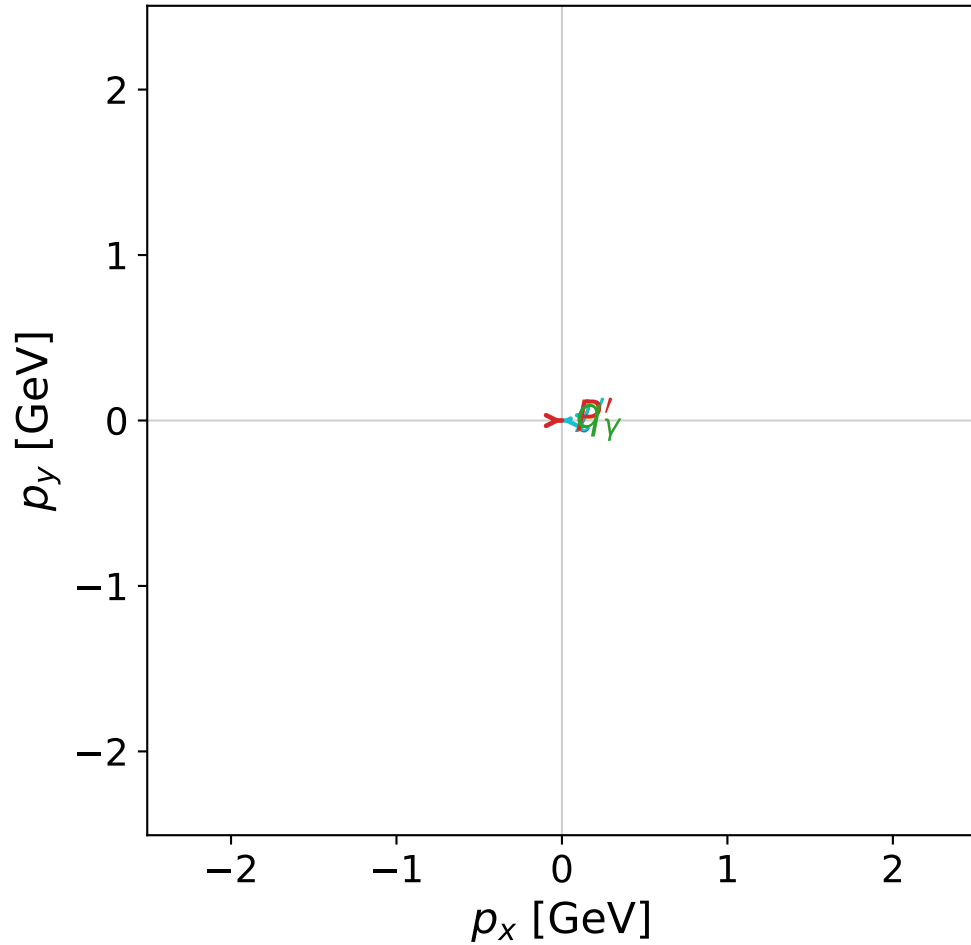


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

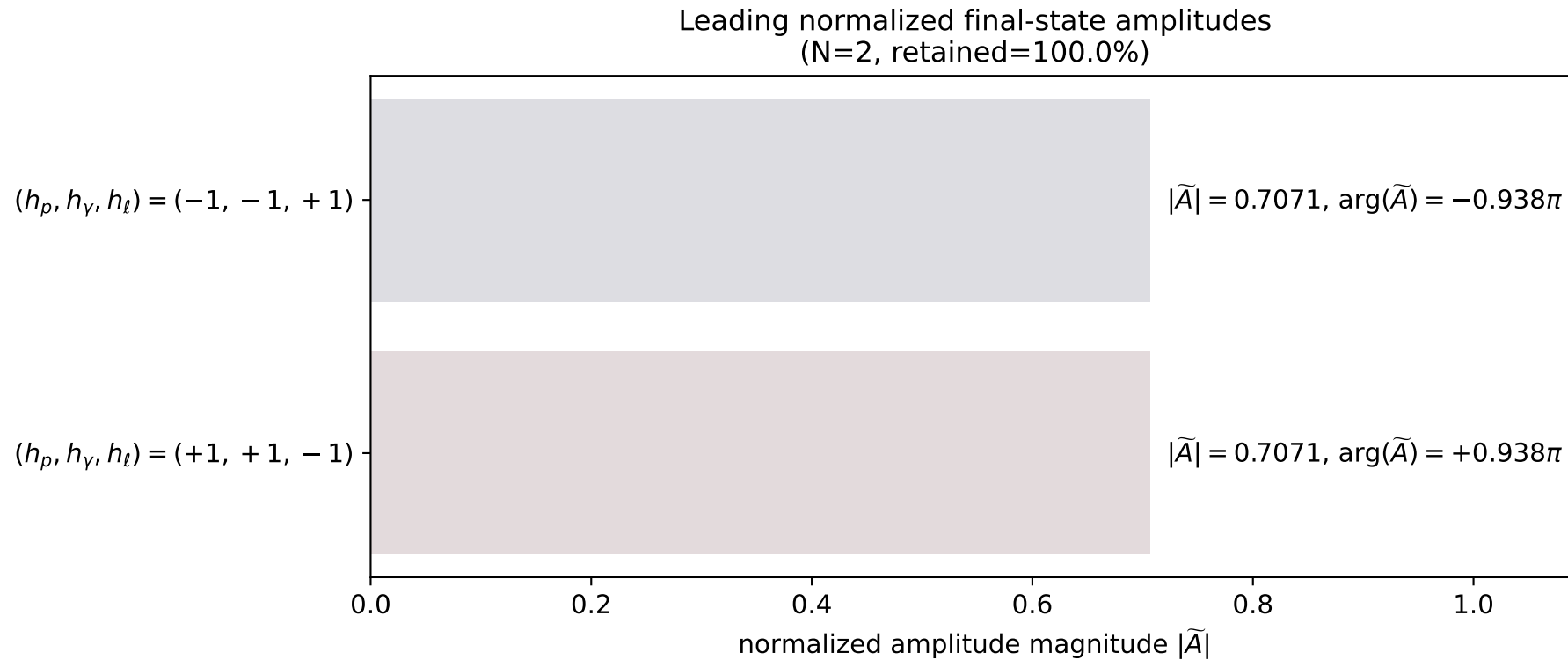


dghz_mixing_angles_polarization_04_local_minimum_108
 $d_{\text{GHZ}}=3.93085\text{e-}08$
 region: polarization_cluster_04_local_minimum_108
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0108
 lepton mixing angle: $\alpha_e=2.3726047$ rad
 incoming lepton: $-0.718615|+\rangle +0.695408|-\rangle$
 proton mixing angle: $\alpha_p=2.3726093$ rad
 incoming proton: $-0.718618|+\rangle +0.695405|-\rangle$

$s=19.1697$, $\sqrt{s}=4.37832$, $|\vec{P}|=2.08868$, $|\vec{P}'|=0.24491$
 $\theta_{P'}=0.000152337$, $\theta_Y=3.14159$, $\phi_{P'}=0.00207501$, $\phi_Y=3.33601$
 $E_Y=1.82689$, $\theta_{\ell'}=2.35835\text{e-}05$, $\phi_{\ell'}=3.14367$
 $Q^2=13.217$, $x_B=0.74867$, $t=-1.65659$, $W^2=5.31684$, $y=0.965237$

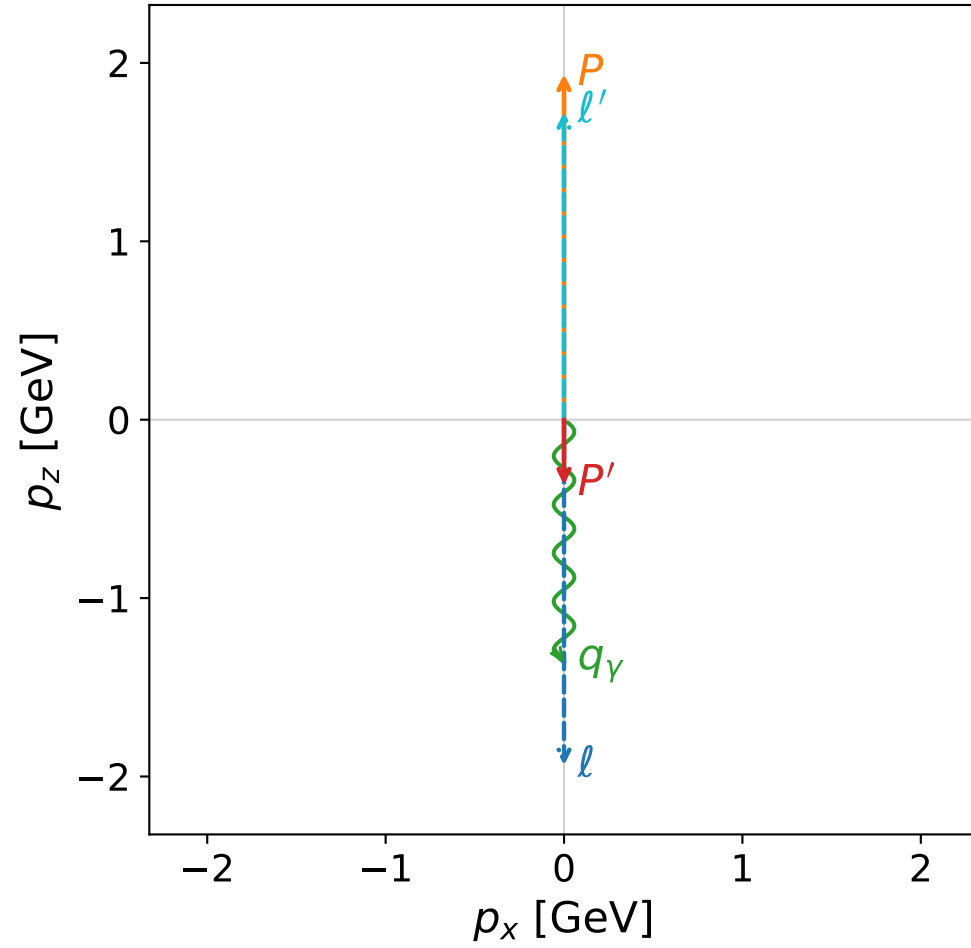
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0887$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0887$
 P $E= 2.2896$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0887$
 ℓ' $E= 1.5820$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5820$
 P' $E= 0.9694$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2449$
 q_Y $E= 1.8269$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8269$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.101\text{e+}17$; $I_h=-1.083\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.682\text{e-}12$; $I_{TL}=+1.110\text{e-}07$

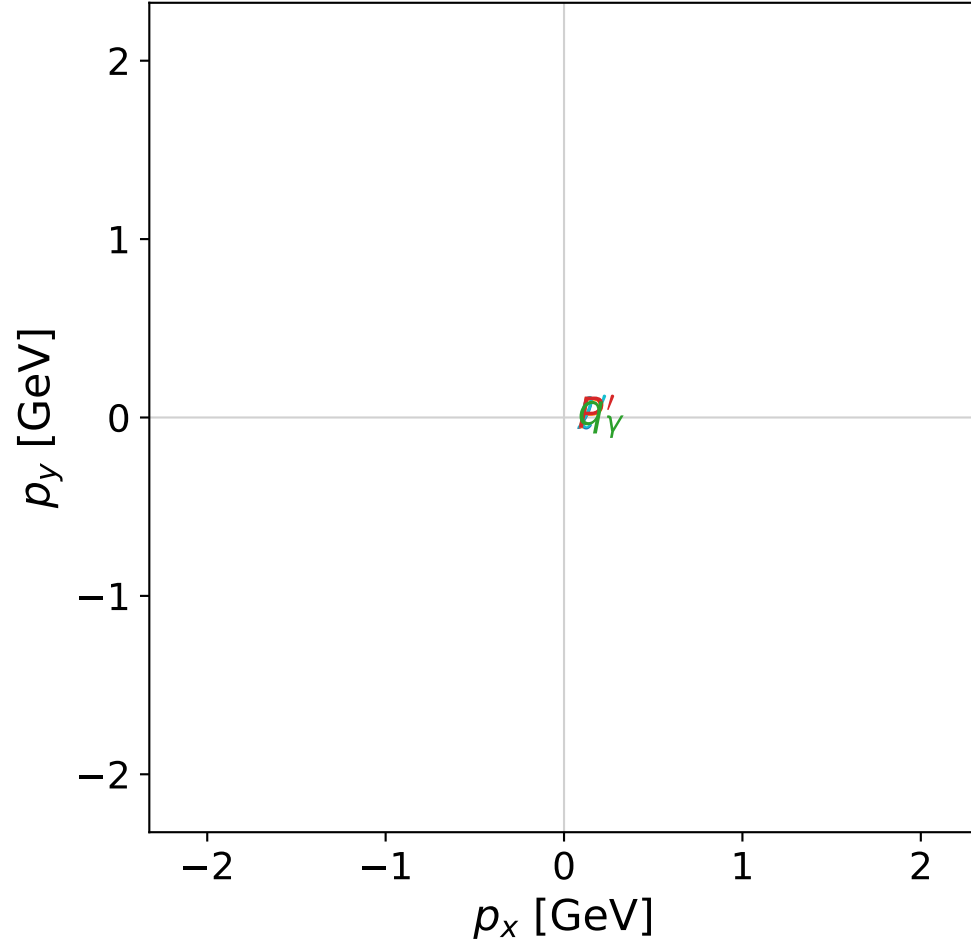


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

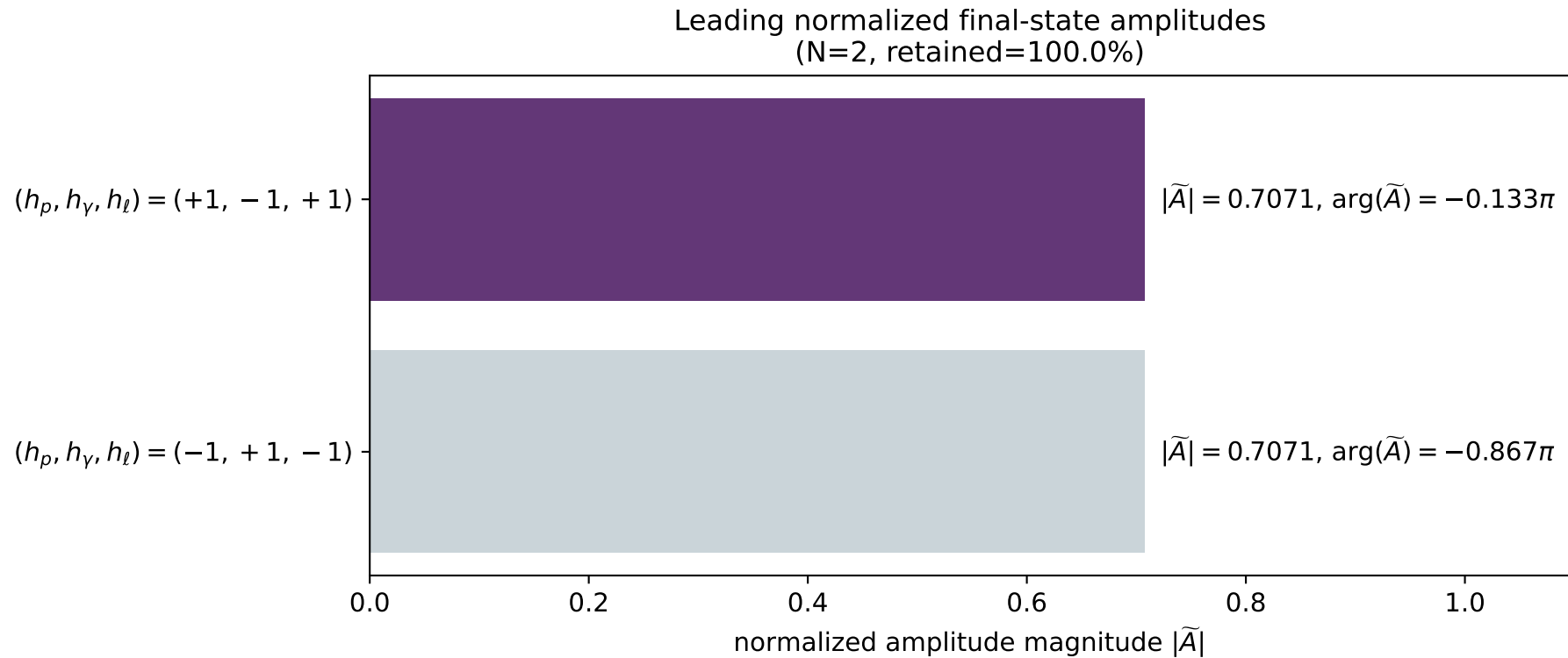


dghz_mixing_angles_polarization_04_local_minimum_109
 $d_{\text{GHZ}}=3.93746\text{e-}08$
 region: polarization_cluster_04_local_minimum_109
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0109
 lepton mixing angle: $\alpha_e=2.4147598$ rad
 incoming lepton: $-0.747283|+\rangle +0.664506|-\rangle$
 proton mixing angle: $\alpha_p=2.4147771$ rad
 incoming proton: $-0.747294|+\rangle +0.664493|-\rangle$

$s=16.7248$, $\sqrt{s}=4.0896$, $|\vec{P}|=1.93723$, $|\vec{P}'|=0.365737$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.18745$, $\phi_\gamma=5.86465$
 $E_\gamma=1.35854$, $\theta_{l'}=0$, $\phi_{l'}=2.56558$
 $Q^2=13.3613$, $x_B=0.884674$, $t=-3.99128$, $W^2=2.62161$, $y=0.953176$

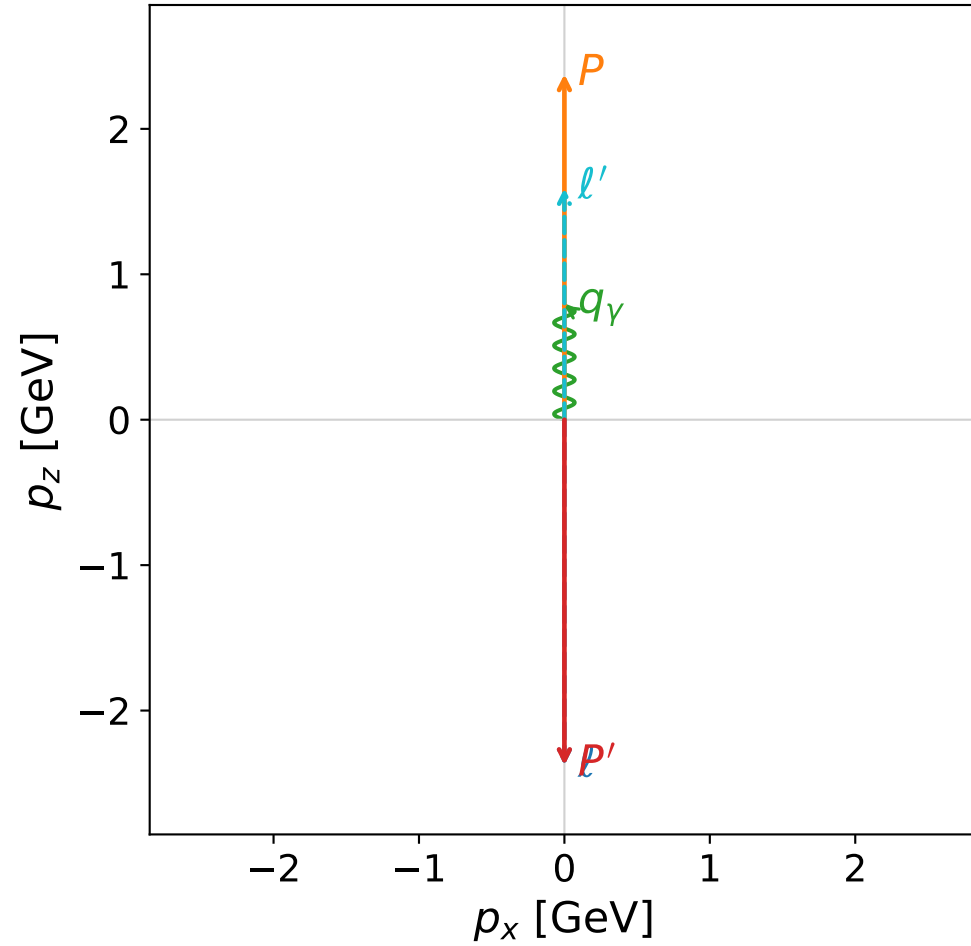
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9372$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9372$
 P $E= 2.1524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9372$
 l' $E= 1.7243$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7243$
 P' $E= 1.0068$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.3657$
 q_γ $E= 1.3585$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3585$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.044\text{e}+24$; $I_h=+4.254\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.247\text{e-}28$; $I_{TL}=-3.513\text{e-}18$

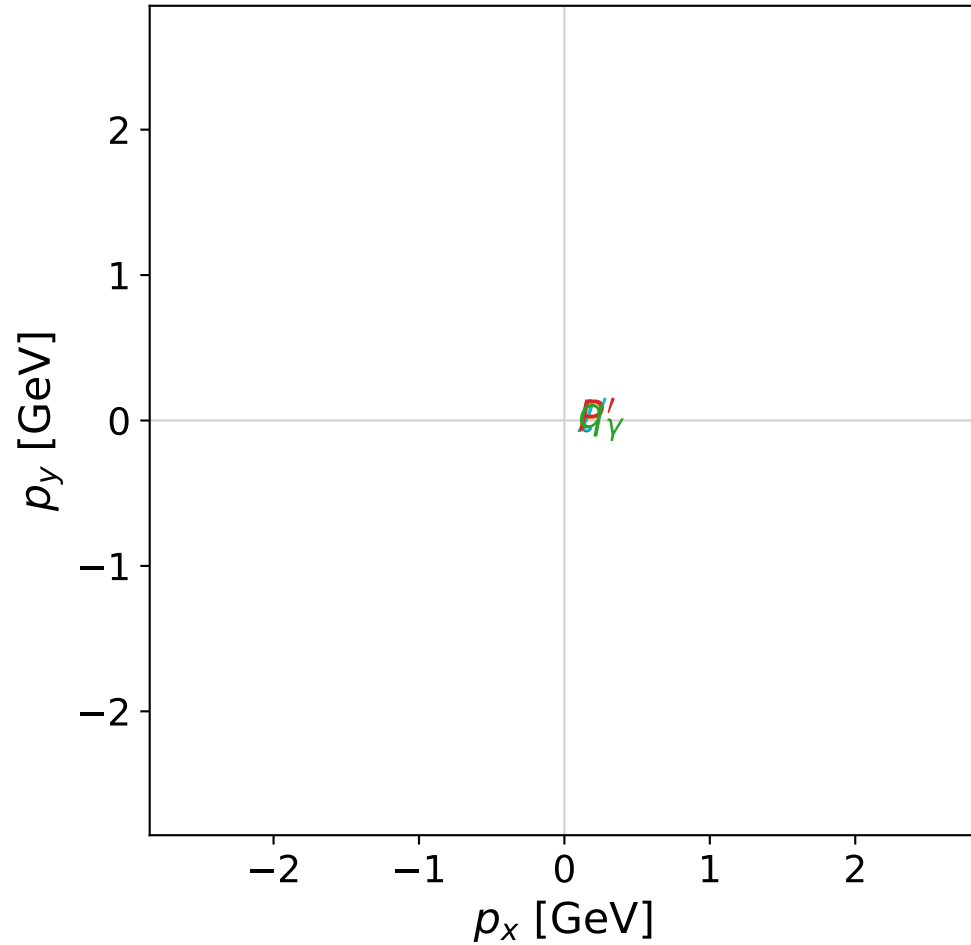


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

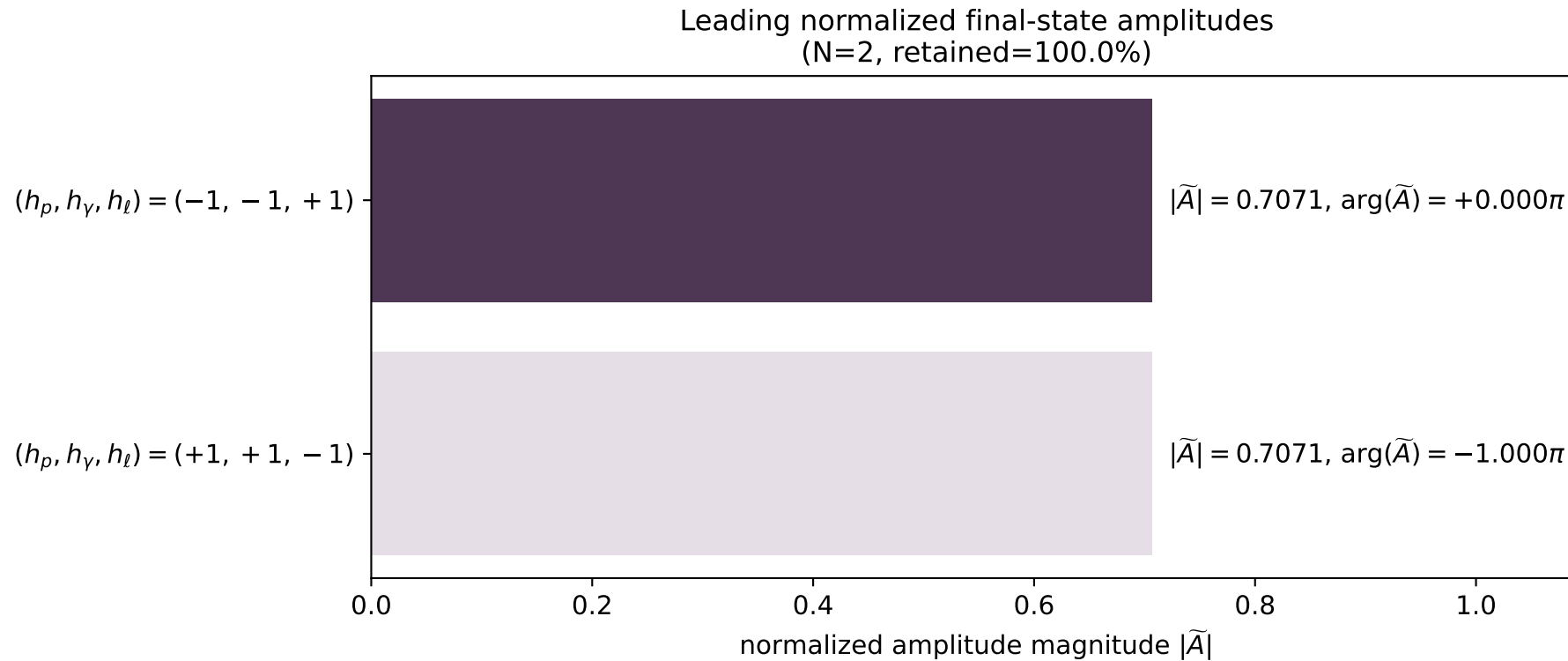


dghz_mixing_angles_polarization_04_local_minimum_117
 $d_{\text{GHZ}}=4.36167\text{e-}08$
 region: polarization_cluster_04_local_minimum_117
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0117
 lepton mixing angle: $\alpha_e=2.0273398$ rad
 incoming lepton: $-0.440848|+\rangle +0.897582|-\rangle$
 proton mixing angle: $\alpha_p=2.6850445$ rad
 incoming proton: $-0.89758|+\rangle +0.440852|-\rangle$

$s=24.3182$, $\sqrt{s}=4.93135$, $|\vec{P}|=2.37646$, $|\vec{P}'|=2.37646$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.60836$, $\phi_\gamma=5.61305$
 $E_\gamma=0.784824$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.466771$
 $Q^2=15.1299$, $x_B=0.66155$, $t=-22.5903$, $W^2=8.62032$, $y=0.975768$

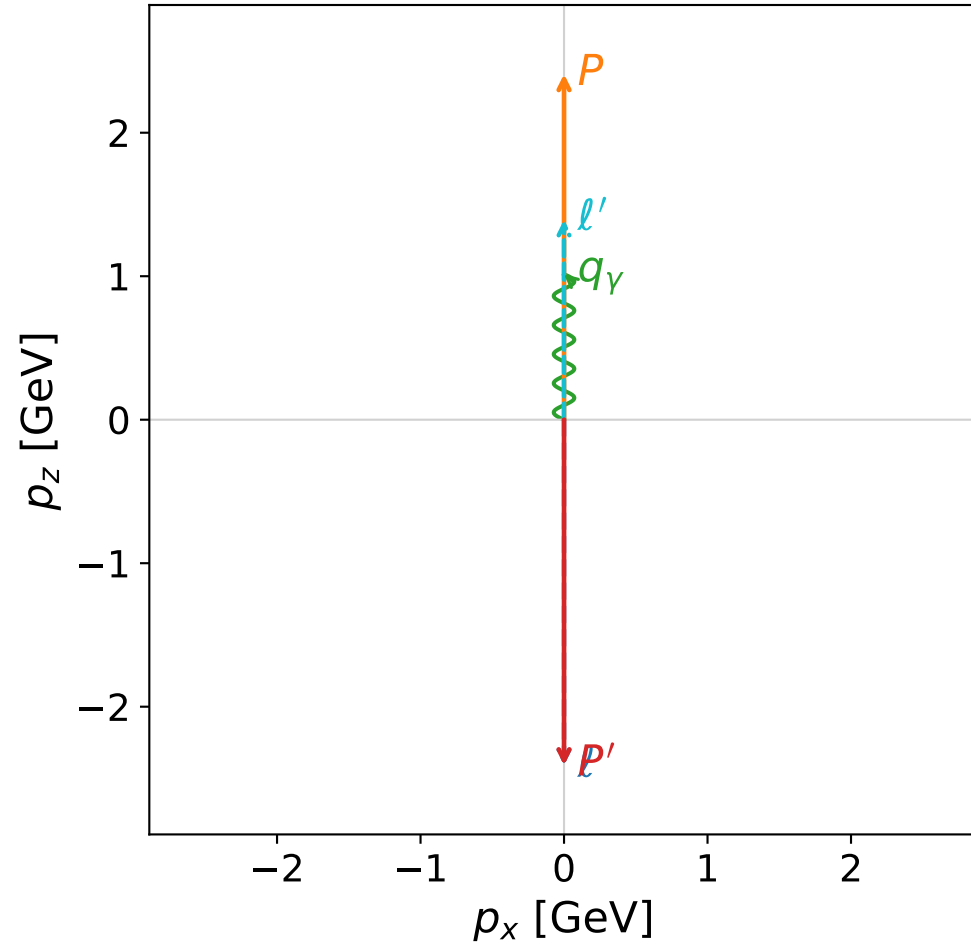
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3765$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3765$
 P $E= 2.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3765$
 ℓ' $E= 1.5916$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5916$
 P' $E= 2.5549$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3765$
 q_γ $E= 0.7848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7848$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.309\text{e+}23$; $I_h=+2.665\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.671\text{e-}28$; $I_{TL}=+2.666\text{e-}14$

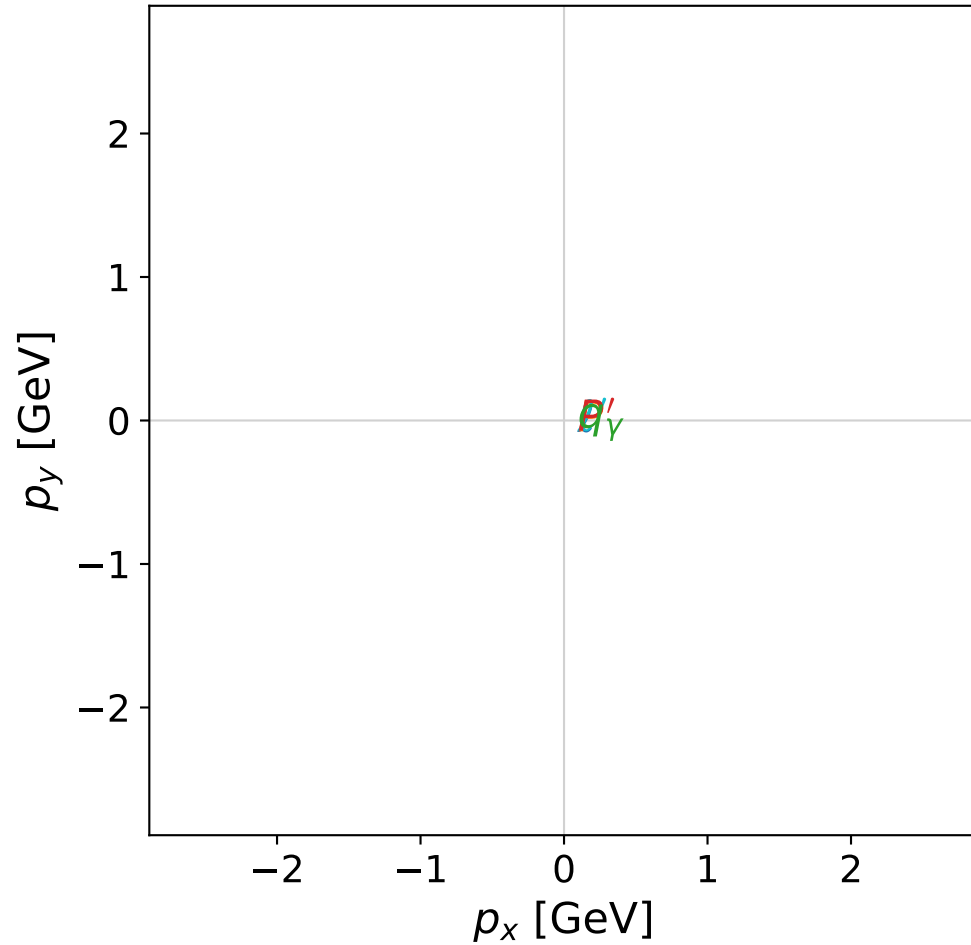


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

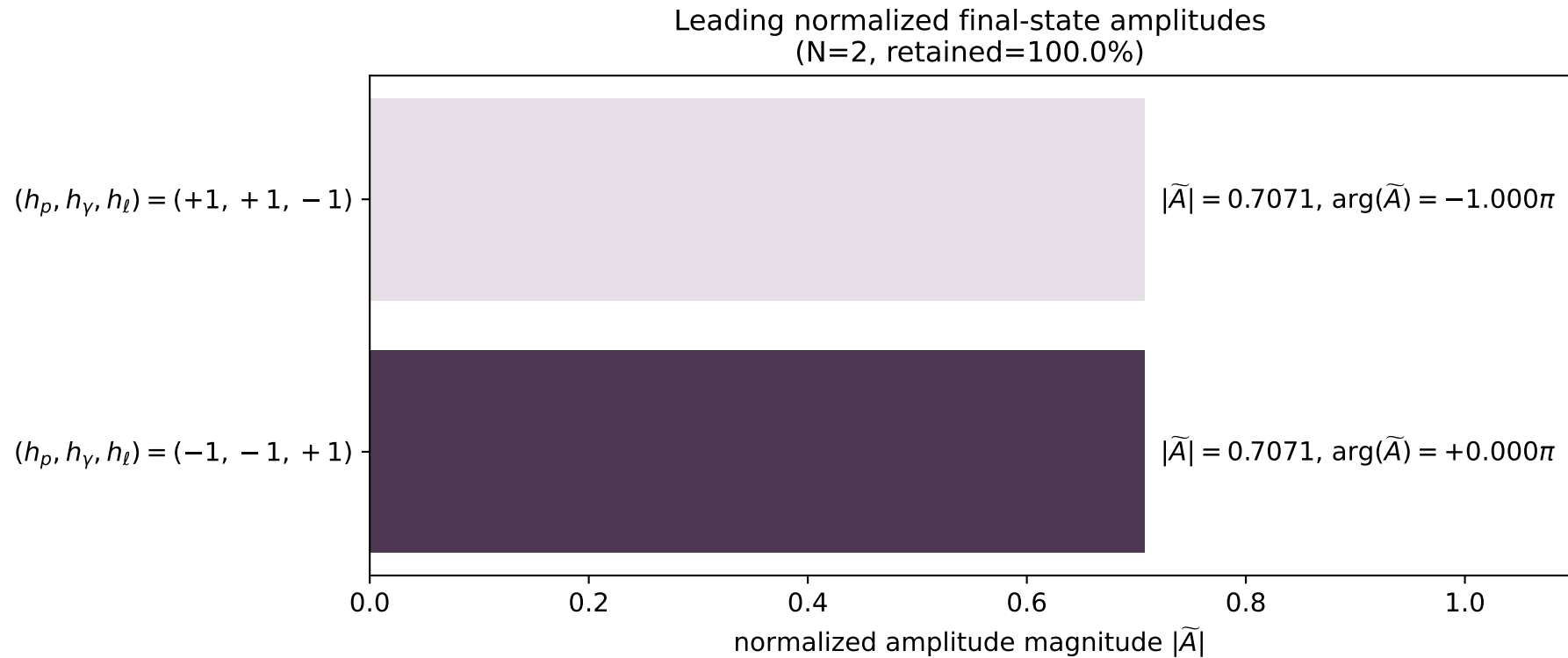


dghz_mixing_angles_polarization_04_local_minimum_119
 $d_{\text{GHZ}}=4.46361\text{e-}08$
 region: polarization_cluster_04_local_minimum_119
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0119
 lepton mixing angle: $\alpha_e=2.099717$ rad
 incoming lepton: $-0.504602|+\rangle +0.863352|-\rangle$
 proton mixing angle: $\alpha_p=2.61268$ rad
 incoming proton: $-0.863356|+\rangle +0.504595|-\rangle$

$s=24.93$, $\sqrt{s}=4.993$, $|\vec{P}|=2.40839$, $|\vec{P}'|=2.40839$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.04535$, $\phi_\gamma=0.354409$
 $E_\gamma=1.01446$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.903761$
 $Q^2=13.4285$, $x_B=0.569998$, $t=-23.2014$, $W^2=11.0102$, $y=0.979573$

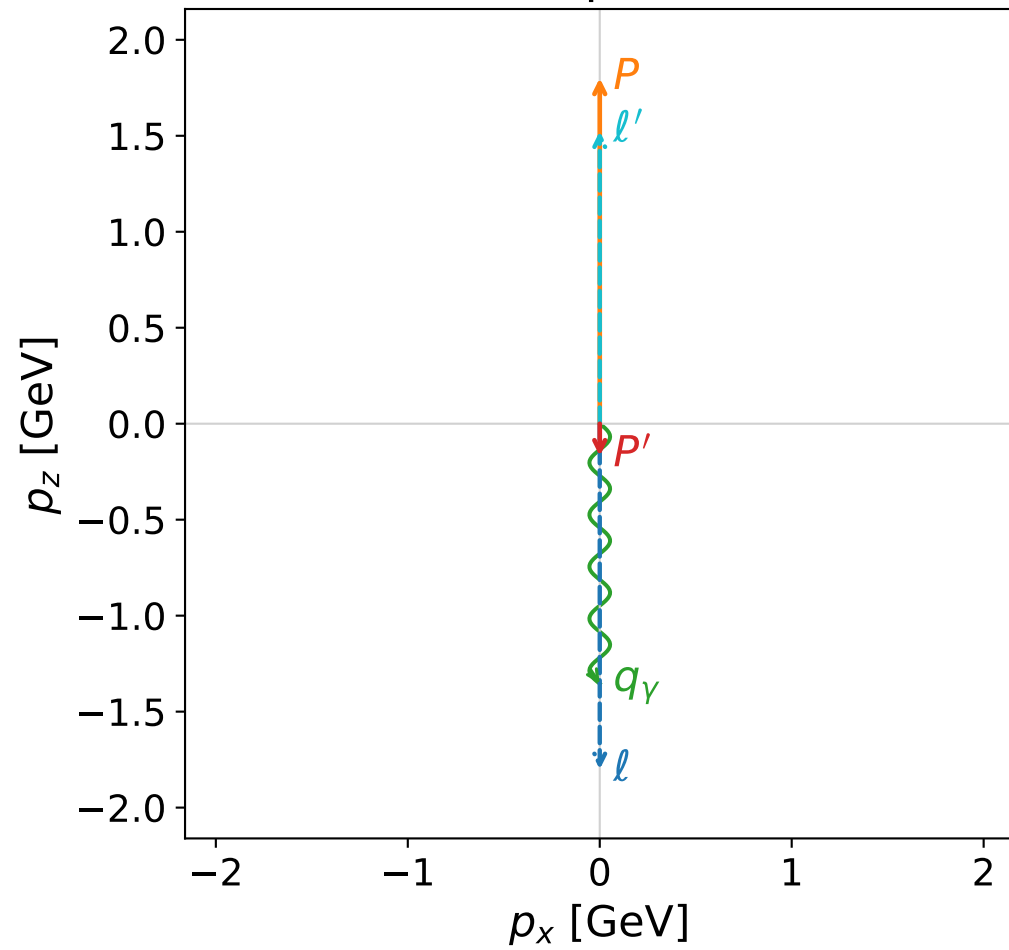
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4084$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4084$
 P $E= 2.5846$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4084$
 ℓ' $E= 1.3939$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3939$
 P' $E= 2.5846$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4084$
 q_γ $E= 1.0145$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0145$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.994\text{e}+23$; $I_h=-1.047\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.224\text{e-}29$; $I_{TL}=-1.047\text{e-}14$

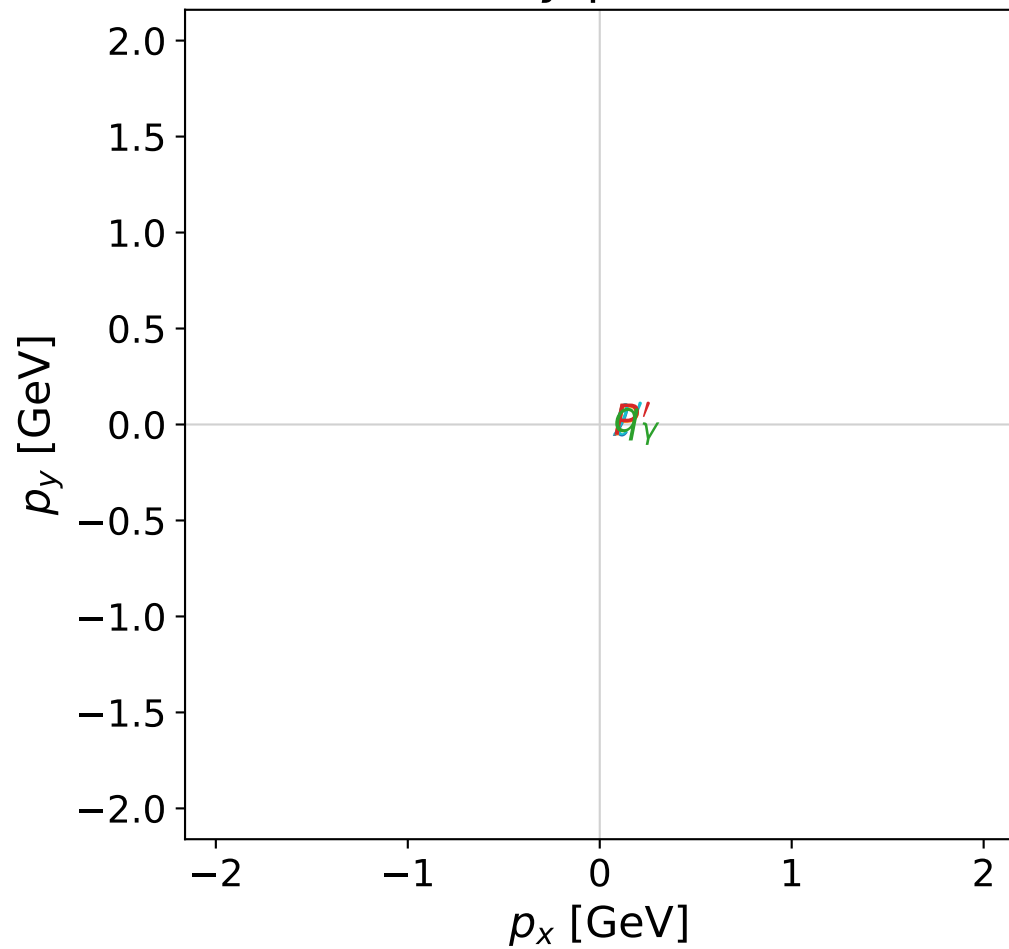


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

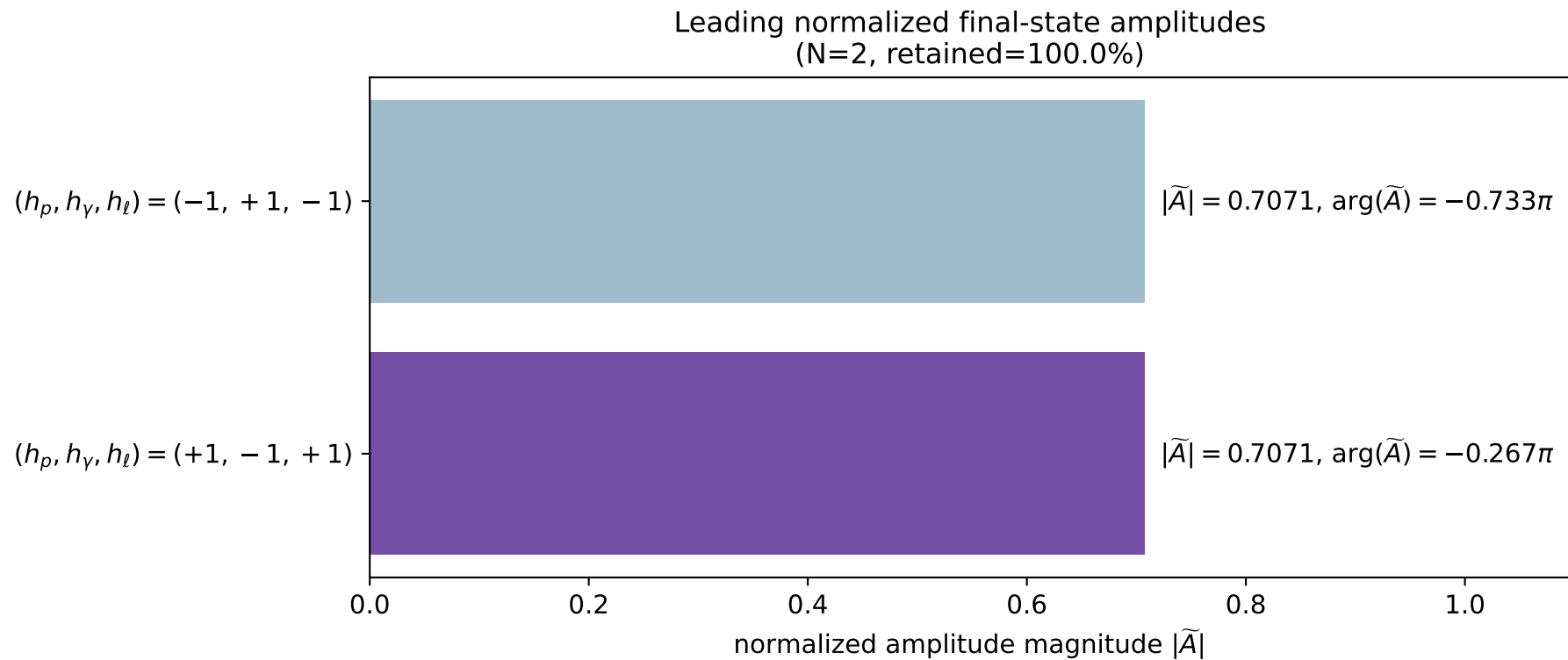


dghz_mixing_angles_polarization_04_local_minimum_122
 $d_{\text{GHZ}}=4.77096\text{e-}08$
 region: polarization_cluster_04_local_minimum_122
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0122
 lepton mixing angle: $\alpha_e=2.3265008$ rad
 incoming lepton: $-0.685802|+\rangle +0.727789|-\rangle$
 proton mixing angle: $\alpha_p=2.3264837$ rad
 incoming proton: $-0.685789|+\rangle +0.7278|-\rangle$

$s=14.673$, $\sqrt{s}=3.83054$, $|\vec{P}|=1.80042$, $|\vec{P}'|=0.16869$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.10573$, $\phi_\gamma=5.44421$
 $E_\gamma=1.3544$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.20483$
 $Q^2=10.9688$, $x_B=0.837731$, $t=-2.71733$, $W^2=3.00452$, $y=0.949273$

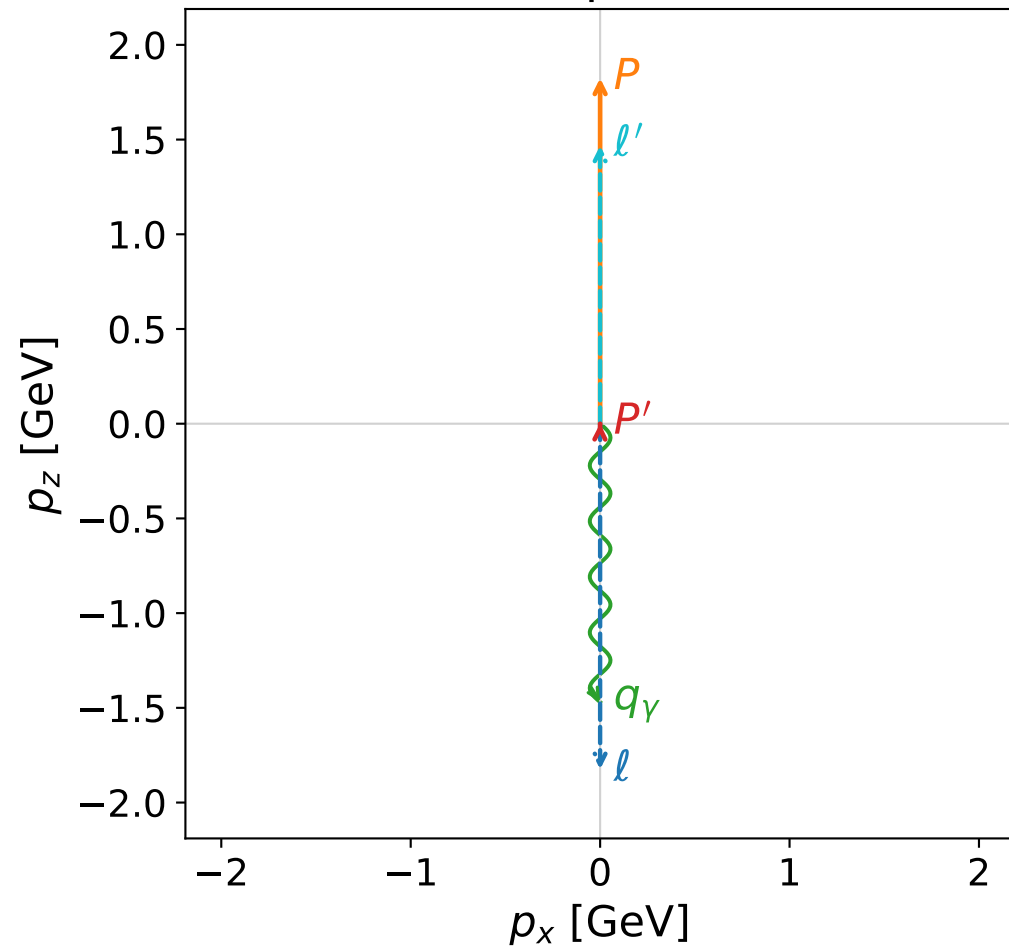
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8004$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8004$
 P $E= 2.0301$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8004$
 ℓ' $E= 1.5231$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5231$
 P' $E= 0.9530$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1687$
 q_γ $E= 1.3544$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3544$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.845\text{e}+24$; $I_h=-1.130\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.673\text{e-}29$; $I_{TL}=+1.745\text{e-}18$

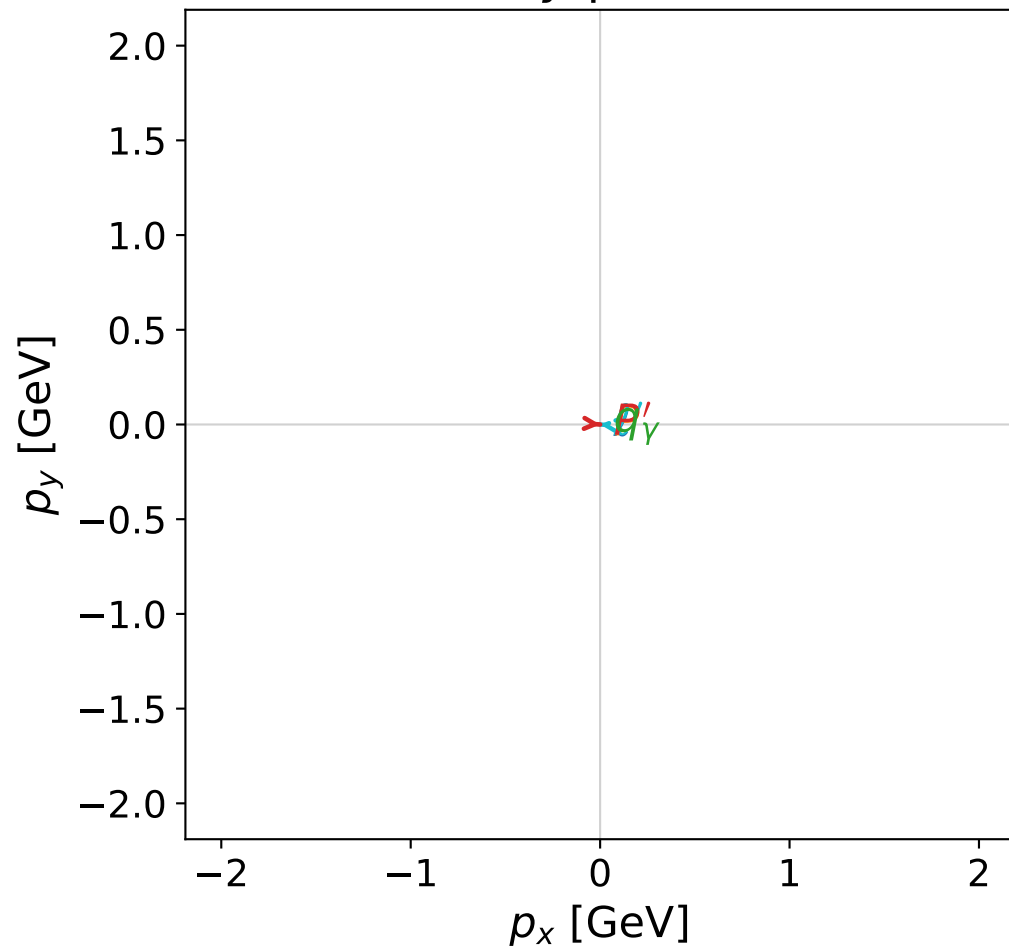


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

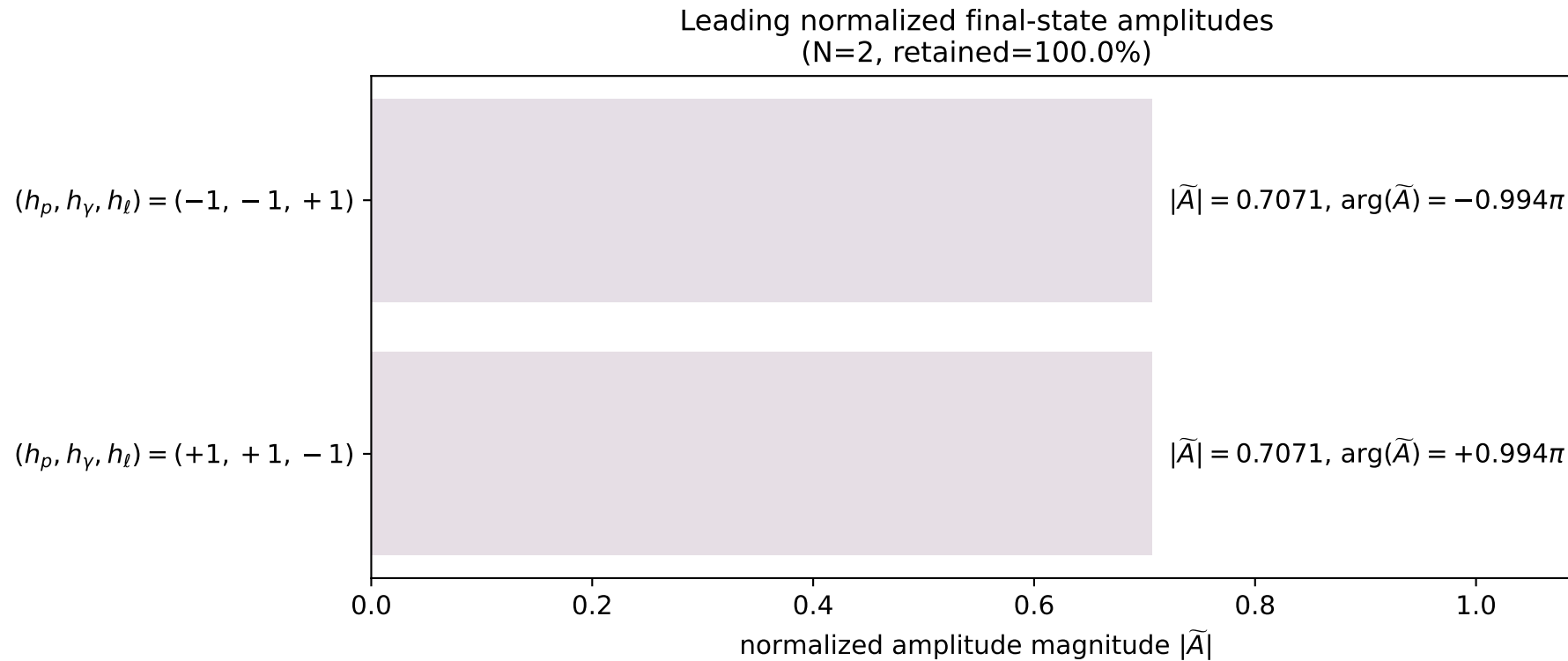


dghz_mixing_angles_polarization_04_local_minimum_123
 $d_{\text{GHZ}}=4.77825\text{e-}08$
 region: polarization_cluster_04_local_minimum_123
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0123
 lepton mixing angle: $\alpha_e=2.3820098$ rad
 incoming lepton: $-0.725123|+\rangle +0.688619|-\rangle$
 proton mixing angle: $\alpha_p=2.3820131$ rad
 incoming proton: $-0.725126|+\rangle +0.688617|-\rangle$

$s=15.0226$, $\sqrt{s}=3.8759$, $|\vec{P}|=1.82445$, $|\vec{P}'|=0.00112548$
 $\theta_{P'}=0.0124896$, $\theta_Y=3.14159$, $\phi_{P'}=6.21424$, $\phi_Y=3.15921$
 $E_Y=1.46951$, $\theta_{\ell'}=9.57262\text{e-}06$, $\phi_{\ell'}=3.07264$
 $Q^2=10.716$, $x_B=0.795185$, $t=-2.08473$, $W^2=3.63996$, $y=0.952862$

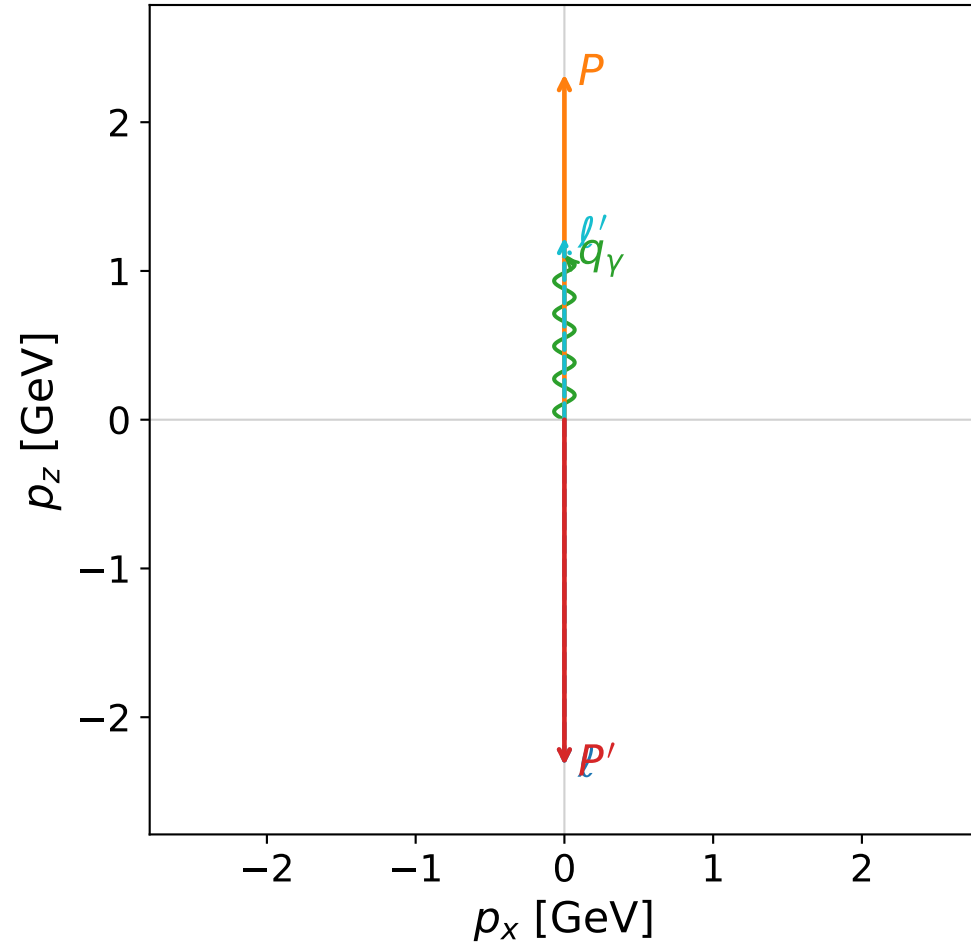
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8244$
 P $E= 2.0515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8244$
 ℓ' $E= 1.4684$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4684$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0011$
 q_Y $E= 1.4695$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4695$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.791\text{e+}17$; $I_h=-8.583\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.399\text{e-}13$; $I_{TL}=+6.678\text{e-}08$

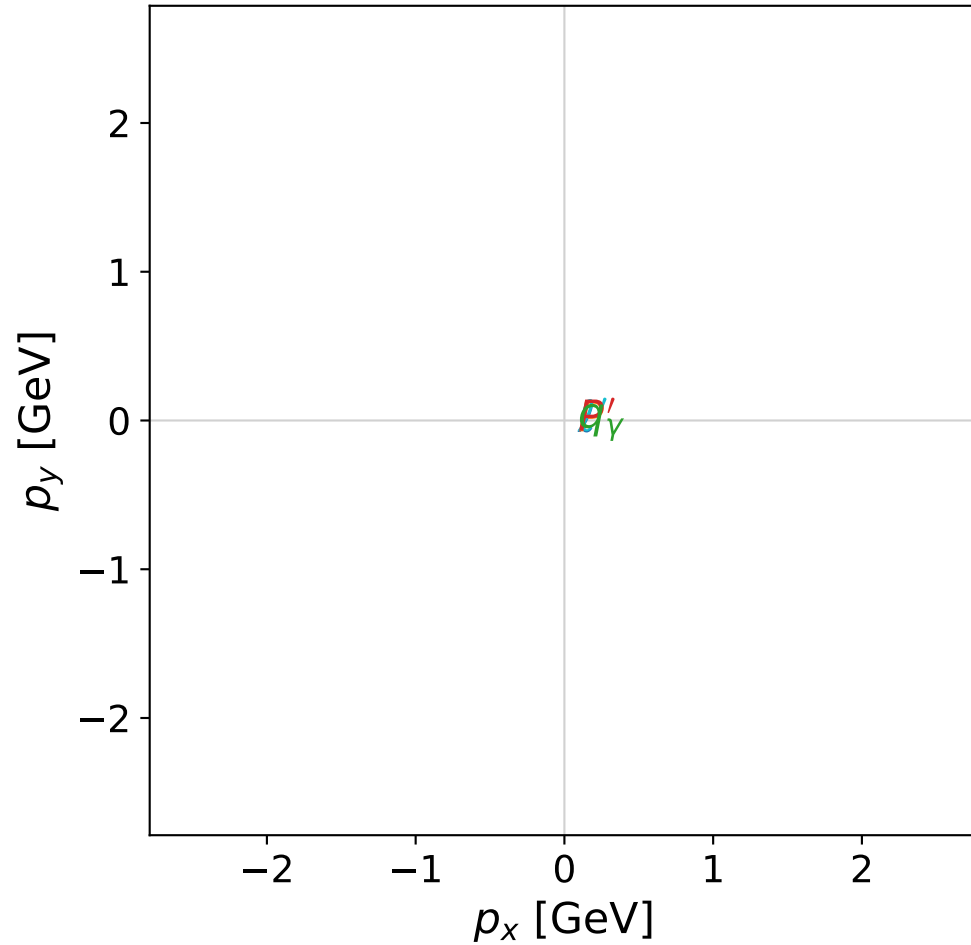


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

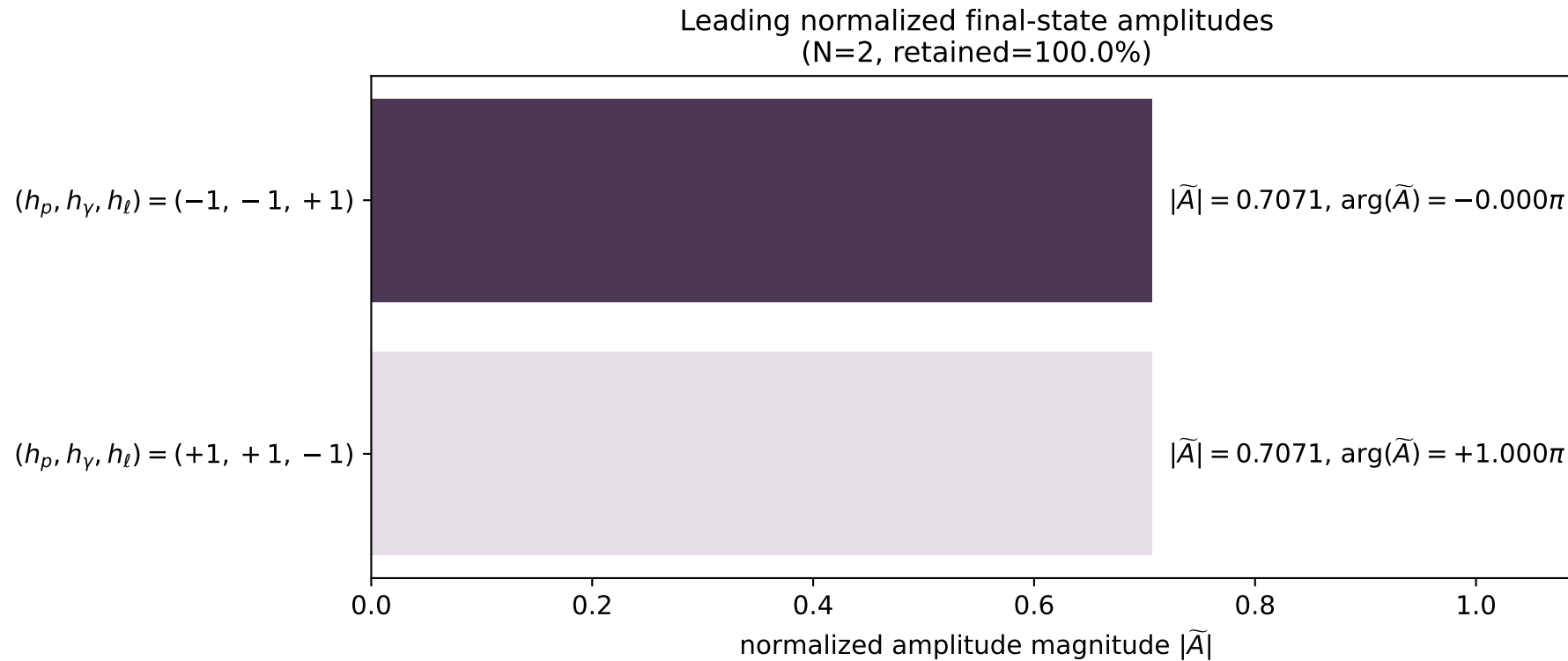


dghz_mixing_angles_polarization_04_local_minimum_125
 $d_{\text{GHZ}}=4.83377\text{e-}08$
 region: polarization_cluster_04_local_minimum_125
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0125
 lepton mixing angle: $\alpha_e=2.1954113$ rad
 incoming lepton: $-0.584785|+\rangle +0.811188|-\rangle$
 proton mixing angle: $\alpha_p=2.5169716$ rad
 incoming proton: $-0.811185|+\rangle +0.58479|-\rangle$

$s=23.3175$, $\sqrt{s}=4.82882$, $|\vec{P}|=2.32331$, $|\vec{P}'|=2.32331$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.30408$, $\phi_\gamma=5.61598$
 $E_\gamma=1.09792$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.44568$
 $Q^2=11.3878$, $x_B=0.517838$, $t=-21.591$, $W^2=11.4831$, $y=0.980098$

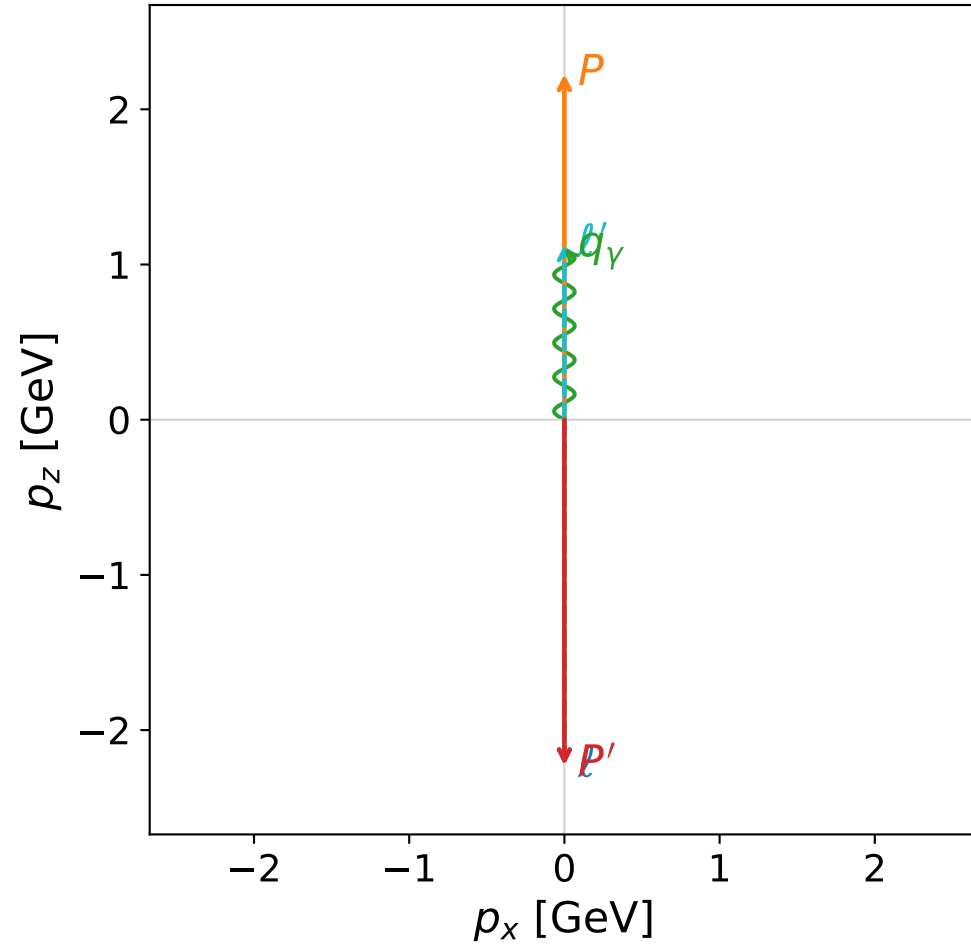
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3233$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 P $E= 2.5055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3233$
 ℓ' $E= 1.2254$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.2254$
 P' $E= 2.5055$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 q_γ $E= 1.0979$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0979$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.643\text{e}+23$; $I_h=-1.878\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.960\text{e-}28$; $I_{TL}=-1.878\text{e-}14$

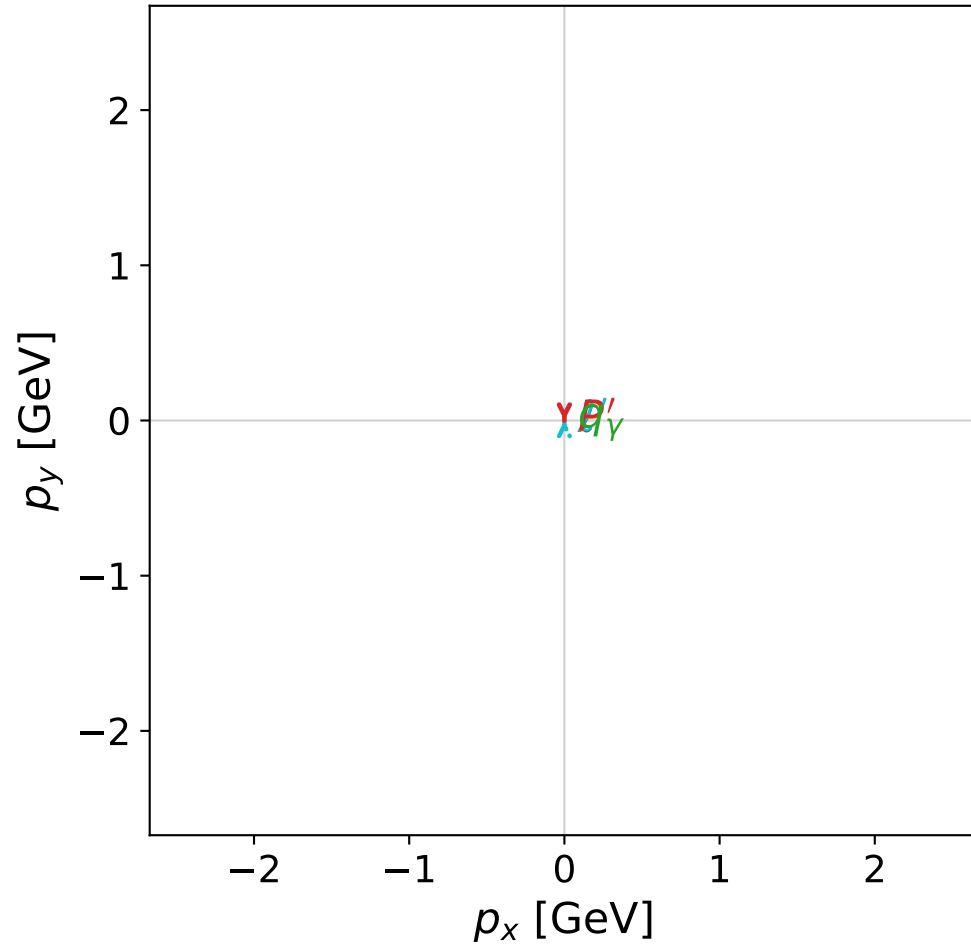


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

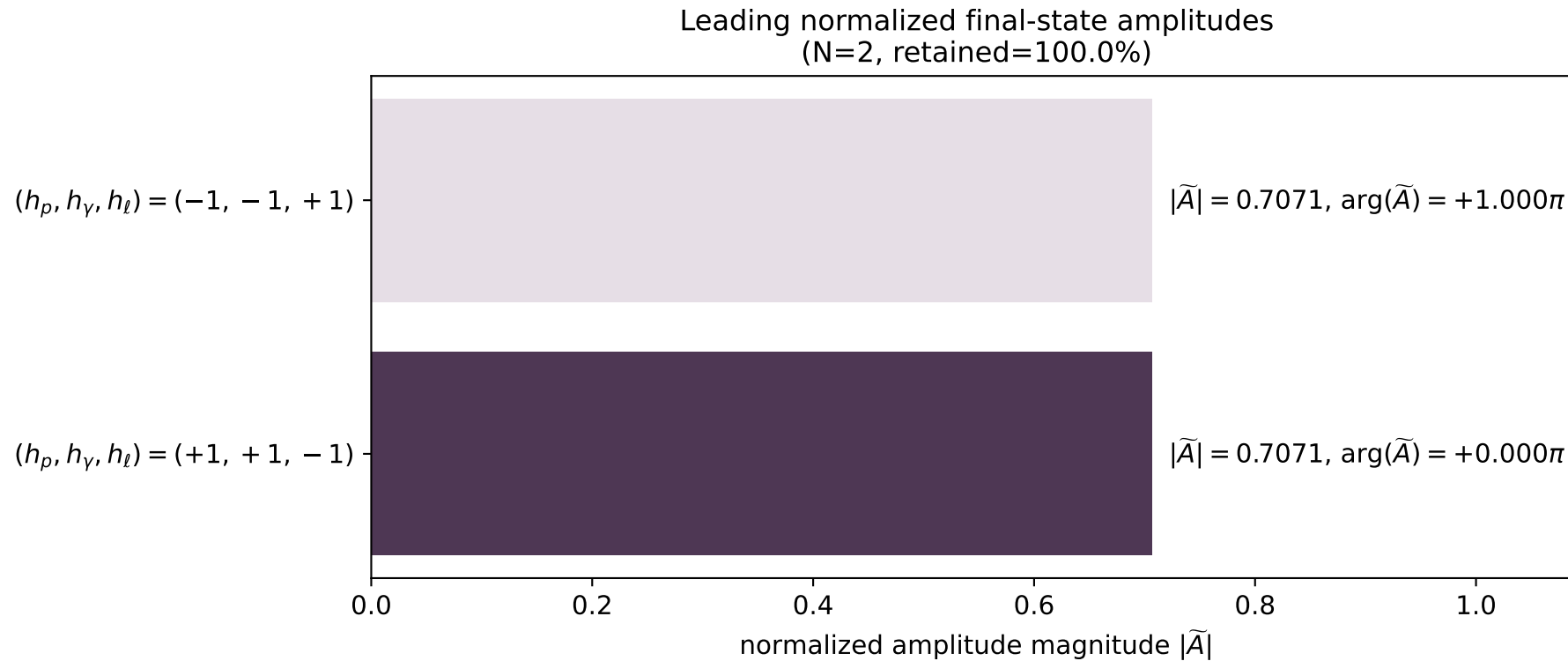


dghz_mixing_angles_polarization_04_local_minimum_133
 $d_{\text{GHZ}}=5.36187\text{e-}08$
 region: polarization_cluster_04_local_minimum_133
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0133
 lepton mixing angle: $\alpha_e=2.2324916$ rad
 incoming lepton: $-0.614455|+\rangle +0.788952|-\rangle$
 proton mixing angle: $\alpha_p=2.4798937$ rad
 incoming proton: $-0.788949|+\rangle +0.614458|-\rangle$

$s=21.5586$, $\sqrt{s}=4.64312$, $|\vec{P}|=2.22681$, $|\vec{P}'|=2.22681$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=4.93115$, $\phi_Y=3.73044$
 $E_Y=1.09901$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.78956$
 $Q^2=10.0456$, $x_B=0.496047$, $t=-19.8348$, $W^2=11.0856$, $y=0.97933$

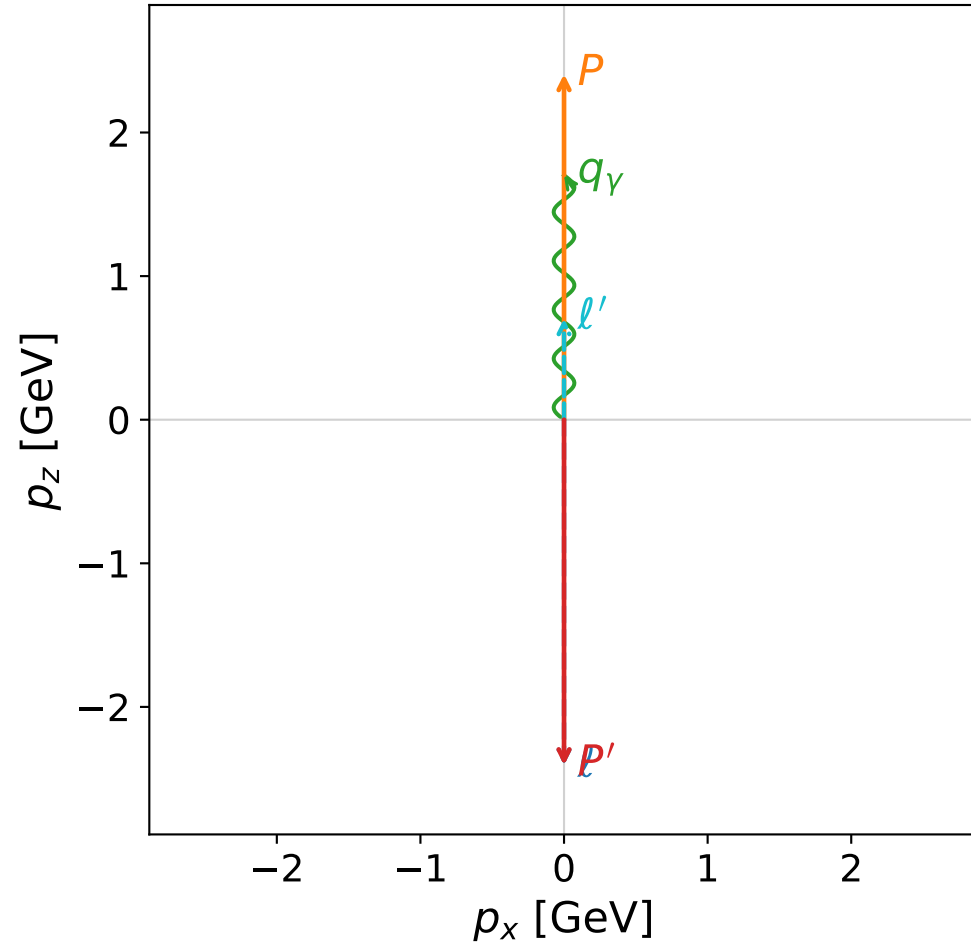
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2268$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2268$
 P $E= 2.4163$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2268$
 ℓ' $E= 1.1278$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.1278$
 P' $E= 2.4163$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.2268$
 q_Y $E= 1.0990$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0990$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.808\text{e}+23$; $I_h=+4.000\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.511\text{e-}28$; $I_{TL}=+4.000\text{e-}14$

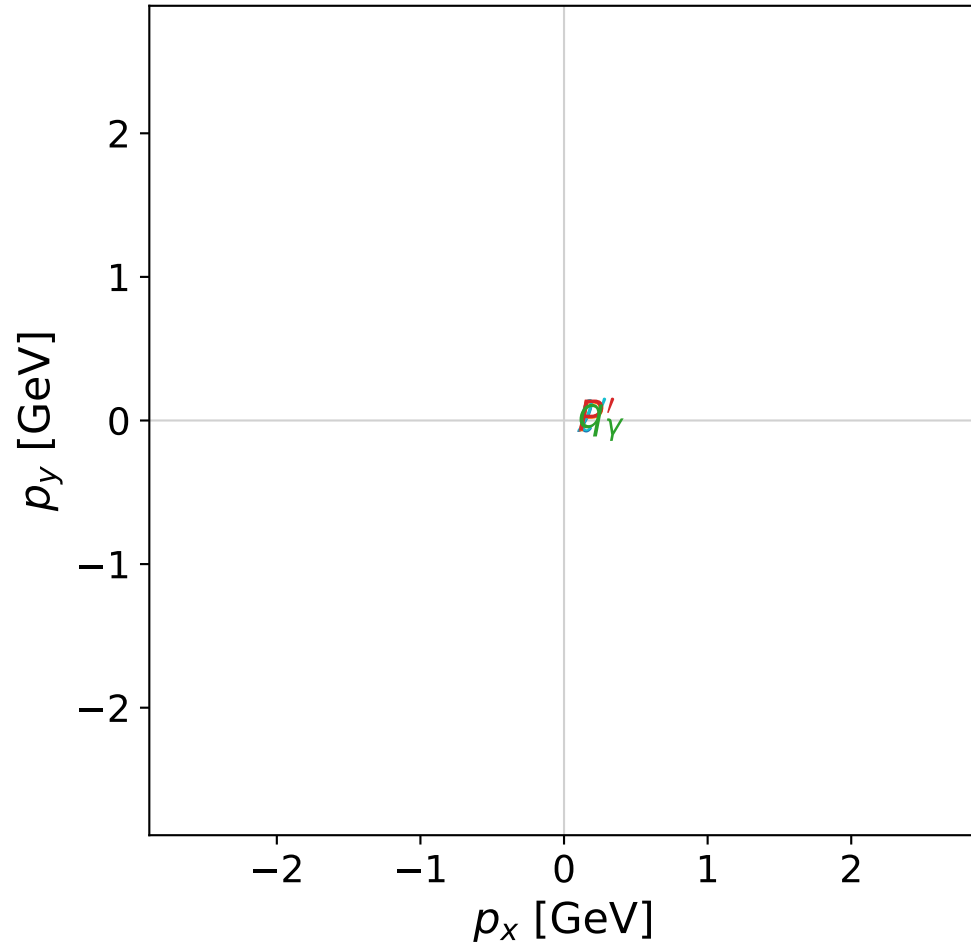


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

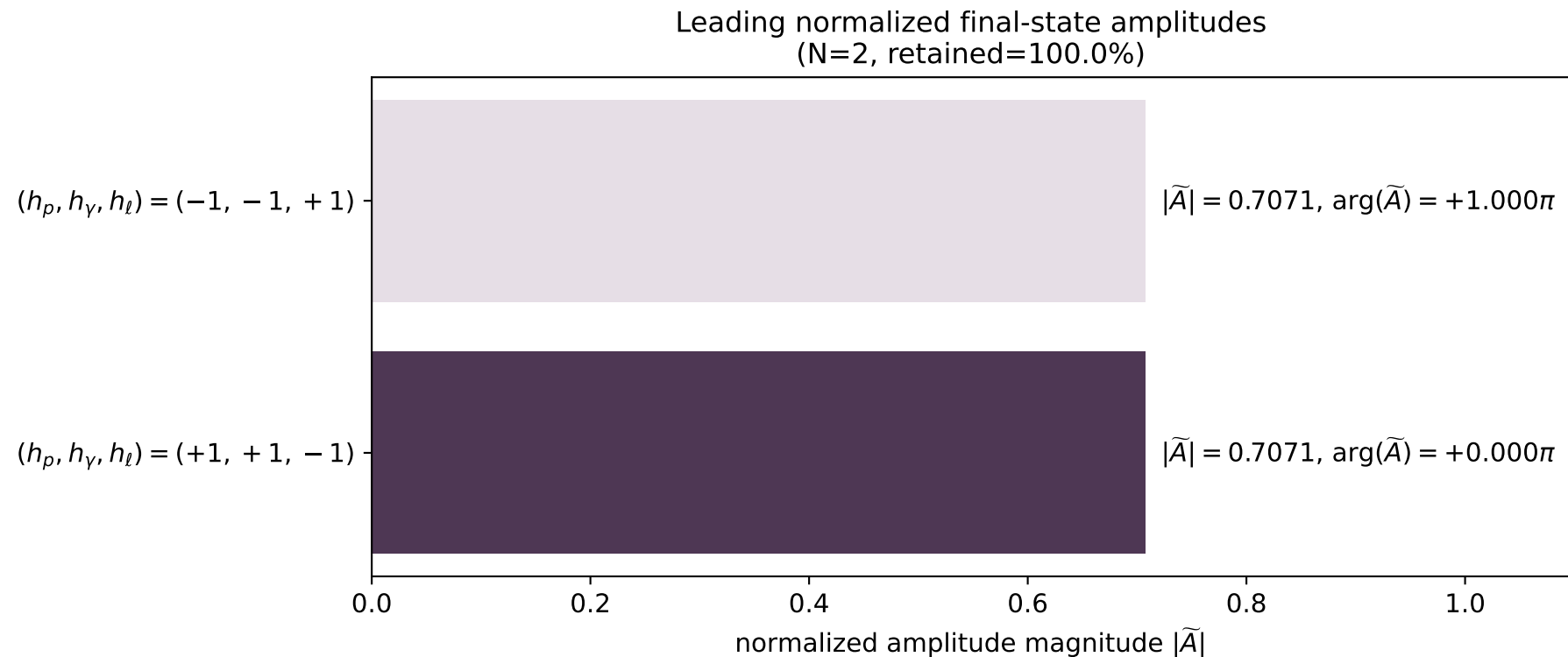


dghz_mixing_angles_polarization_04_local_minimum_145
 $d_{\text{GHZ}}=7.88665\text{e-}08$
 region: polarization_cluster_04_local_minimum_145
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0145
 lepton mixing angle: $\alpha_e=2.4645339$ rad
 incoming lepton: $-0.779419|+\rangle +0.626503|-\rangle$
 proton mixing angle: $\alpha_p=2.2478416$ rad
 incoming proton: $-0.626493|+\rangle +0.779427|-\rangle$

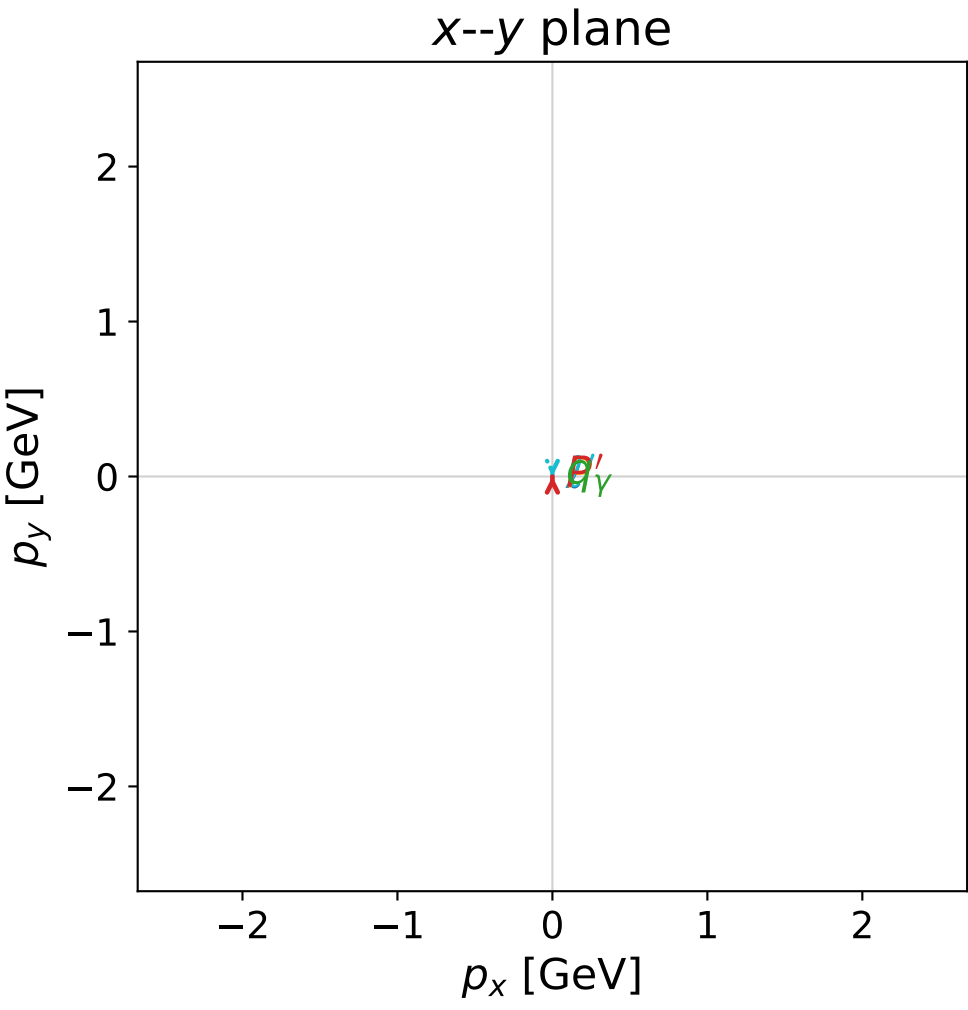
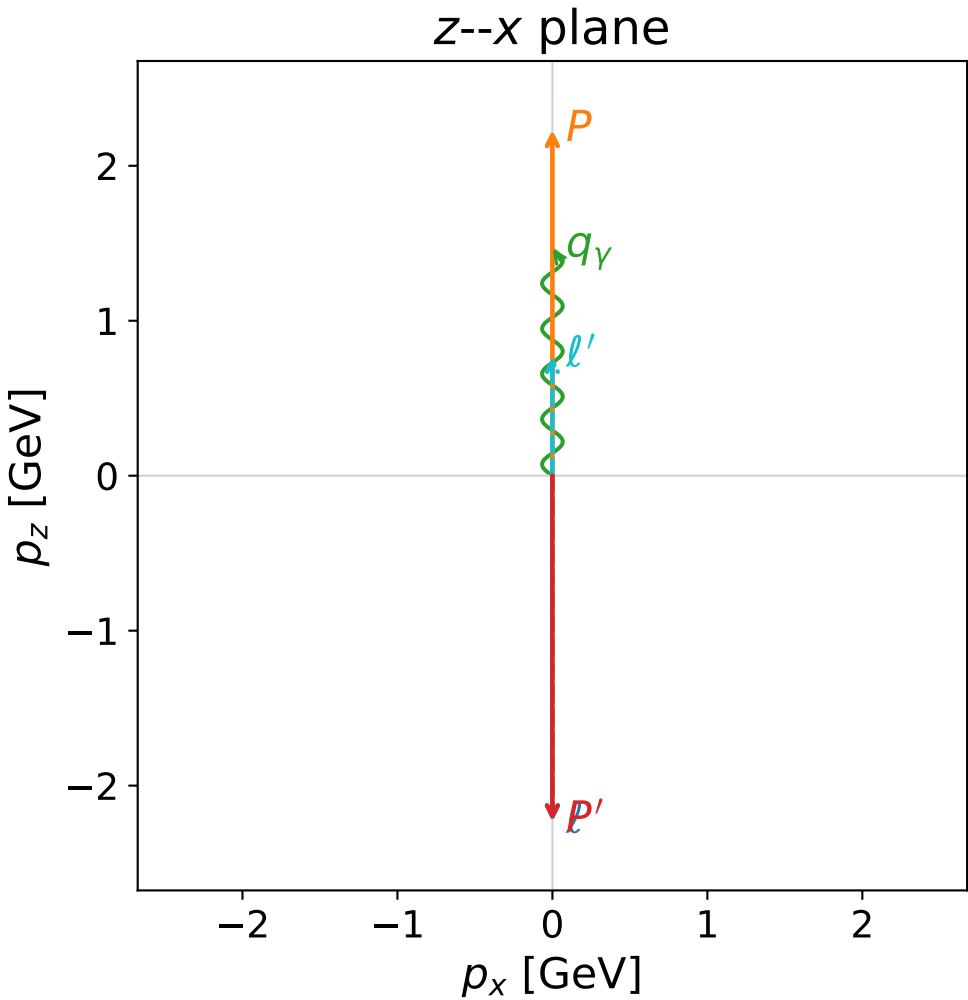
$s=24.8957$, $\sqrt{s}=4.98956$, $|\vec{P}|=2.40661$, $|\vec{P}'|=2.40661$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.62423$, $\phi_\gamma=3.19394$
 $E_\gamma=1.70223$, $\theta_{l'}=0$, $\phi_{l'}=5.76582$
 $Q^2=6.78069$, $x_B=0.285293$, $t=-23.1671$, $W^2=17.8666$, $y=0.989656$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4066$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 P $E= 2.5829$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4066$
 l' $E= 0.7044$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7044$
 P' $E= 2.5829$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 q_γ $E= 1.7022$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7022$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.387\text{e}+24$; $I_h=+3.120\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.103\text{e-}28$; $I_{TL}=+3.120\text{e-}14$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

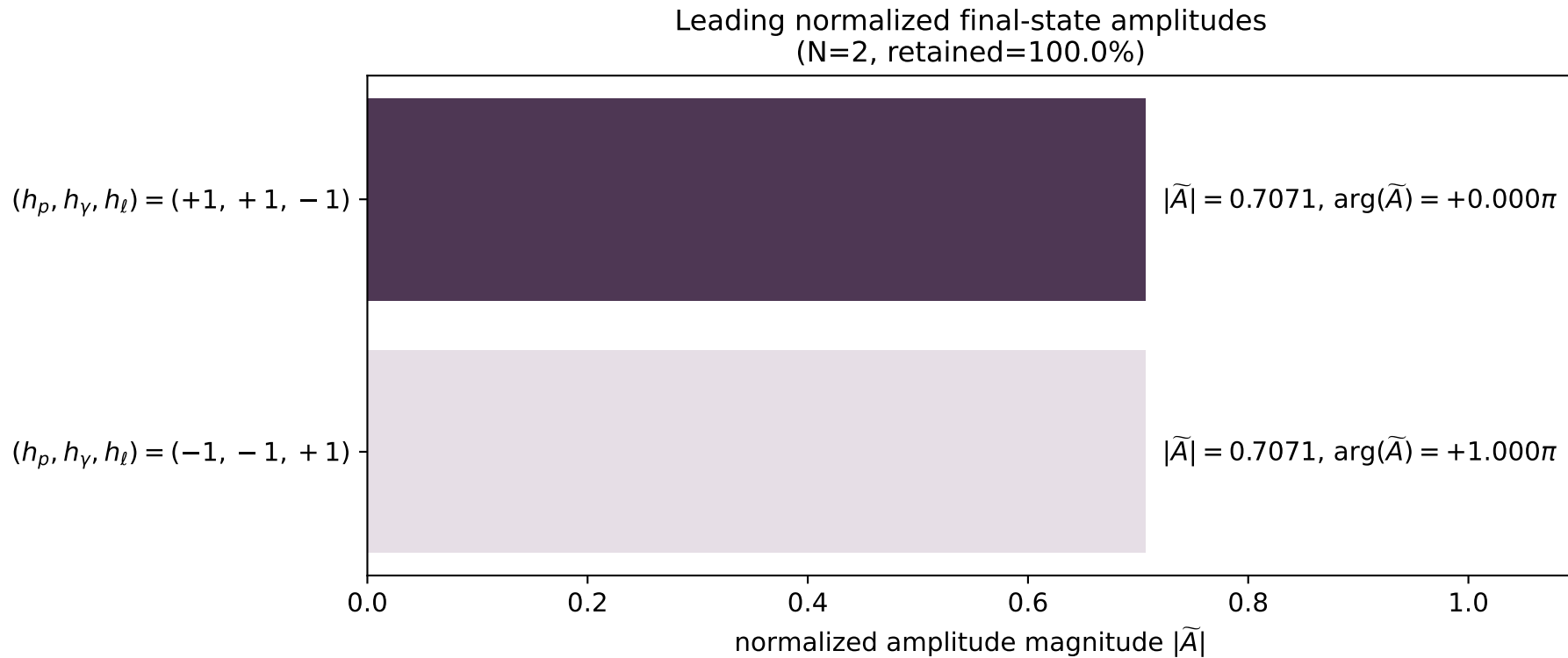


dghz_mixing_angles_polarization_04_local_minimum_153
 $d_{\rm GHZ}=9.35668\text{e-}08$
region: polarization_cluster_04_local_minimum_153
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0153
lepton mixing angle: $\alpha_e=2.6674747$ rad
incoming lepton: $-0.889696|+\rangle +0.456554|-\rangle$
proton mixing angle: $\alpha_p=2.0449223$ rad
incoming proton: $-0.456561|+\rangle +0.889692|-\rangle$

$s=21.6182$, $\sqrt{s}=4.64954$, $|\vec{P}|=2.23015$, $|\vec{P}'|=2.23015$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.53792$, $\phi_\gamma=3.63958$
 $E_\gamma=1.45997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.67951$
 $Q^2=6.8705$, $x_B=0.336017$, $t=-19.8944$, $W^2=14.4563$, $y=0.985945$

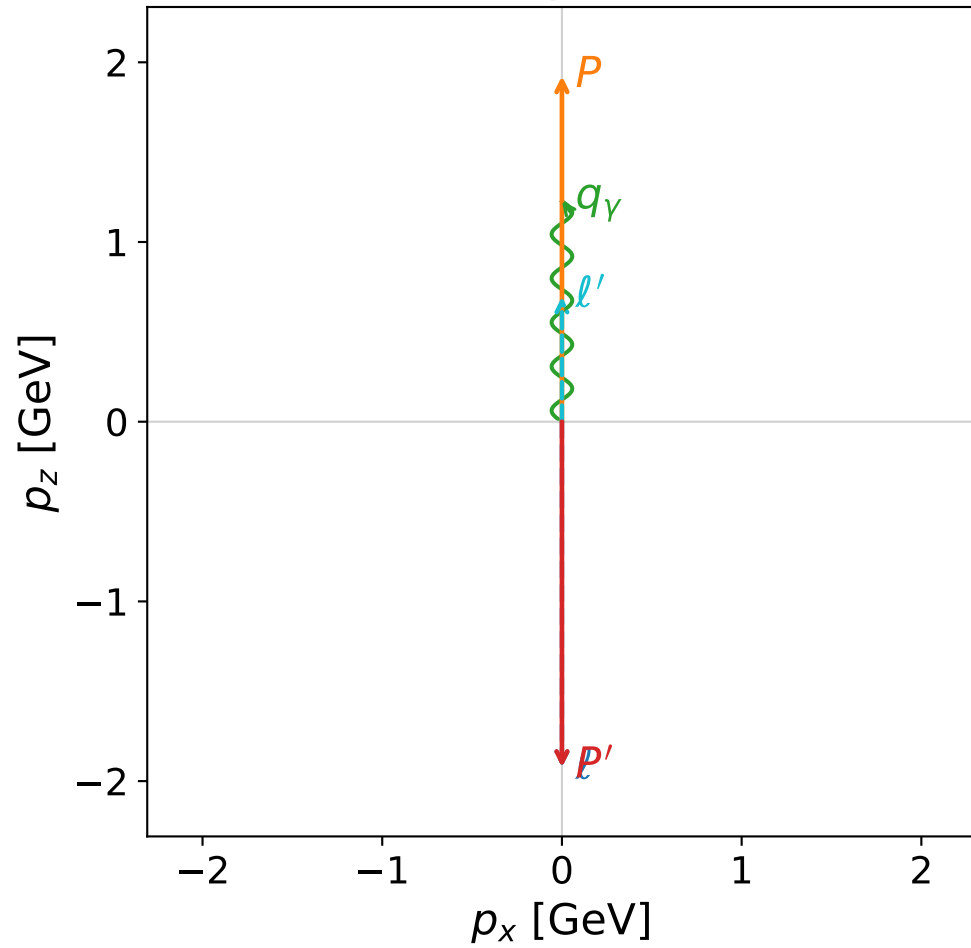
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2302$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 P $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2302$
 ℓ' $E= 0.7702$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7702$
 P' $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 q_γ $E= 1.4600$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4600$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve= $1.340\text{e}+24$; I_h= $-1.922\text{e-}14$
virtual-photon powers: T 100.000%, L 0.000%
R_L/T= $2.797\text{e-}28$; I_TL= $-1.922\text{e-}14$

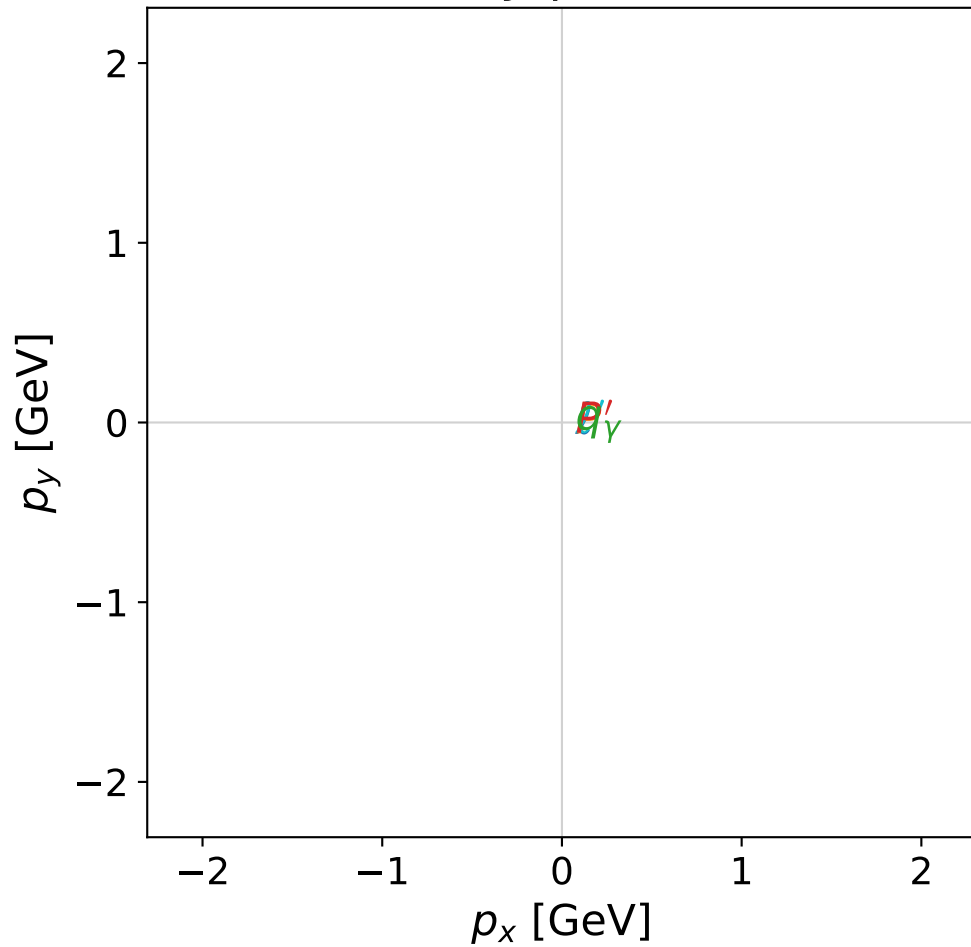


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

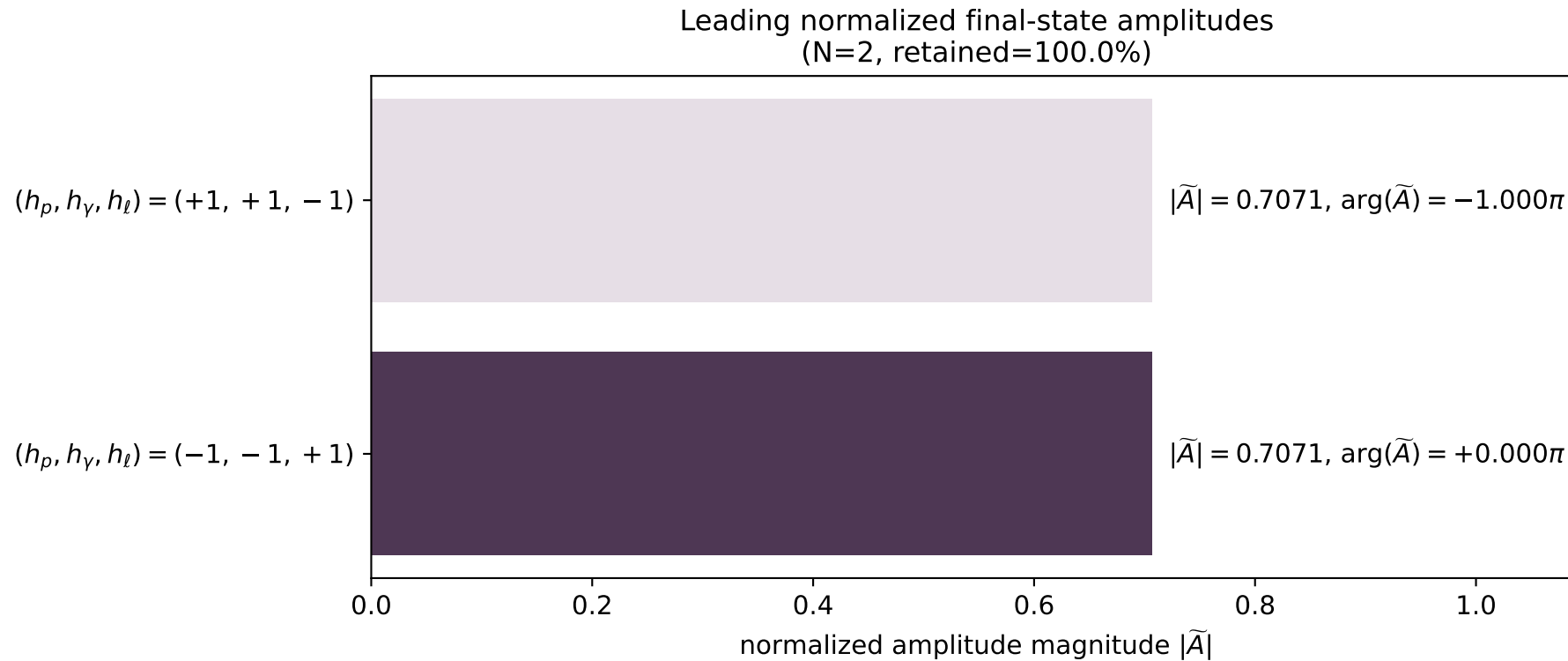


dghz_mixing_angles_polarization_04_local_minimum_155
 $d_{\text{GHZ}}=9.76885\text{e-}08$
 region: polarization_cluster_04_local_minimum_155
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0155
 lepton mixing angle: $\alpha_e=2.379922$ rad
 incoming lepton: $-0.723684|+\rangle +0.690131|-\rangle$
 proton mixing angle: $\alpha_p=2.3324913$ rad
 incoming proton: $-0.690149|+\rangle +0.723667|-\rangle$

$s=16.5062$, $\sqrt{s}=4.06278$, $|\vec{P}|=1.92311$, $|\vec{P}'|=1.92311$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.19286$, $\phi_\gamma=0.182475$
 $E_\gamma=1.22715$, $\theta_{l'}=0$, $\phi_{l'}=1.05127$
 $Q^2=5.35358$, $x_B=0.34934$, $t=-14.7934$, $W^2=10.8511$, $y=0.98071$

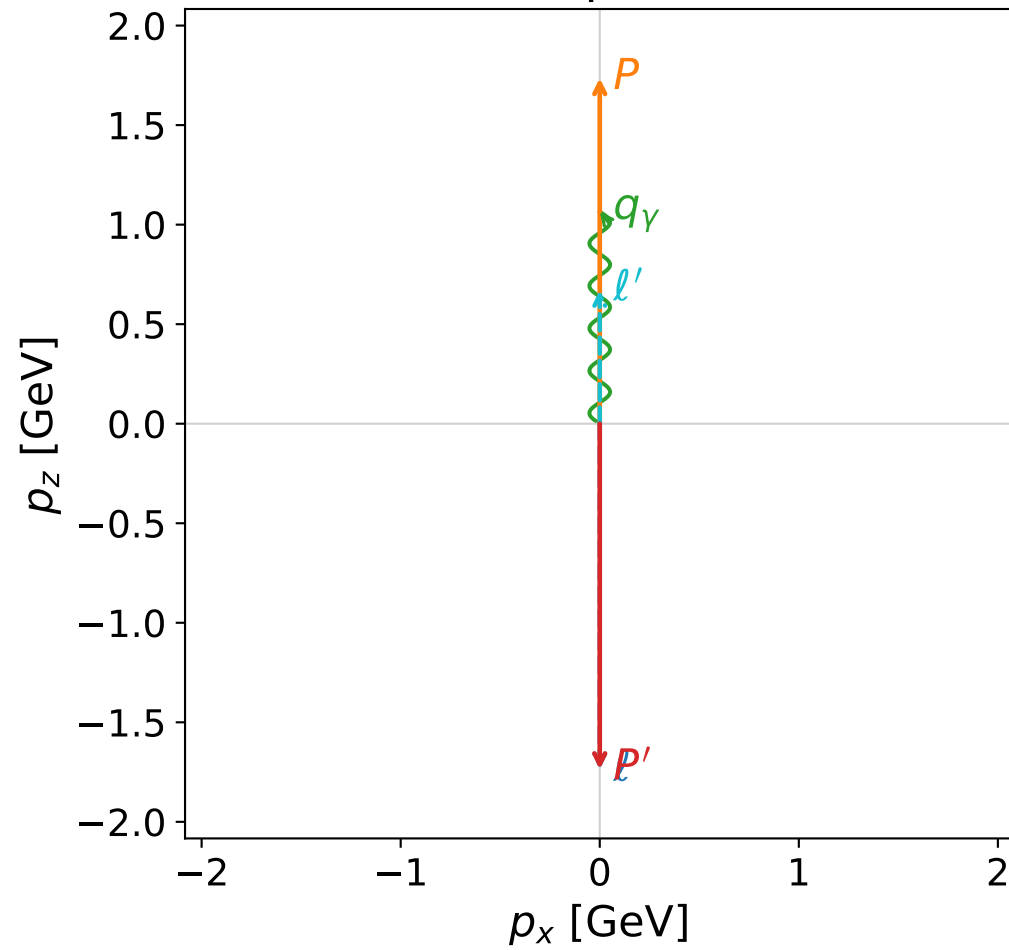
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9231$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9231$
 P $E= 2.1397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9231$
 ℓ' $E= 0.6960$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.1397$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9231$
 q_γ $E= 1.2272$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2272$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.695\text{e}+24$; $I_h=+2.010\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.024\text{e-}30$; $I_{TL}=+2.009\text{e-}15$

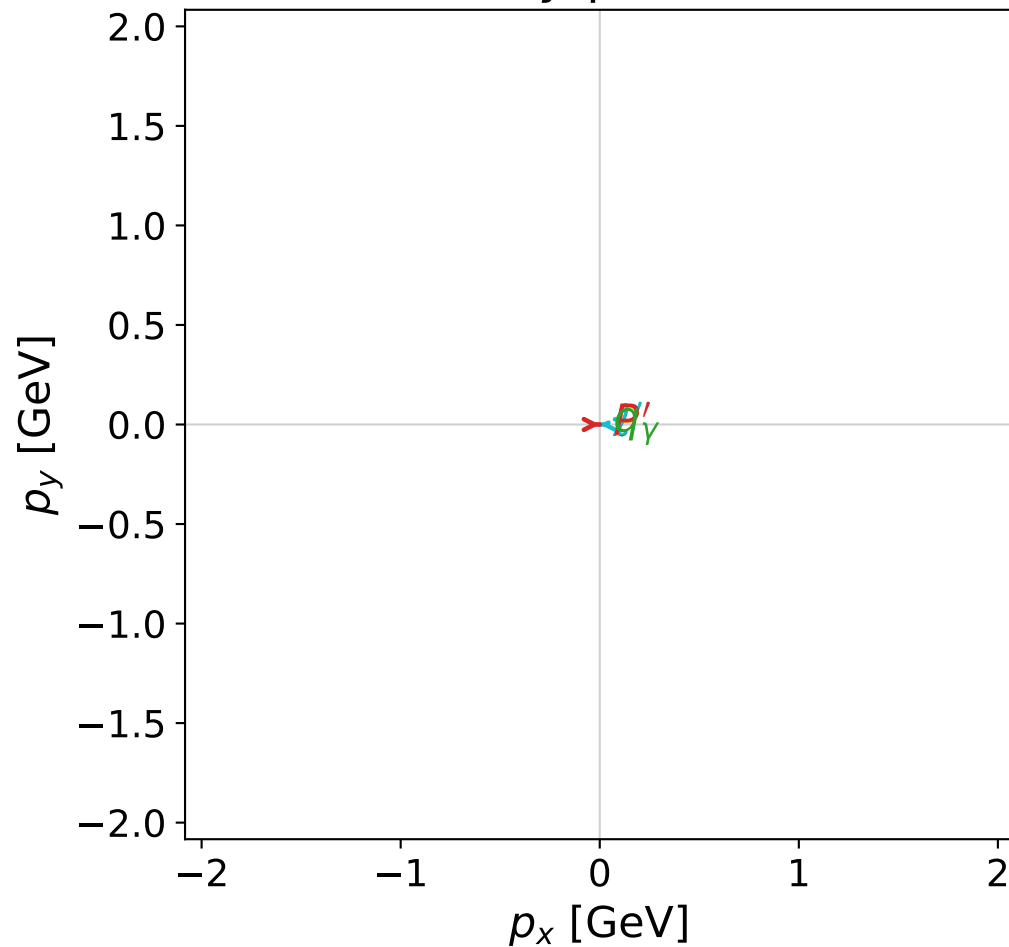


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

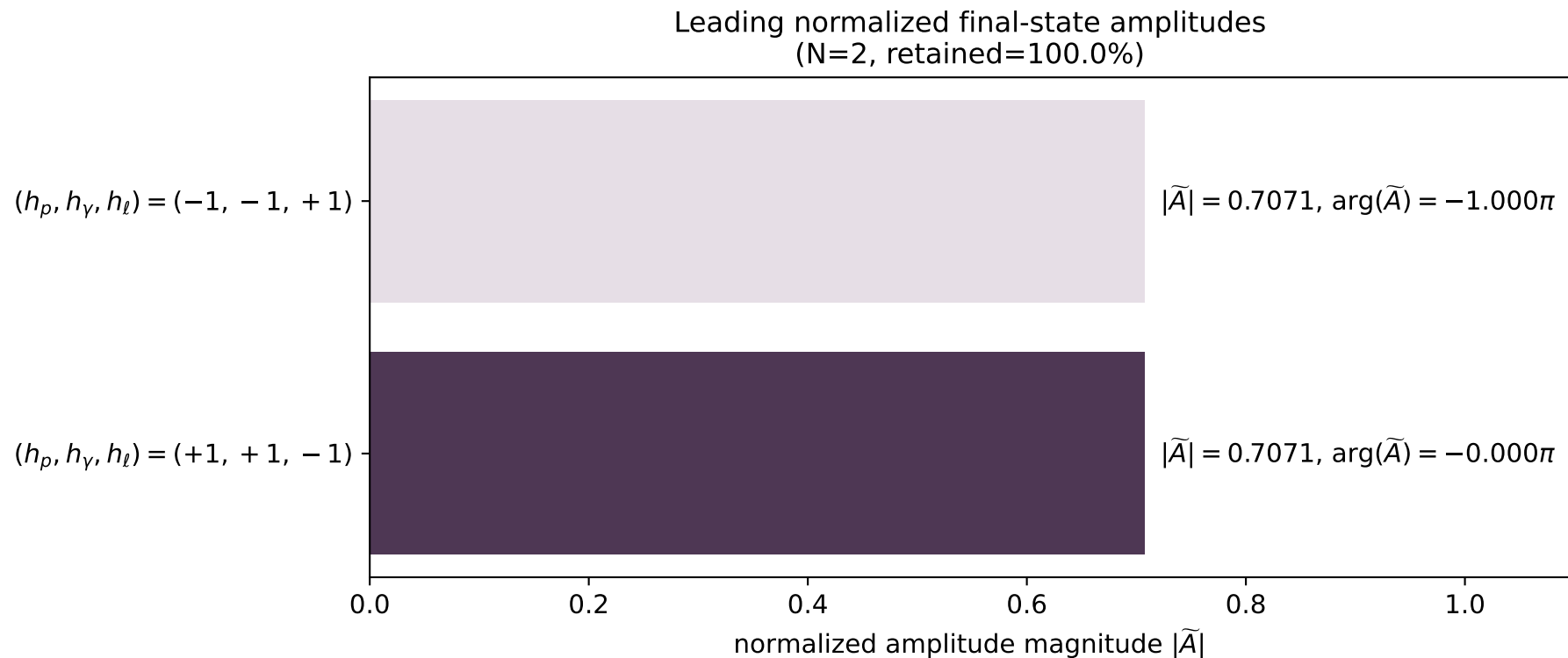


dghz_mixing_angles_polarization_04_local_minimum_162
 $d_{\text{GHZ}}=1.14455\text{e-}07$
 region: polarization_cluster_04_local_minimum_162
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0162
 lepton mixing angle: $\alpha_e=2.254464$ rad
 incoming lepton: $-0.631641|+\rangle +0.775261|-\rangle$
 proton mixing angle: $\alpha_p=2.4579197$ rad
 incoming proton: $-0.775258|+\rangle +0.631645|-\rangle$

$s=13.7588$, $\sqrt{s}=3.70928$, $|\vec{P}|=1.73604$, $|\vec{P}'|=1.73604$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.143001$, $\phi_\gamma=2.52234$
 $E_\gamma=1.06513$, $\theta_{l'}=0$, $\phi_{l'}=3.28459$
 $Q^2=4.65894$, $x_B=0.370915$, $t=-12.0554$, $W^2=8.78157$, $y=0.975287$

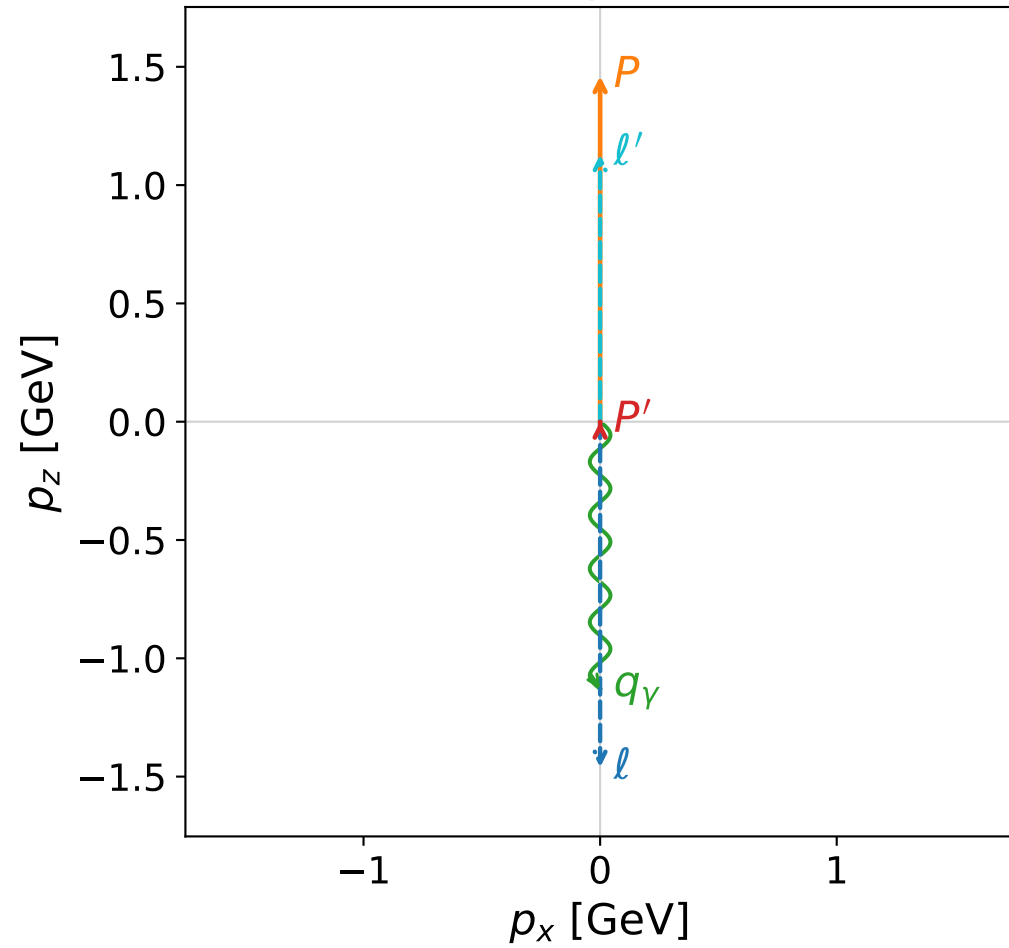
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7360$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 P $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7360$
 ℓ' $E= 0.6709$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6709$
 P' $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 q_γ $E= 1.0651$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0651$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.892\text{e}+24$; $I_h=-6.399\text{e-}16$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.134\text{e-}31$; $I_{TL}=-6.400\text{e-}16$

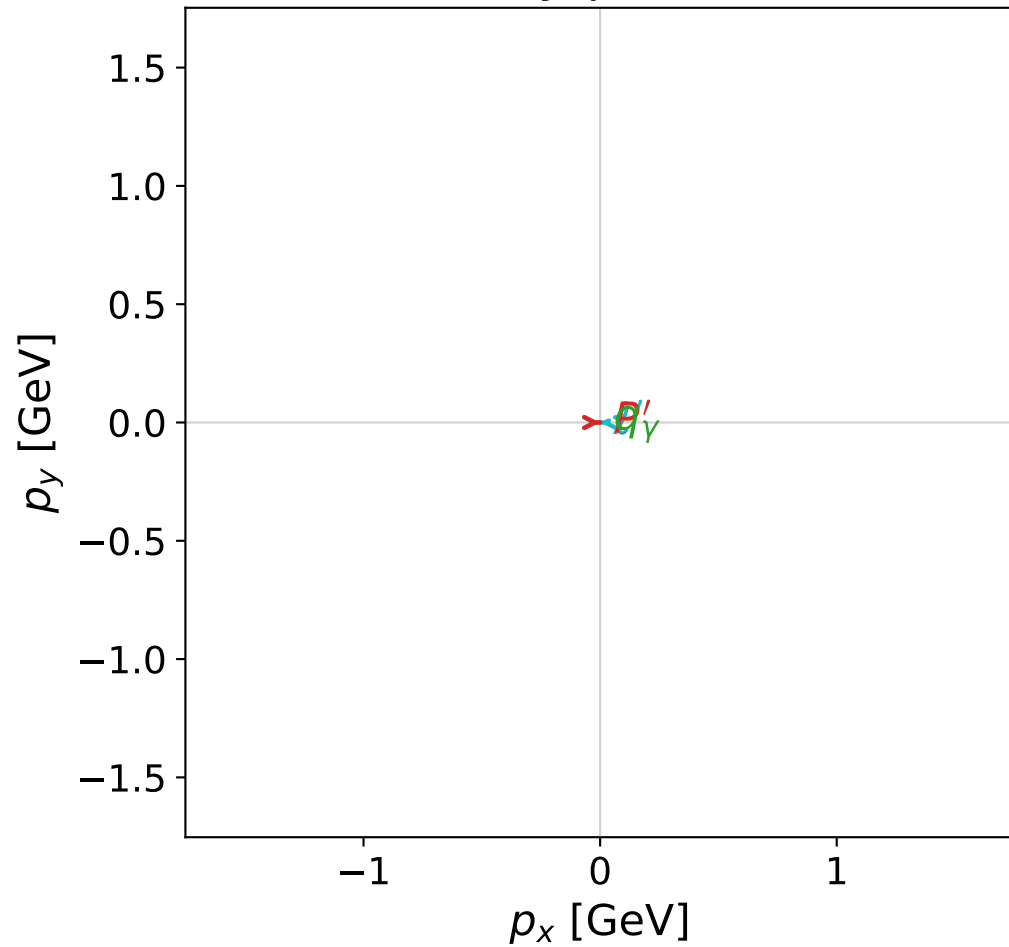


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

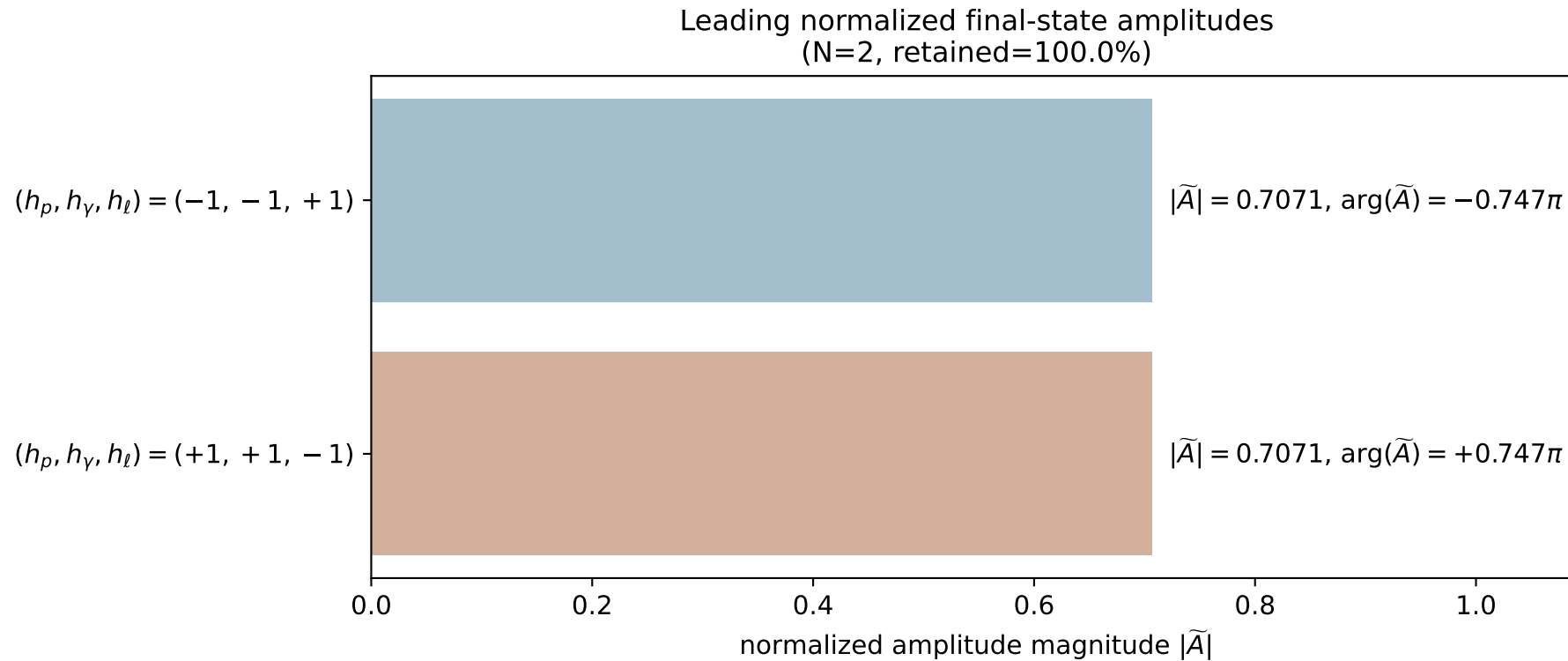


dghz_mixing_angles_polarization_04_local_minimum_166
 $d_{\text{GHZ}}=1.3556\text{e-}07$
 region: polarization_cluster_04_local_minimum_166
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0166
 lepton mixing angle: $\alpha_e=1.8819764$ rad
 incoming lepton: $-0.306182|+\rangle +0.951973|-\rangle$
 proton mixing angle: $\alpha_p=1.8819874$ rad
 incoming proton: $-0.306193|+\rangle +0.95197|-\rangle$

$s=10.2184$, $\sqrt{s}=3.19662$, $|\vec{P}|=1.46069$, $|\vec{P}'|=0.000294288$
 $\theta_{P'}=0.000613329$, $\theta_Y=3.14159$, $\phi_{P'}=0.00132073$, $\phi_Y=3.93698$
 $E_Y=1.12946$, $\theta_{\ell'}=1.59797\text{e-}07$, $\phi_{\ell'}=3.14291$
 $Q^2=6.59744$, $x_B=0.75685$, $t=-1.49606$, $W^2=2.99937$, $y=0.933439$

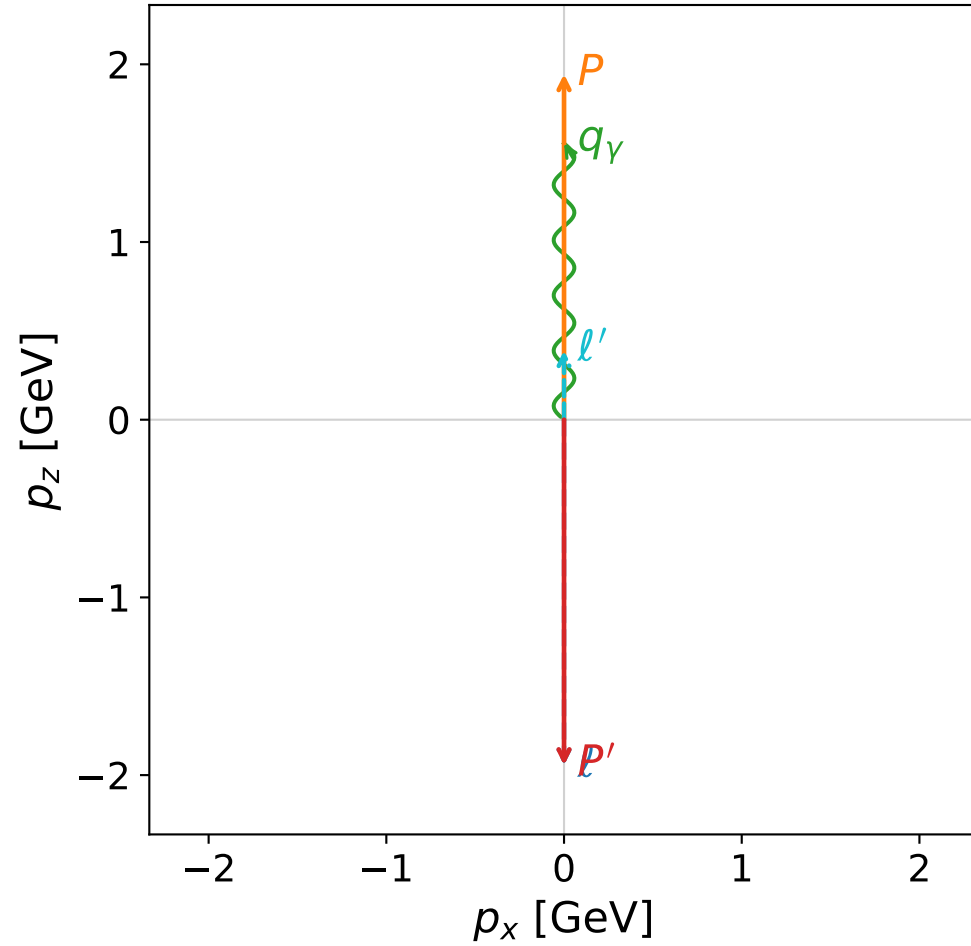
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4607$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4607$
 P $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4607$
 ℓ' $E= 1.1292$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1292$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0003$
 q_Y $E= 1.1295$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.1295$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.785\text{e+}20$; $I_h=+5.060\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.283\text{e-}15$; $I_{TL}=-4.117\text{e-}08$

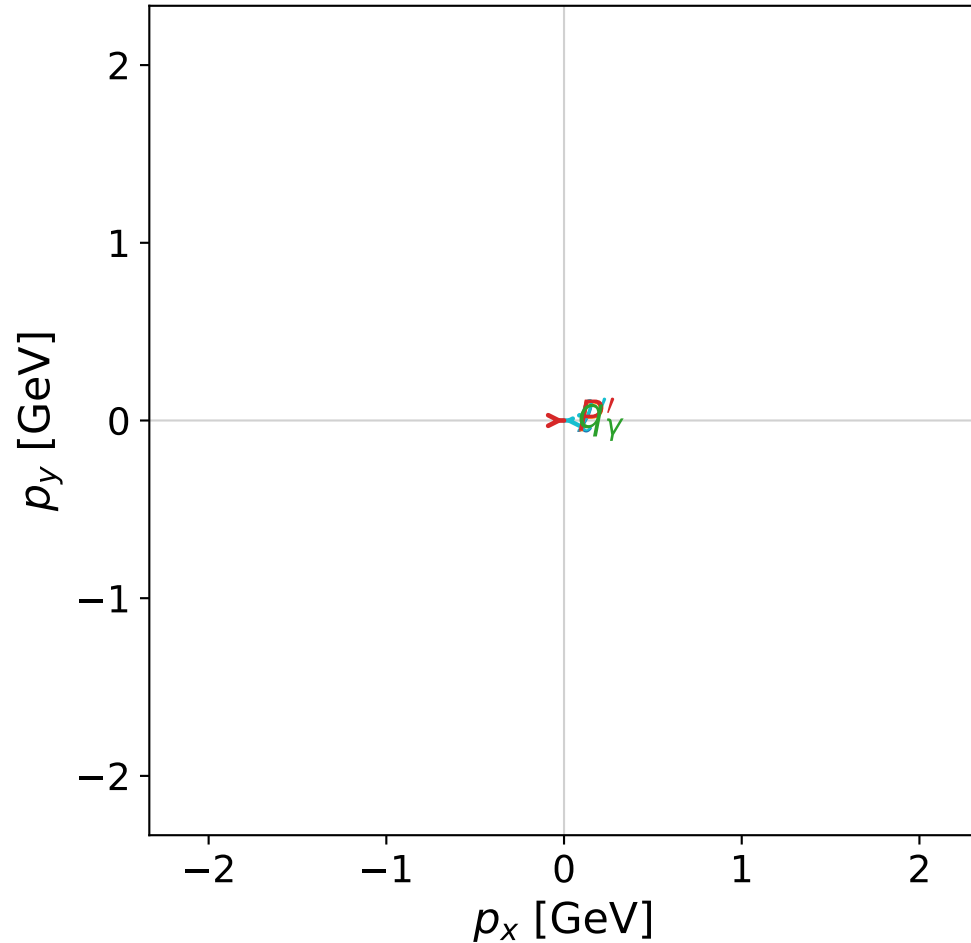


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

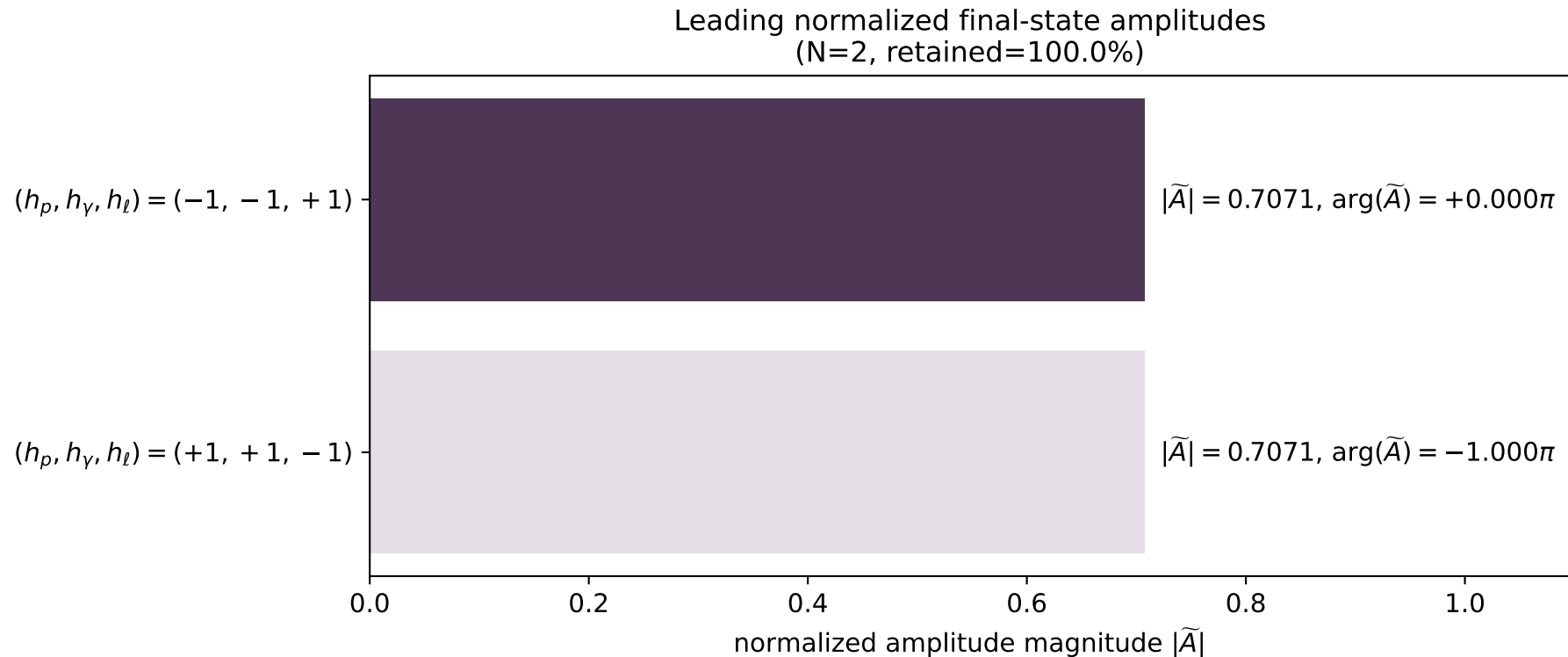


dghz_mixing_angles_polarization_04_local_minimum_169
 $d_{\text{GHZ}}=1.74025\text{e-}07$
 region: polarization_cluster_04_local_minimum_169
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0169
 lepton mixing angle: $\alpha_e=2.2827776$ rad
 incoming lepton: $-0.653335|+\rangle +0.757069|-\rangle$
 proton mixing angle: $\alpha_p=2.4296097$ rad
 incoming proton: $-0.757068|+\rangle +0.653336|-\rangle$

$s=16.843$, $\sqrt{s}=4.10403$, $|\vec{P}|=1.94482$, $|\vec{P}'|=1.94482$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=6.13063$, $\phi_\gamma=0.266108$
 $E_\gamma=1.55486$, $\theta_{l'}=0$, $\phi_{l'}=2.98904$
 $Q^2=3.03359$, $x_B=0.192048$, $t=-15.1293$, $W^2=13.6422$, $y=0.989526$

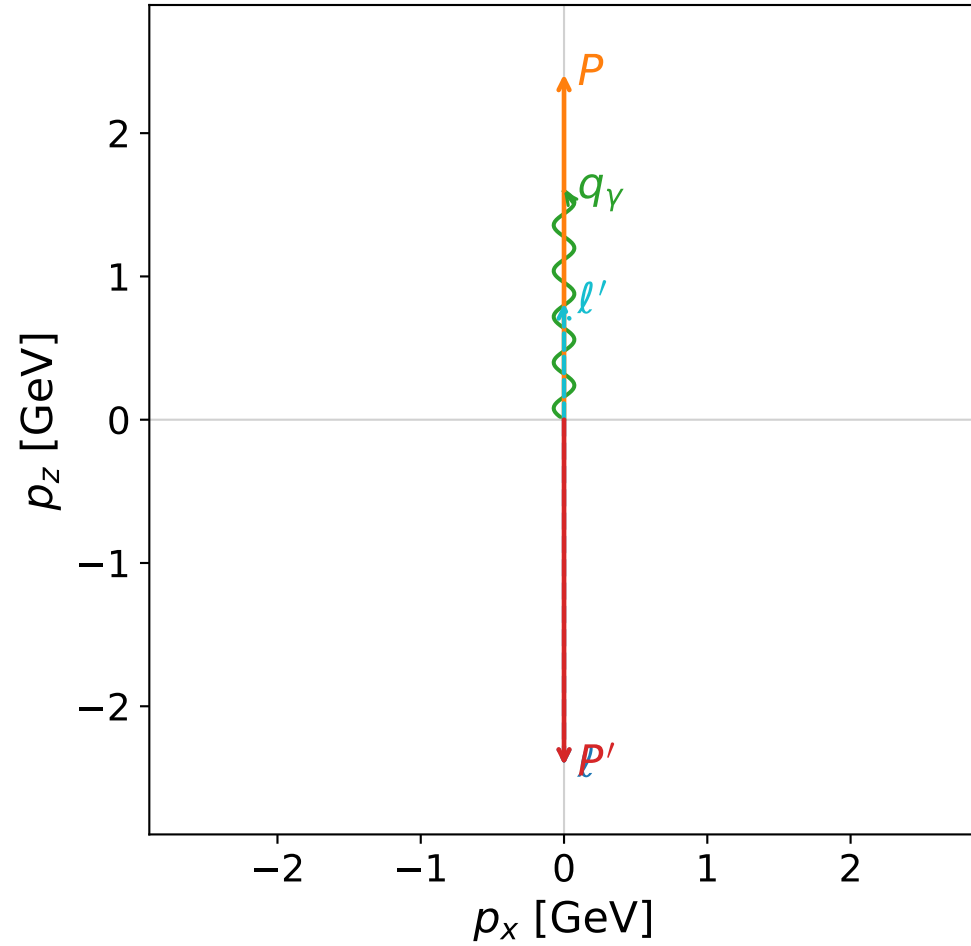
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9448$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9448$
 P $E= 2.1592$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9448$
 ℓ' $E= 0.3900$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3900$
 P' $E= 2.1592$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9448$
 q_γ $E= 1.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5549$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.828\text{e}+24$; $I_h=-2.810\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.034\text{e-}30$; $I_{TL}=-2.810\text{e-}15$

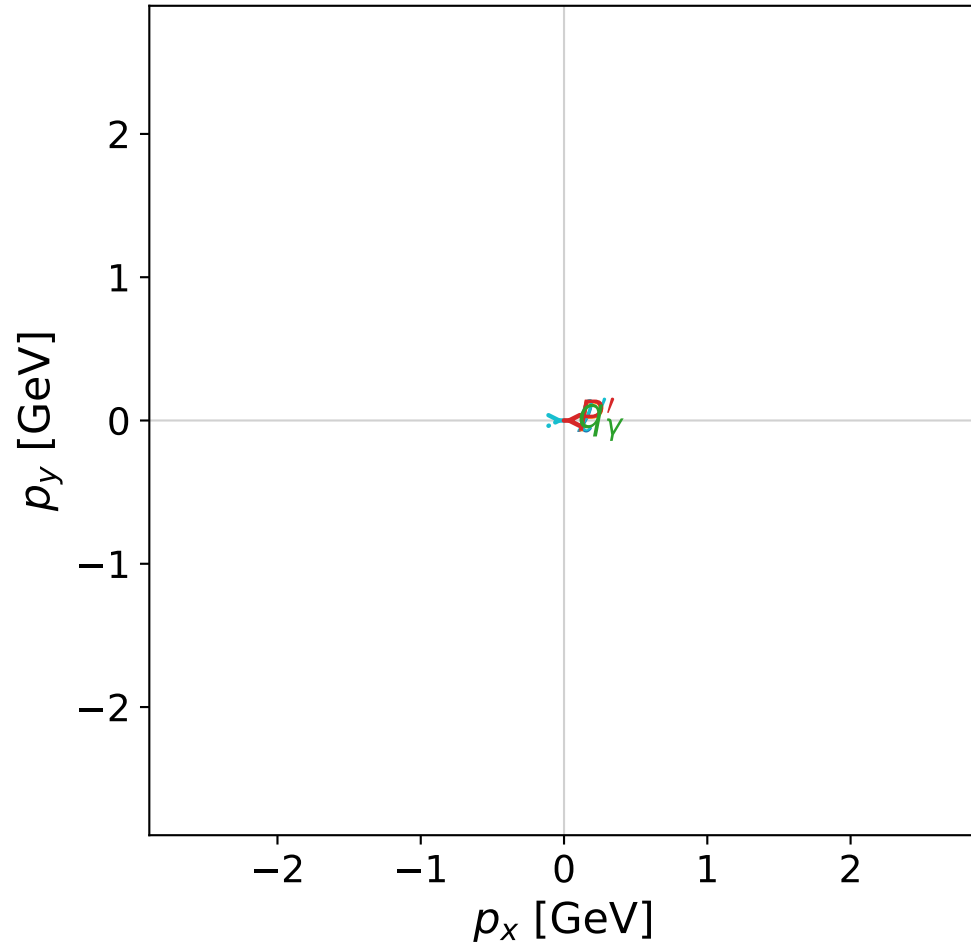


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

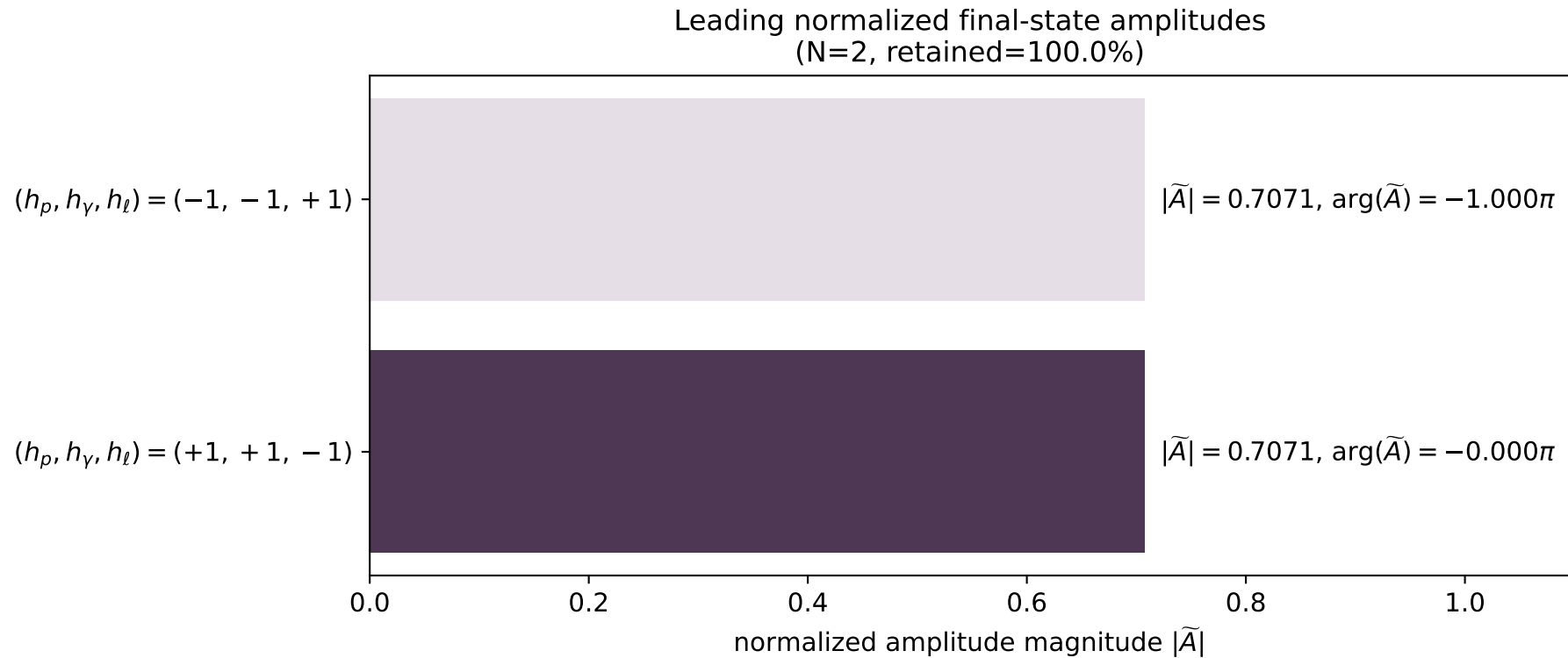


dghz_mixing_angles_polarization_04_local_minimum_176
 $d_{\text{GHZ}}=3.49411\text{e-}07$
 region: polarization_cluster_04_local_minimum_176
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0176
 lepton mixing angle: $\alpha_e=3.0459519$ rad
 incoming lepton: $-0.99543|+\rangle +0.095495|-\rangle$
 proton mixing angle: $\alpha_p=1.6664366$ rad
 incoming proton: $-0.0954945|+\rangle +0.99543|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.3165$, $\phi_\gamma=2.32512$
 $E_\gamma=1.59717$, $\theta_{l'}=0$, $\phi_{l'}=0.17491$
 $Q^2=7.86165$, $x_B=0.329859$, $t=-23.2713$, $W^2=16.8516$, $y=0.988111$

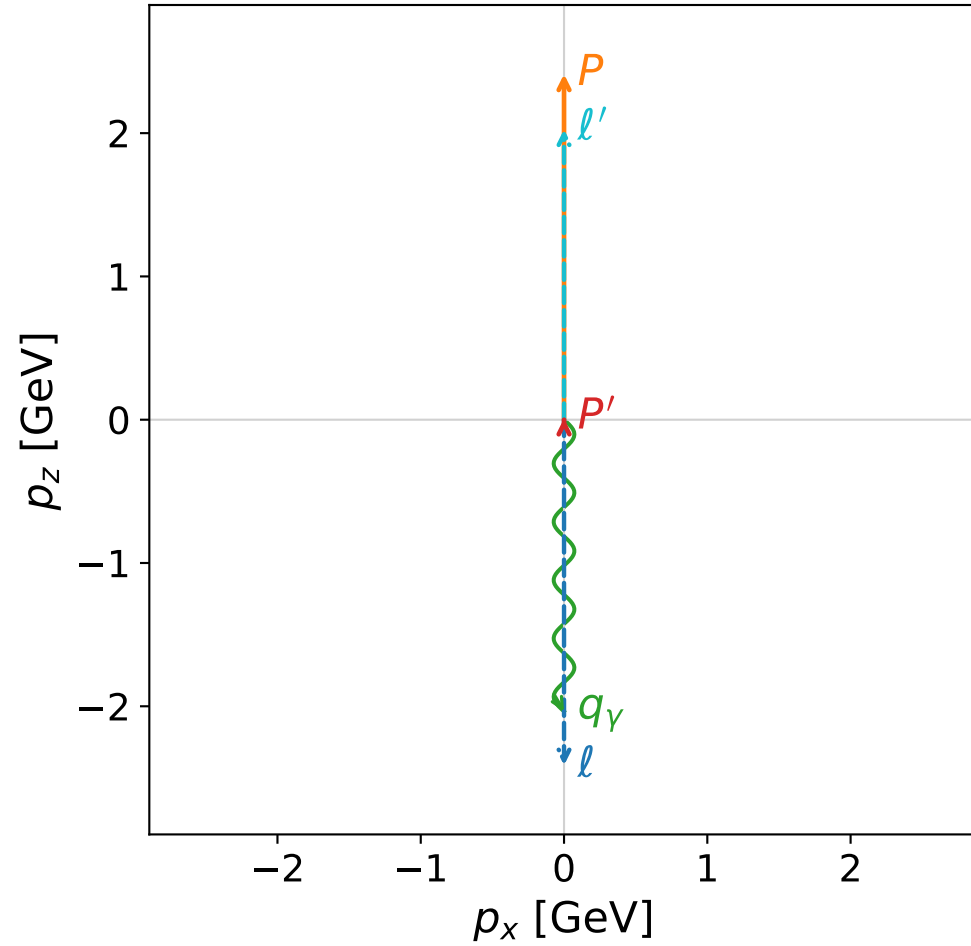
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.8148$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8148$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 1.5972$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5972$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.211\text{e+}24$; $I_h=+1.847\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.719\text{e-}27$; $I_{TL}=+1.844\text{e-}14$

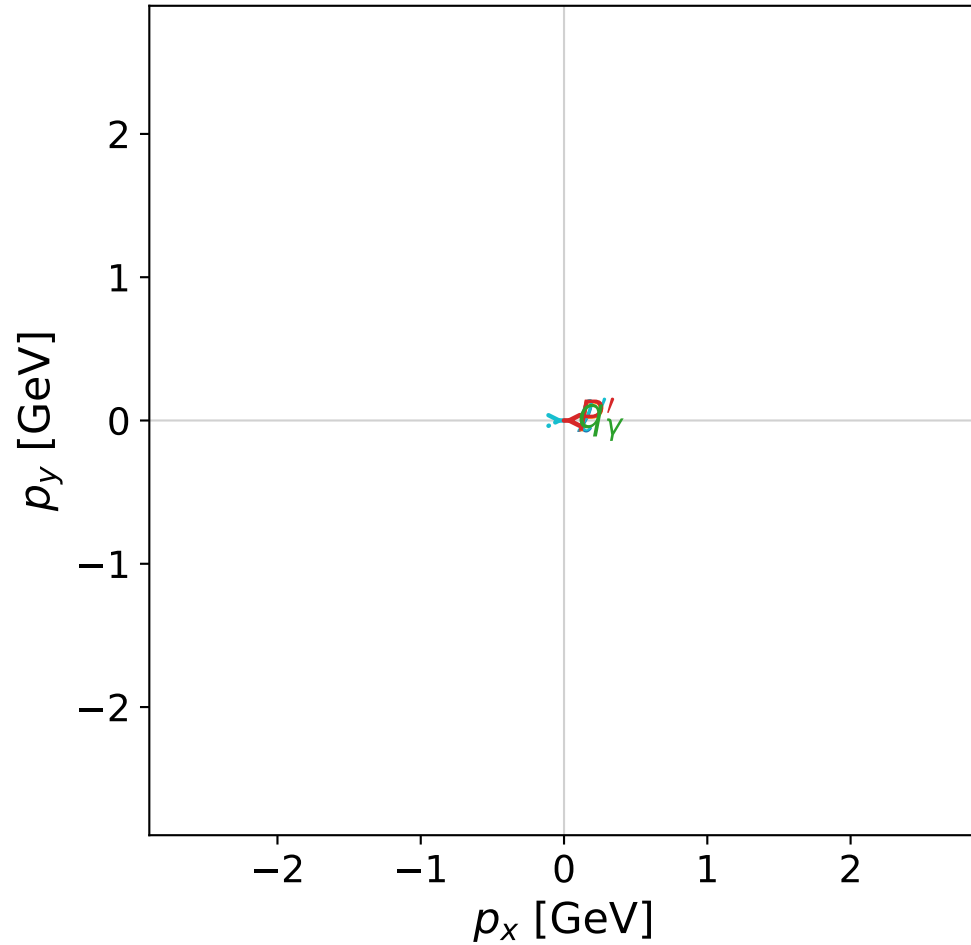


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

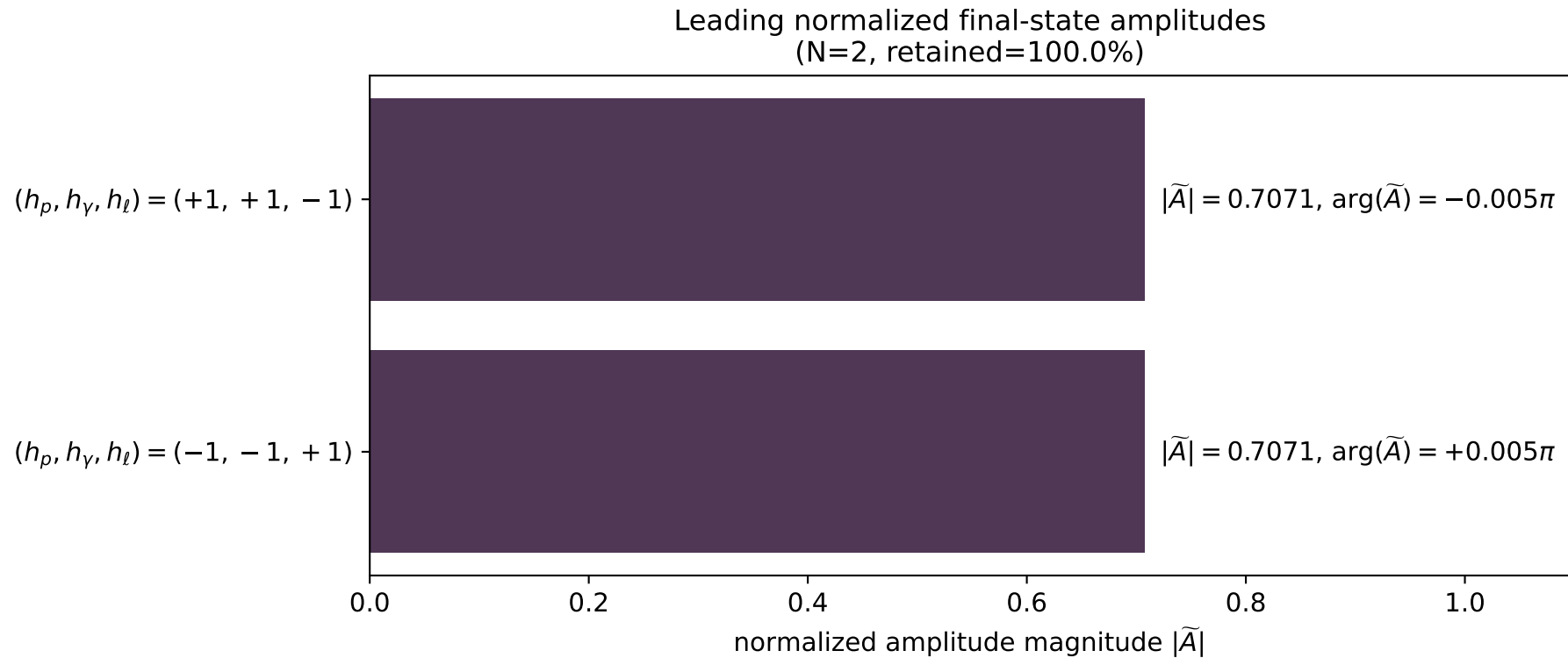


dghz_mixing_angles_polarization_04_local_minimum_177
 $d_{\text{GHZ}}=4.14871\text{e-}07$
 region: polarization_cluster_04_local_minimum_177
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0177
 lepton mixing angle: $\alpha_e=3.1093625$ rad
 incoming lepton: $-0.999481|+\rangle +0.0322245|-\rangle$
 proton mixing angle: $\alpha_p=3.1093621$ rad
 incoming proton: $-0.999481|+\rangle +0.0322249|-\rangle$

$s=24.9987$, $\sqrt{s}=4.99987$, $|\vec{P}|=2.41195$, $|\vec{P}'|=0.00931426$
 $\theta_{P'}=0.00233638$, $\theta_Y=3.14159$, $\phi_{P'}=3.14277$, $\phi_Y=0.0168325$
 $E_Y=2.03557$, $\theta_{\ell'}=1.07398\text{e-}05$, $\phi_{\ell'}=0.00117805$
 $Q^2=19.5489$, $x_B=0.835218$, $t=-3.05056$, $W^2=4.73668$, $y=0.970433$

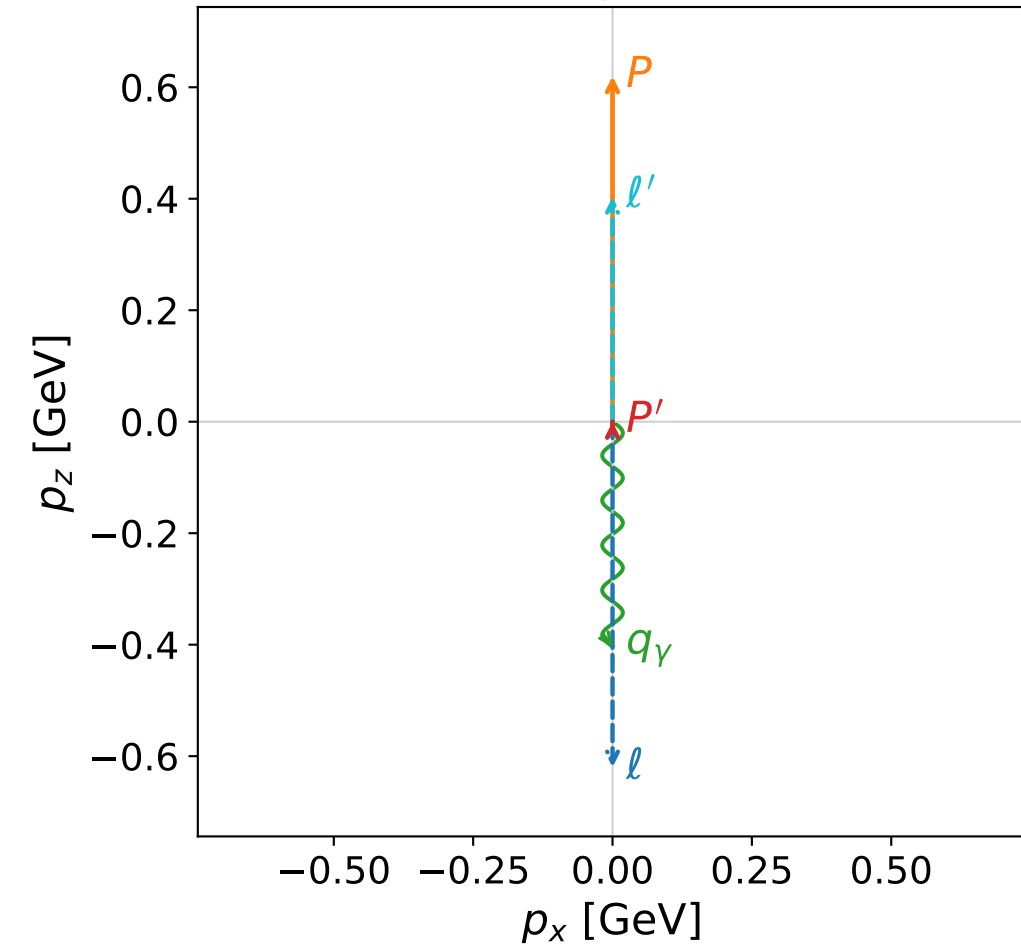
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0263$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0093$
 q_γ $E= 2.0356$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0356$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.283\text{e+}15$; $I_h=+2.089\text{e-}08$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.062\text{e-}10$; $I_{TL}=-1.454\text{e-}05$

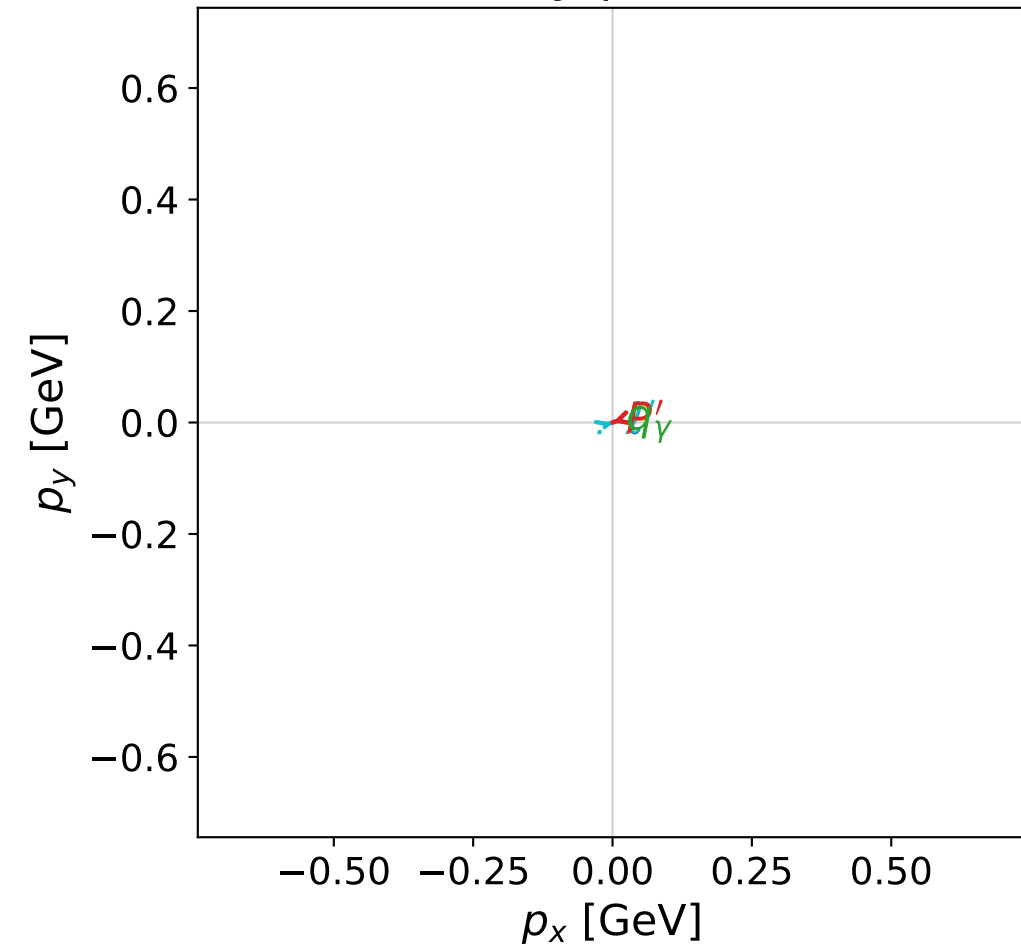


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_183
 $d_{\text{GHZ}}=1.00228\text{e-}06$
 region: polarization_cluster_04_local_minimum_183
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0183
 lepton mixing angle: $\alpha_e=1.8447186$ rad
 incoming lepton: $-0.27051|+\rangle +0.962717|-\rangle$
 proton mixing angle: $\alpha_p=1.8447094$ rad
 incoming proton: $-0.270501|+\rangle +0.96272|-\rangle$

$s=3.04272$, $\sqrt{s}=1.74434$, $|\vec{P}|=0.61997$, $|\vec{P}'|=1.00646\text{e-}06$
 $\theta_{p'}=0.00076699$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.44967$, $\phi_\gamma=6.28171$
 $E_\gamma=0.40317$, $\theta_{l'}=0$, $\phi_{l'}=0.308074$
 $Q^2=0.999811$, $x_B=0.569317$, $t=-0.349628$, $W^2=1.63619$, $y=0.811955$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.6200$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6200$
 P $E= 1.1244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6200$
 l' $E= 0.4032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4032$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.4032$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4032$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.291\text{e+}23$; $I_h=-2.157\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.829\text{e-}18$; $I_{TL}=+1.556\text{e-}09$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

